SHREE VENKATESHWARA

HI-TECH ENGINEERING COLLEGE (AUTONOMOUS)

GOBI-638 455

NAME OF THE LAB: 23CSE15-BUSINESS ANALYTICS LAB

University Register No:______________________________

Name:______________________Roll No:______________

Branch:-______________________Batch:__________

Certified that this is bonafide record of work done by the above student during the year 20 to 20 .

LAB IN-CHARGE **HEAD OF THE DEPARTMENT**

Submitted for the practical Examination held on______________

INTERNAL EXAMINER **EXTERNAL EXAMINER**

23CSE15- BUSINESS ANALYTICS LAB

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8.		APPLY AND EXPLORE VARIOUS PLOTTING FUNCTIONS ON THE DATA SET			
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10.		PREPARE & LOAD DATA			
11.		DEVELOP THE DATA MODEL			
12.		PERFORM DAX CALCULATIONS			
13.		DESIGN A REPORT			
14.		CREATE A DASHBOARD AND PERFORM DATA ANALYSIS			

23CSE15 BUSINESS ANALYTICS LABORATORY L T P C 2 0 2 3

OBJECTIVES:

- To understand the Analytics Life Cycle.
- To comprehend the process of acquiring Business Intelligence.
- To understand various types of analytics for Business Forecasting
- To model the supply chain management for Analytics.
- To apply analytics for different functions of a business

LIST OF EXPERIMENTS:

1. Explore The Features Of Ms- Excel
2. Numerical And Import Export Operations In Excel
3. Statistical Operations
4. Perform Z-Test, T-Test & Anova
5. Perform Data Pre- Processing
6. Perform Dimensionality Reduction Operation Using Pca,Kpca & Svd
7. Perform Bivariate And Multivariate Analysis On The Dataset
8. Apply And Explore Various Plotting Functions On The Data Set
9. Explore The Features Of Power Bi Desktop
10. Prepare & Load Data
11. Develop The Data Model
12. Perform Dax Calculations
13. Design A Report
14. Create A Dashboard And Perform Data Analysis

OUTCOMES:

At the end of this course, the students will be able to:

CO1: Explain the real world business problems and model with analytical solutions.
CO2: Identify the business processes for extracting Business Intelligence
CO3 : Apply predictive analytics for business fore-casting
CO4: Apply analytics for supply chain and logistics management
CO5: Use analytics for marketing and sales.

Ex.No.1	EXPLORE THE FEATURES OF MS-EXCEL
DATE:	

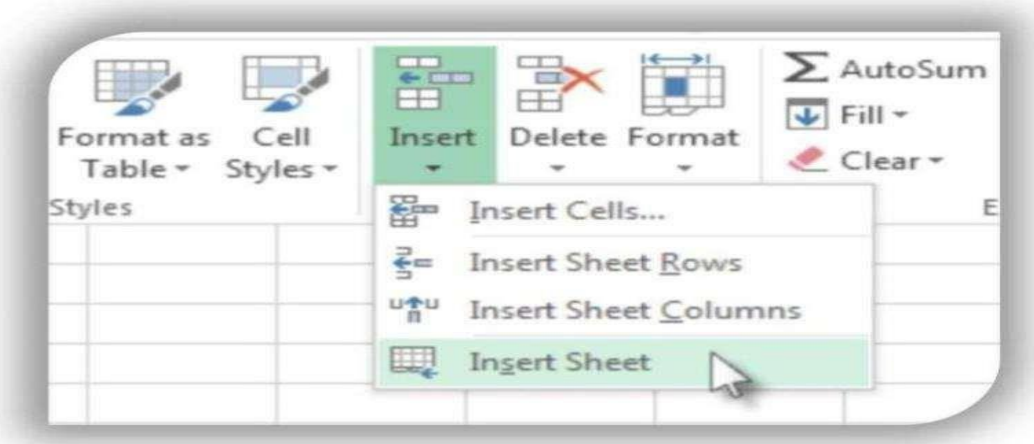
AIM: To explore the amazing features of MS-Excel.

Most professionals use Microsoft Excel to store their data and represent it understandably. It is similar to Google Sheets. To know all the features of Excel which are helpful and time-saving. Here are the top 10 unique features of Microsoft Excel. To analyze the data quickly. Microsoft Excel is used in any type of device like Windows, macOS, Android, and iOS.

PROCEDURE:

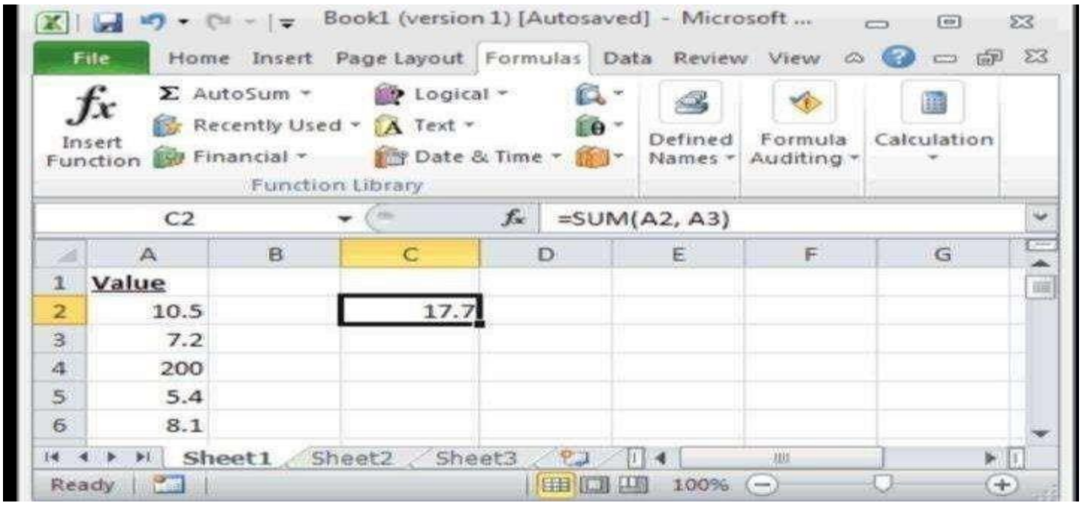
1. Inserting a Spreadsheet

By default, you will have 1-3 sheets at the beginning. You may need multiple spreadsheets for your data. Inserting and deleting new spreadsheets is quite simple in MS Excel. Click on the "+" button to insert a new sheet. You can also use the shortcut Shift+F11 to insert a new spreadsheet.



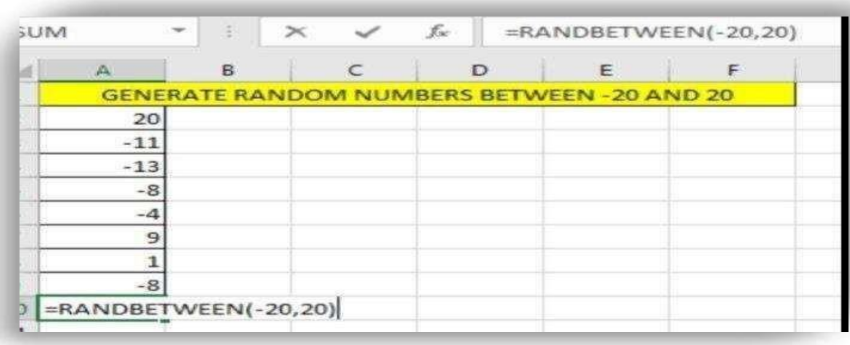
2. Sum Of Numbers :

To enter a list of numbers in a column and find the sum. Don't search for a Calculator or pen and paper to perform addition. MS Excel has a cool feature to find the sum easily. Select the cell of which you want to find the sum, and press the shortcut Alt+=. Tadaa! The sum is displayed automatically.



3. **Inserting Random Numbers:**

You can find plenty of unique features and functions in Excel. One of them is "RANDBETWEEN." You can use this function to insert random numbers into the sheet. It takes two arguments. The first one is the least number you are going to insert into your sheet, and the second one is the largest number. With this feature, you don't have to waste time guessing the numbers. You can finish this within a fraction of a second.

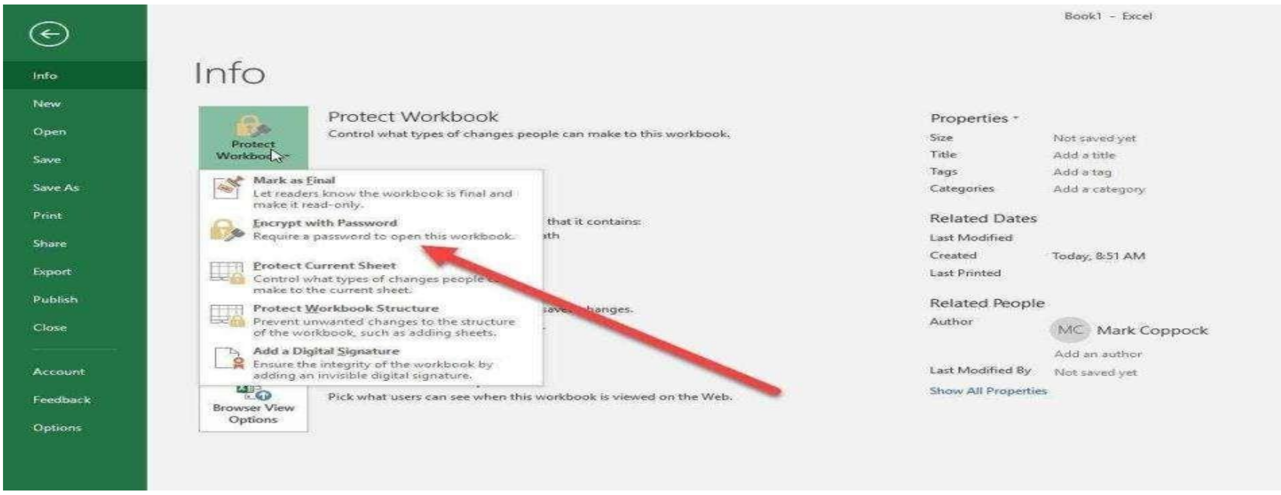


4. **Shortcut Keys**

Excel has made it easy to analyze data. Now it is easier to access the data when you know the shortcut keys. You can perform certain operations without touching the mouse in a fraction of a second. Some of them are

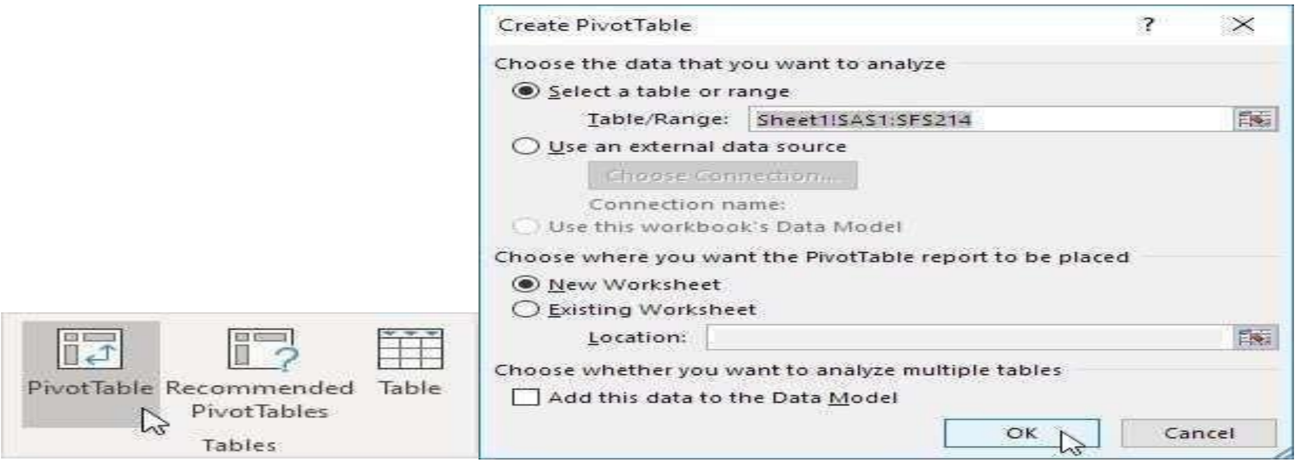
5.Password.Protection

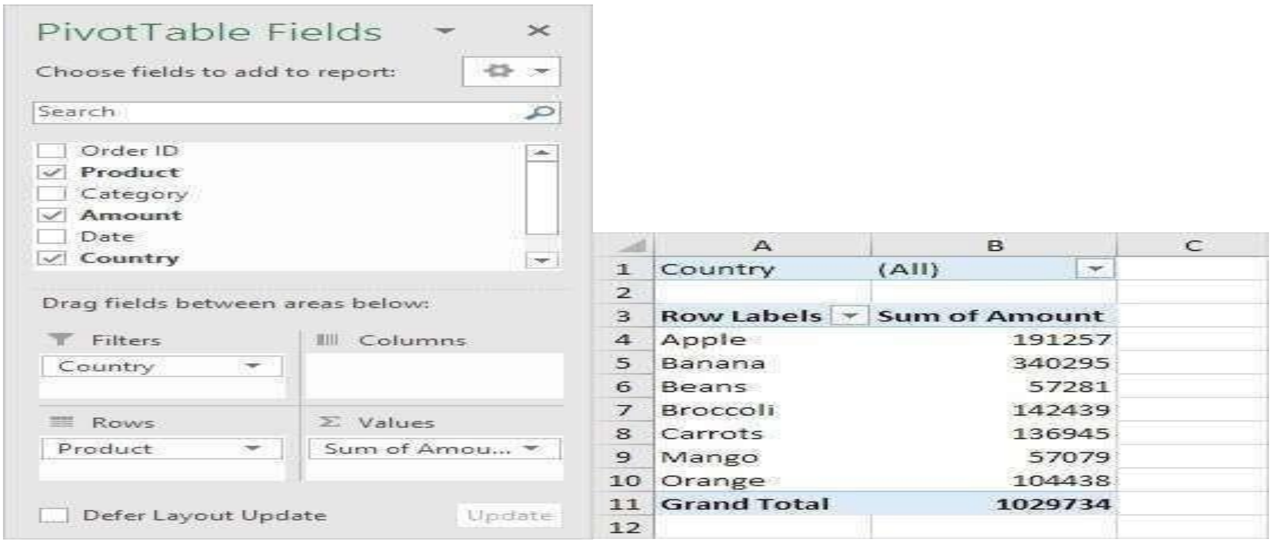
You may have some private data that you want to keep secured. If you are worried about unauthorized access, you can use password protection to ensure your safety. After enabling this feature, no one can open your document without the password.



6.Pivot Tables

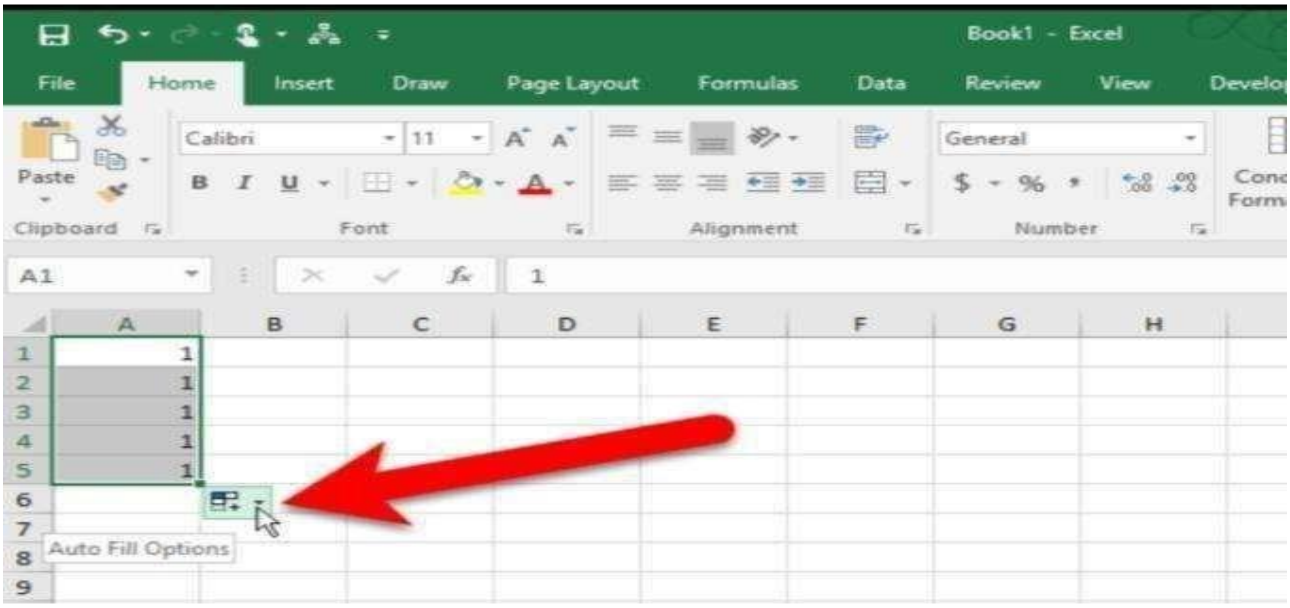
Analysing the data in Excel is quite interesting. If you want to create a table that is clear to understand, you can use the pivot table feature.





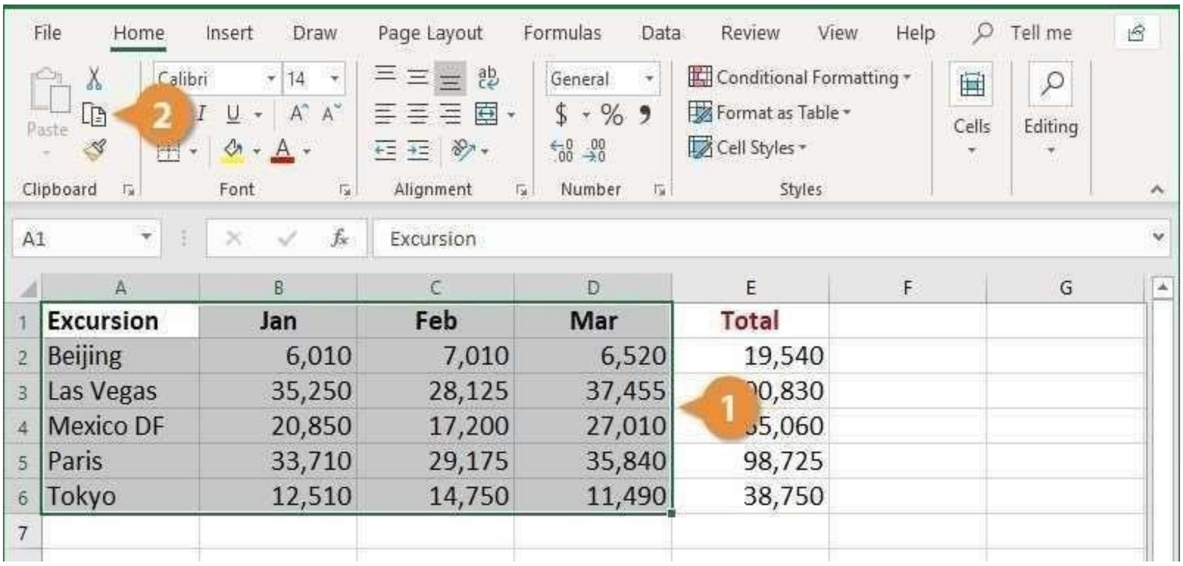
7. Auto-Fill

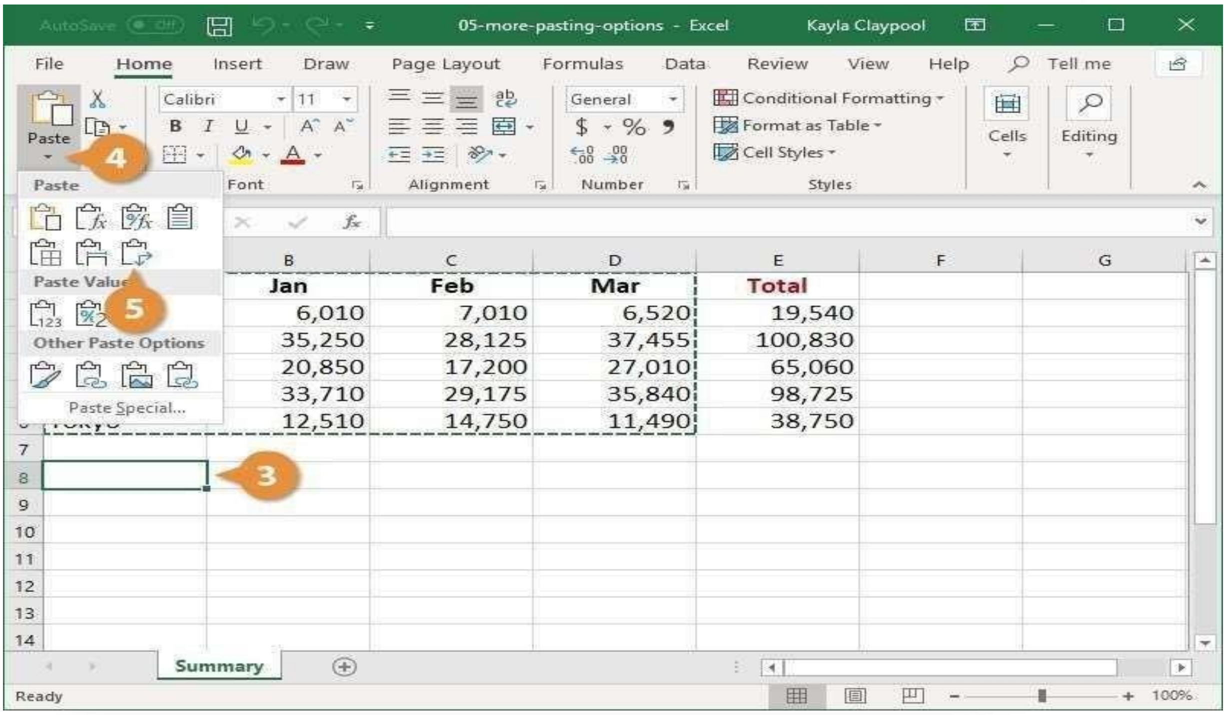
Auto-fill is a simple but useful feature in Excel. With this feature, you can fill the data in series. You can save a lot of time with this feature.



8. Paste Special:

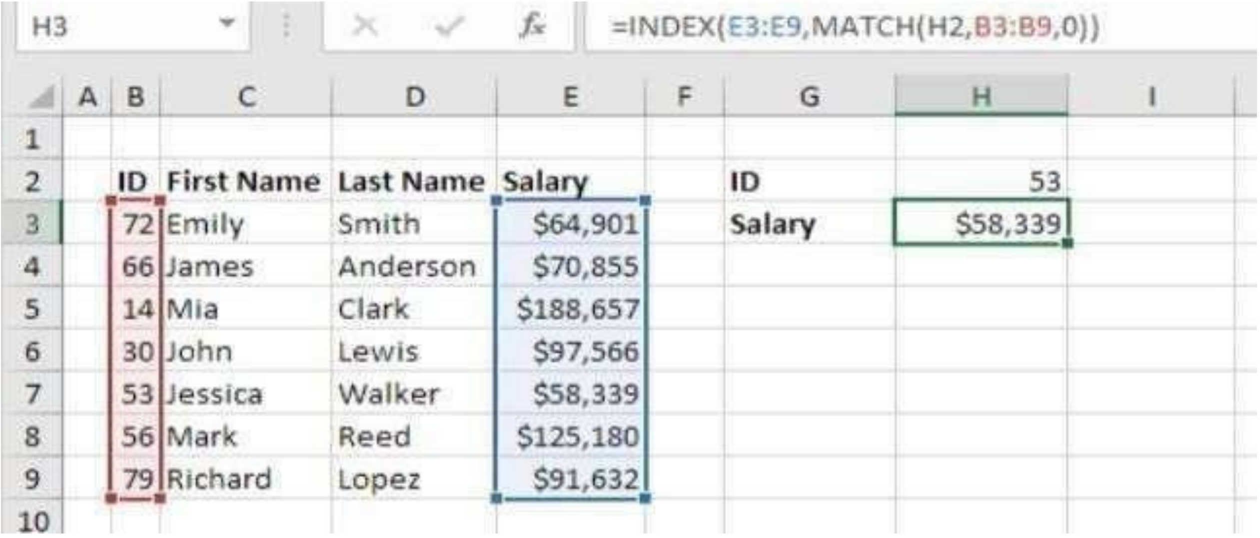
When we started to use the copy and paste option, we didn't have to type everything. But sometimes, you may copy formulae, values, and comments. Sometimes you may not want to copy all of them. In such situations, you can use Paste Special feature. It is also a time-saving feature.





9.Index-Match:

Index-Match future comes in handy when you have to handle a massive database with vast data. If you want to find a particular data, you may feel hard. It will seem, like searching for a drop of blood in an ocean. But Index-Match future will find and get you the desired information instantly. But remember the index should be unique like passport number or roll number.

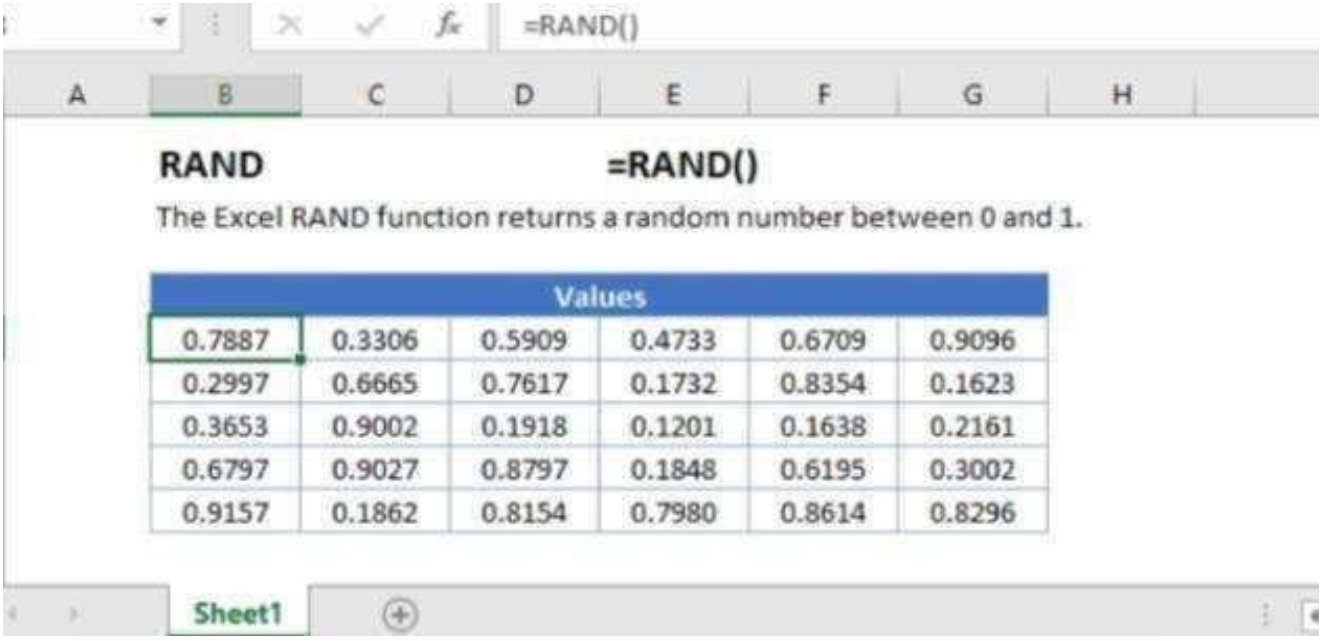


H3 =INDEX(E3:E9,MATCH(H2,B3:B9,0))

ID	First Name	Last Name	Salary		ID	53
72	Emily	Smith	$64,901		Salary	$58,339
66	James	Anderson	$70,855			
14	Mia	Clark	$188,657			
30	John	Lewis	$97,566			
53	Jessica	Walker	$58,339			
56	Mark	Reed	$125,180			
79	Richard	Lopez	$91,632			

10. Rand Function

We can enter random values with the RANDBETWEEN function. But what to do if we want to enter random fraction values? Then, we can use the RAND function. Using this function, we can enter values between 1 and 0. It does not have any parameters. Just put the parentheses. Note that RANDBETWEEN and RAND functions will vary when we change the sheet.



=RAND()

RAND =RAND()

The Excel RAND function returns a random number between 0 and 1.

Values					
0.7887	0.3306	0.5909	0.4733	0.6709	0.9096
0.2997	0.6665	0.7617	0.1732	0.8354	0.1623
0.3653	0.9002	0.1918	0.1201	0.1638	0.2161
0.6797	0.9027	0.8797	0.1848	0.6195	0.3002
0.9157	0.1862	0.8154	0.7980	0.8614	0.8296

Sheet1

Result:

Thus Microsoft excel features was explored successfully.

Ex.No.2	NUMERICAL AND IMPORT EXPORT OPERATIONS IN EXCEL
DATE:	

a) **Get the input from user and perform numerical operations (MAX, MIN, AVG, SUM, SQRT, ROUND)**

AIM

To perform numerical operations using Excel such as MAX, MIN, AVG, SUM, SQRT, ROUND.

PROCEDURE:

Get the input values from user

	A	B
1	Product	Price(Rs)
2	Honey	20
3	Milk	15
4	sugar	13
5	Bread	11
6	Egg	16
7	Banana	12
8	Butter	19

The MAX function returns the highest value in a set of data.

Firstly, in Cell B12 type the formula given below-

=MAX(B2:B8)

B12 =MAX(B2:B8)

	A	B	C	D
1	Product	Price(Rs)		
2	Honey	20		
3	Milk	15		
4	sugar	13		
5	Bread	11		
6	Egg	16		
7	Banana	12		
8	Butter	19		
9				
10				
11				
12	MAX	20		

The MIN function returns the smallest value from the numbers provided.

Firstly, in Cell B13 type the formula given below- =MIN(B2:B8)

B13 =MIN(B2:B8)

	A	B	C	D
1	Product	Price(Rs)		
2	Honey	20		
3	Milk	15		
4	sugar	13		
5	Bread	11		
6	Egg	16		
7	Banana	12		
8	Butter	19		
9				
10				
11				
12	MAX	20		
13	MIN	11		

The AVERAGE function calculates the average (arithmetic mean) of a group of numbers.

Firstly, in Cell B14 type the formula given below-
=AVERAGE(B2:B8)

B14 =AVERAGE(B2:B8)

	A	B	C	D
1	Product	Price(Rs)		
2	Honey	20		
3	Milk	15		
4	sugar	13		
5	Bread	11		
6	Egg	16		
7	Banana	12		
8	Butter	19		
9				
10				
11				
12	MAX	20		
13	MIN	11		
14	AVERAGE	15.14285714		

The SUM function calculates the SUM of a group of numbers in Excel.
in Cell B15 type the formula given below

B15 =SUM(B2:B8)

	A	B	C	D
1	Product	Price(Rs)		
2	Honey	20		
3	Milk	15		
4	sugar	13		
5	Bread	11		
6	Egg	16		
7	Banana	12		
8	Butter	19		
9				
10				
11				
12	MAX	20		
13	MIN	11		
14	AVERAGE	15.14285714		
15	SUM	106		

The Excel SQRT function returns the square root of positive number.

B16 =SQRT(B6)

	A	B	C	D
1	Product	Price(Rs)		
2	Honey	20		
3	Milk	15		
4	sugar	13		
5	Bread	11		
6	Egg	16		
7	Banana	12		
8	Butter	19		
9				
10				
11				
12	MAX	20		
13	MIN	11		
14	AVERAGE	15.14285714		
15	SUM	106		
16	SQRT	4		

The ROUND function rounds a number to a specified number of digits.

B17 =ROUND(B14,2)

	A	B	C	D
1	Product	Price(Rs)		
2	Honey	20		
3	Milk	15		
4	sugar	13		
5	Bread	11		
6	Egg	16		
7	Banana	12		
8	Butter	19		
9				
10				
11				
12	MAX	20		
13	MIN	11		
14	AVERAGE	15.14285714		
15	SUM	106		
16	SQRT	4		
17	ROUND	15.14		

B18 =ROUNDUP(B14,0)

	A	B	C	D
1	Product	Price(Rs)		
2	Honey	20		
3	Milk	15		
4	sugar	13		
5	Bread	11		
6	Egg	16		
7	Banana	12		
8	Butter	19		
9				
10				
11				
12	MAX	20		
13	MIN	11		
14	AVERAGE	15.14285714		
15	SUM	106		
16	SQRT	4		
17	ROUND	15.14		
18	ROUND UP	16		

B19 =ROUNDDOWN(B17,0)

	A	B	C	D	E
1	Product	Price(Rs)			
2	Honey	20			
3	Milk	15			
4	sugar	13			
5	Bread	11			
6	Egg	16			
7	Banana	12			
8	Butter	19			
9					
10					
11					
12	MAX	20			
13	MIN	11			
14	AVERAGE	15.14285714			
15	SUM	106			
16	SQRT	4			
17	ROUND	15.14			
18	ROUND UP	16			
19	ROUND DOWN	15			

b) Perform data import/export operations for different file formats.

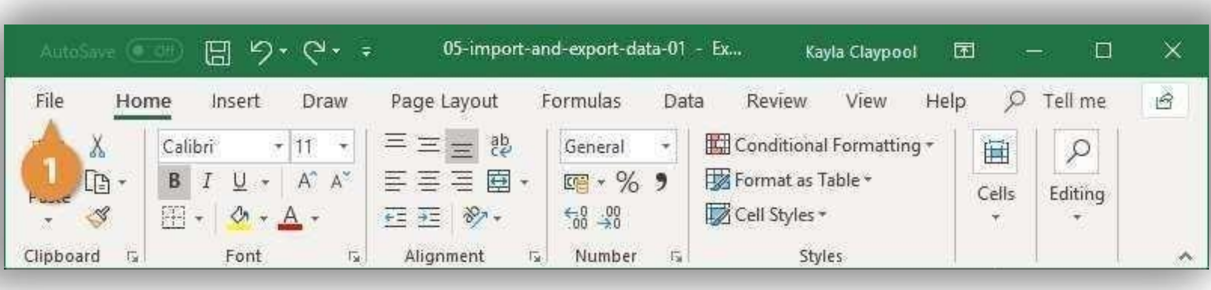
Aim:

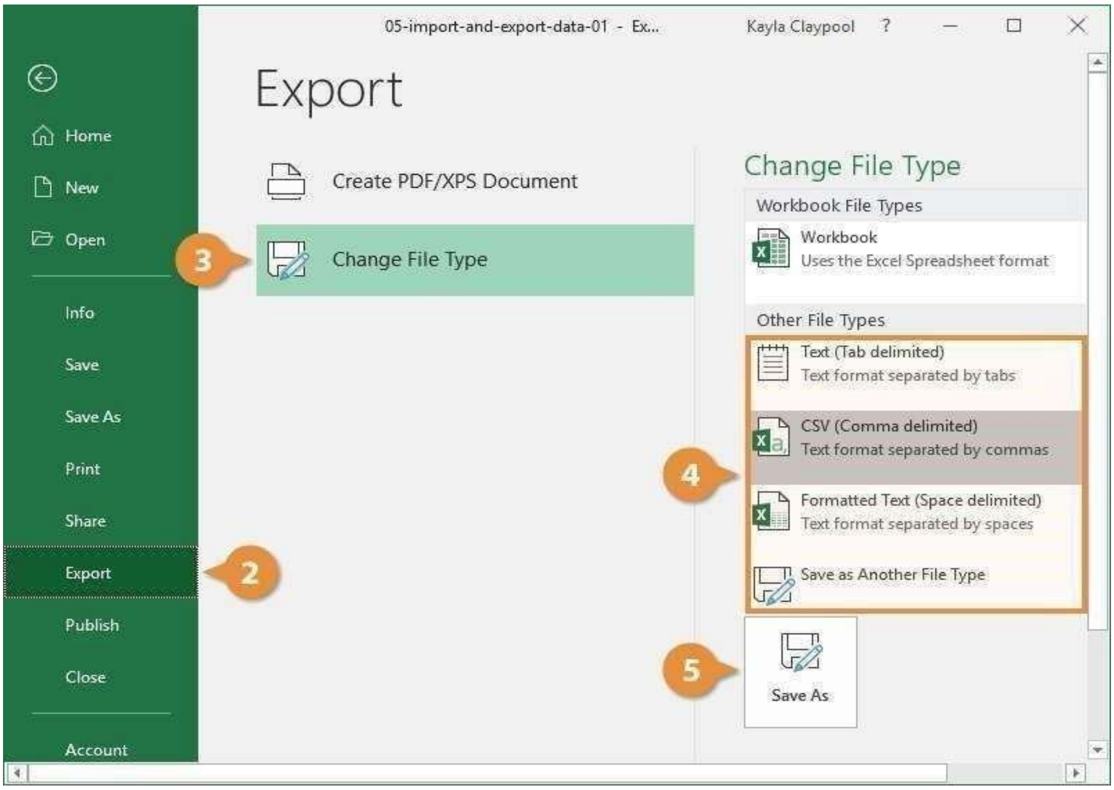
Perform data import/export operations for different file formats.

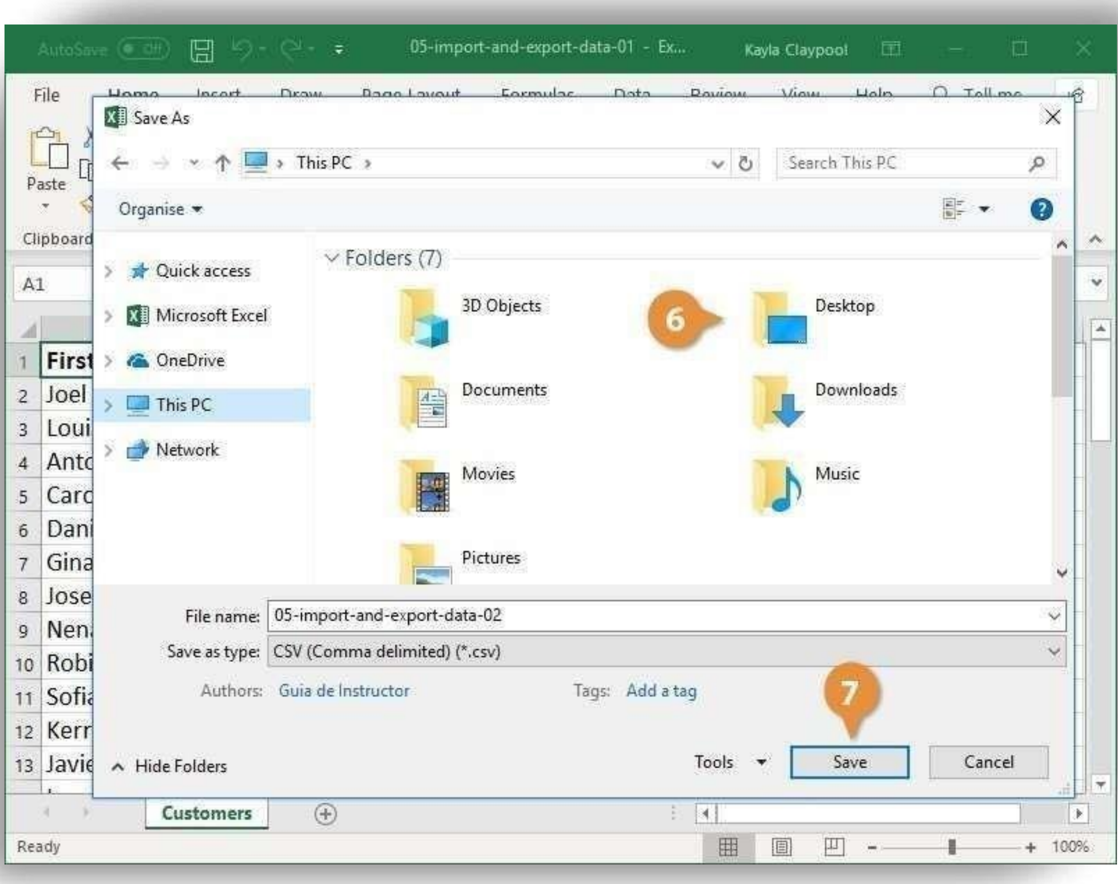
Import Text File into Excel by drag the text file into Excel.

Procedure:

Export Data When you have data that needs to be transferred to another system, export it from Excel in a format that can be interpreted by other programs, such as a text or CSV file.



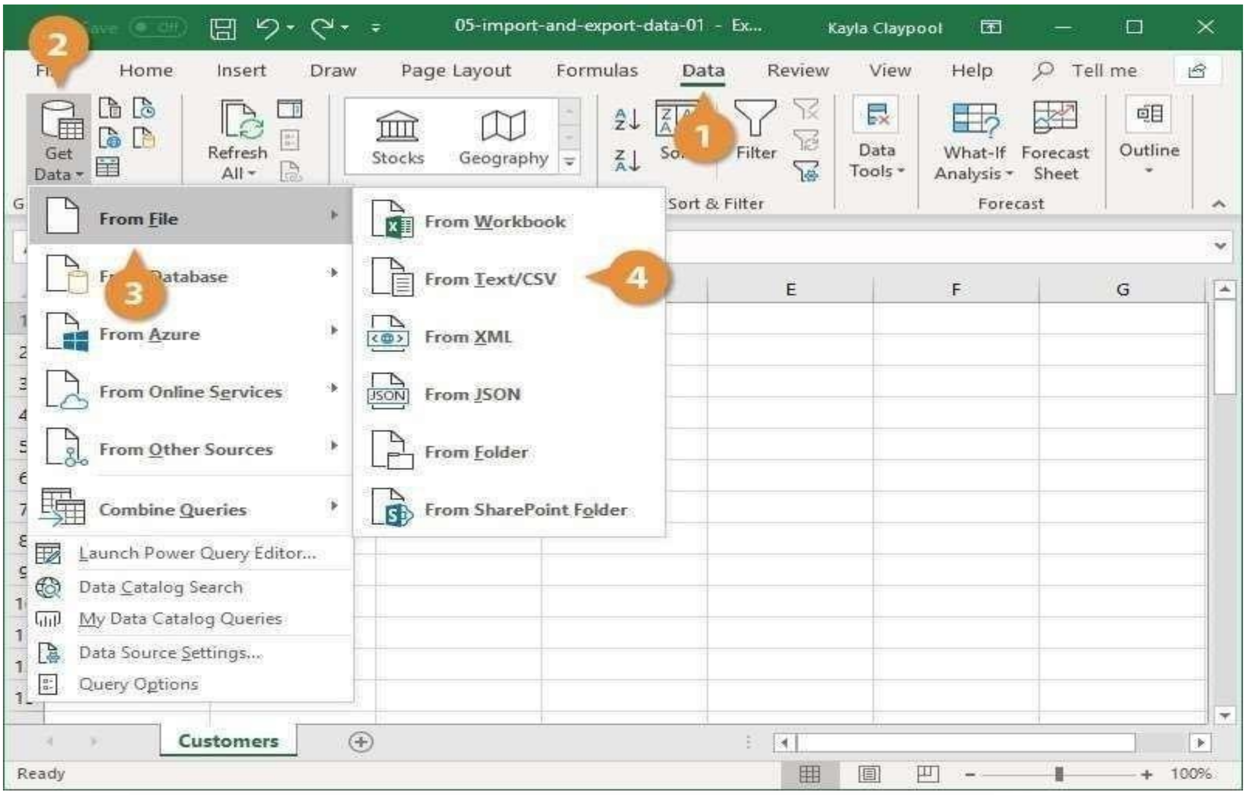


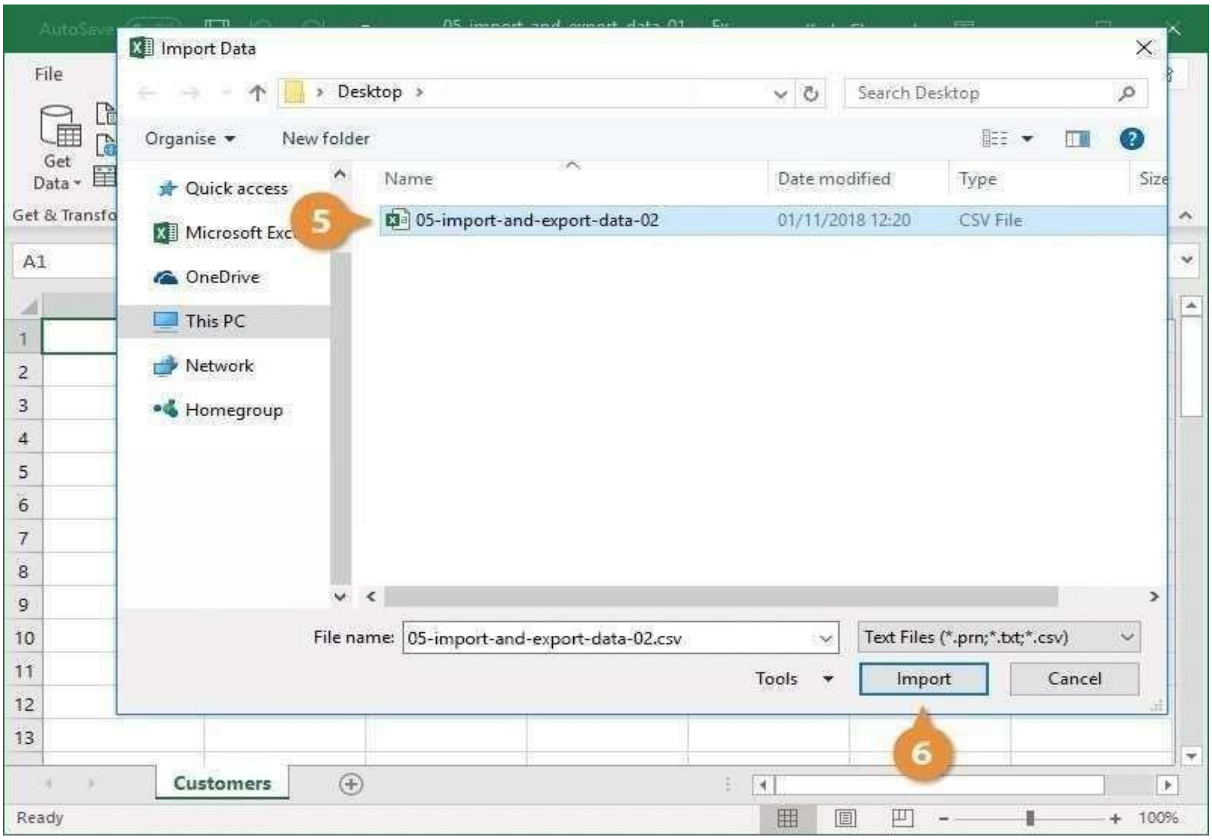


Import Data

Excel can import data from external data sources including other files, databases, or web pages.

- Click the Data tab on the Ribbon.
- Click the Get Data button.
- Some data sources may require special security access, and the connection process can often be very complex. Enlist the help of your organization's technical support staff for assistance. Select From File.
- Select From Text/CSV.





Verify the preview looks correct.

Result:

Thus numerical and import export operations was explored successfully.

Ex.No.3	STATISTICAL OPERATIONS
DATE:	

AIM:

To perform statistical operations like Mean, Median, Mode and Standard deviation, Variance, Skewness, kurtosis.

PROCEDURE:

Mean, median, and mode: Mean, median, and mode are different measures of center in a numerical data set. They each try to summarize a dataset with a single number to represent a "typical" data point from the dataset.

Mean: The "average" number; found by adding all data points and dividing by the number of data points.

Example: The mean of 444, 111, and 777 is (4+1+7)/3=12/3=4(4+1+7)/3=12/3=4left parenthesis, 4, plus, 1, plus, 7, right parenthesis, slash, 3, equals, 12, slash, 3, equals, 4.

Median: The middle number; found by ordering all data points and picking out the one in the middle (or if there are two middle numbers, taking the mean of those two numbers).

Example: The median of 444, 111, and 777 is 444 because when the numbers are put in order (1(1left parenthesis, 1, 444, 7)7)7, right parenthesis, the number 444 is in the middle.

Mode: The most frequent number—that is, the number that occurs the highest number of times.

	A	B	C	D
1	**Job**	**Wage**		
2	Electrician	$20		
3	Nurse	$26		
4	Police officer	$47		
5	Sales manager	$54		
6	Manufacturing engineer	$63		
7	Celebrity	$14,500		
8				
9	**Typical wage**			
10	**Mean**	$2,451.67	=AVERAGE(B2:B7)	
11	**Median**	$50.50	=MEDIAN(B2:B7)	

	A	B	C	D
1	Item	Status	Amount	
2	Banana	Delivered	$70	
3	Apple	Cancelled	$90	
4	Banana	In transit	$90	
5	Cherry	Delivered	$100	
6	Cherry	In transit	$115	
7	Apple	Delivered	$130	
8	Banana	Delivered	$250	
9				
10	Mode	$90	=MODE(C2:C8)	

Variance

According to layman's words, the variance is a measure of how far a set of data are dispersed out from their mean or average value. It is denoted as 'σ^2'.

Properties of Variance

- It is always non-negative since each term in the variance sum is squared and therefore the result is either positive or zero.
- Variance always has squared units. For example, the variance of a set of weights estimated in kilograms will be given in kg squared. Since the population variance is squared, we cannot compare it directly with the mean or the data themselves.

Standard Deviation

The spread of statistical data is measured by the standard deviation. Distribution measures the deviation of data from its mean or average position. The degree of dispersion is computed by the method of estimating the deviation of data points. It is denoted by the symbol, 'σ'.

Properties of Standard Deviation

- It describes the square root of the mean of the squares of all values in a data set and is also called the root-mean-square deviation.
- The smallest value of the standard deviation is 0 since it cannot be negative.
- When the data values of a group are similar, then the standard deviation will be very low or close to zero. But when the data values vary with each other, then the standard variation is high or far from zero.

VAR function in Excel

- It is the oldest Excel function to estimate variance based on a sample. The VAR function is available in all versions of Excel 2000 to 2019.

Note. In Excel 2010, the VAR function was replaced with VAR.S that provides improved accuracy. Although VAR is still available for backward compatibility, it is recommended to use VAR.S in the current versions of Excel.

VAR.S function in Excel

It is the modern counterpart of the Excel VAR function. Use the VAR.S function to find sample variance in Excel 2010 and later.

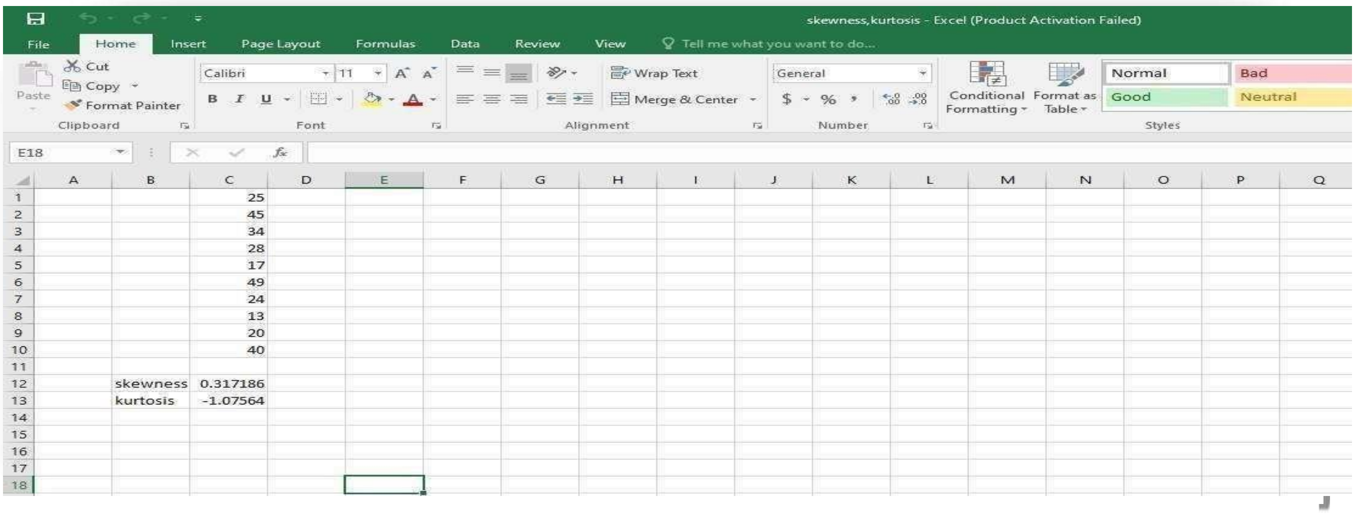
VARA function in Excel

The Excel VARA function returns a sample variance based on a set of numbers, text, and logical values as shown in this table.

=VARA (B2:B7)

As shown in the screenshot, all the formulas return the same result (rounded to 2 decimal places):

	A	B	C	D	E	F	G
1	Student	Score			Sample variance		
2	Daniela	85		VAR	34.97	=VAR(B2:B7)	
3	Tommy	79		VAR.S	34.97	=VAR.S(B2:B7)	
4	Edward	90		VARA	34.97	=VARA(B2:B7)	
5	Julia	88					
6	Timothy	88					
7	Peter	75					



Result:

Thus the statistical operations was successfully explored.

Ex.No.4	PERFORM T test, Z test, and ANOVA
DATE:	

a) t-Test

AIM:

To perform T test,Z test,andANOVA test using Excel.

The t-Test is used to test the null hypothesis that the means of two populations are equal. Below you can find the study hours of 6 female students and 5 male students.

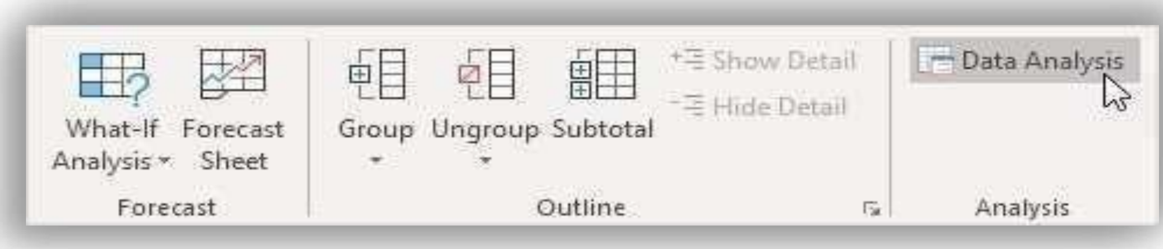
H0: $\mu 1 - \mu 2 = 0$

H1: $\mu 1 - \mu 2 \neq 0$

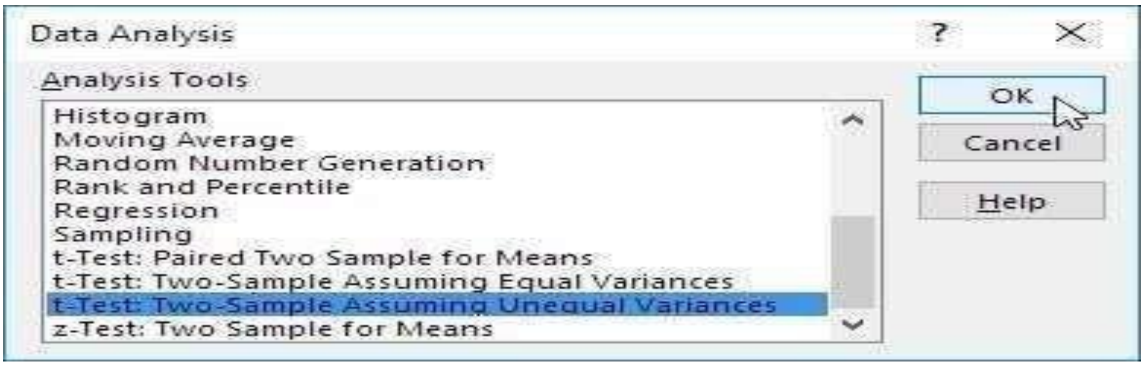
	A	B	C
1	Female	Male	
2	26	23	
3	25	30	
4	43	18	
5	34	25	
6	18	28	
7	52		
8			

To perform a t-Test, execute the following steps.

1. First, perform an F-Test to determine if the variances of the two populations are equal. is not the case.
2. On the Data tab, in the Analysis group, click Data Analysis.



3. Select t-Test: Two-Sample Assuming Unequal Variances and click OK.



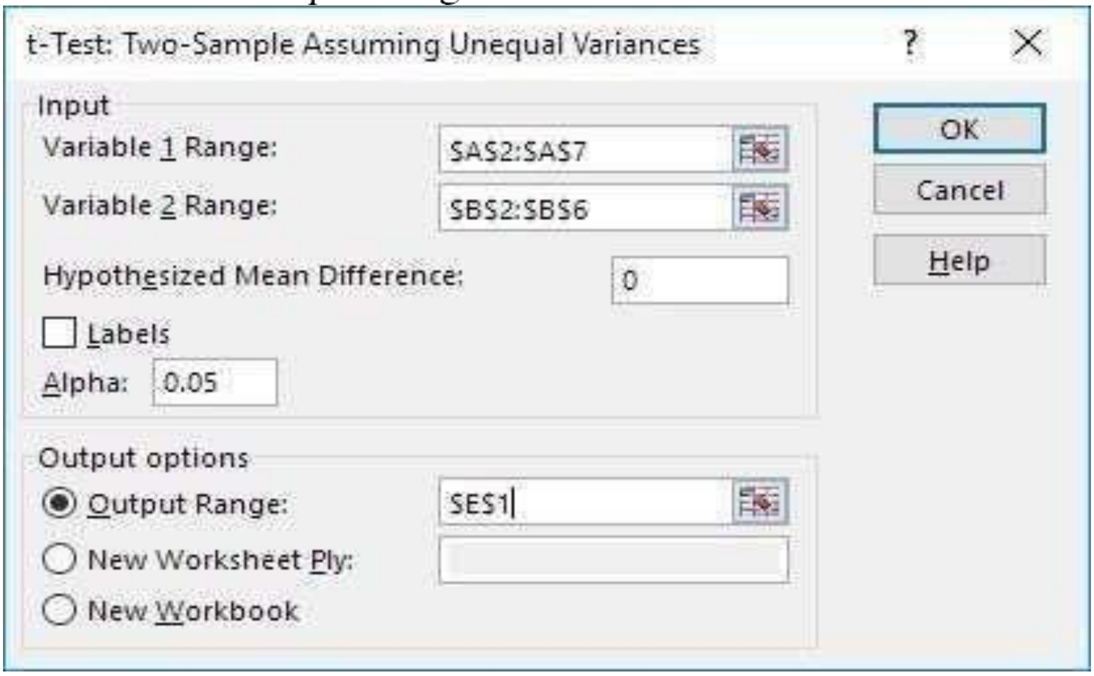
4. Click in the Variable 1 Range box and select the range A2:A7.

5. Click in the Variable 2 Range box and select the range B2:B6.

6. Click in the Hypothesized Mean Difference box and type 0 (H0: µ1

-

7. Click in the Output Range box and select cell E1.



8. Click OK.

Result:

E	F	G
t-Test: Two-Sample Assuming Unequal Variances		
	Variable 1	*Variable 2*
Mean	33	24.8
Variance	160	21.7
Observations	6	5
Hypothesized Mean Difference	0	
df	7	
t Stat	1.47260514	
P(T<=t) one-tail	0.092170202	
t Critical one-tail	1.894578605	
P(T<=t) two-tail	0.184340405	
t Critical two-tail	2.364624252	

Z Test in Excel

The Z.TEST function is one such hypothesis test function. It tests the mean of the two sample data sets when the variance is known and the sample size is large. The sample size should be >= 30. Otherwise, we need to use T-TEST. To Z.TEST, we need to have two independent data points that are not related to each other or do not affect each other data points. We should normally distributed the data.

Syntax

The Z.TEST is the built-in function in Excel. Below is the formula of the Z.TEST function in Excel.

Example #1 – Using the Z.Test Formula

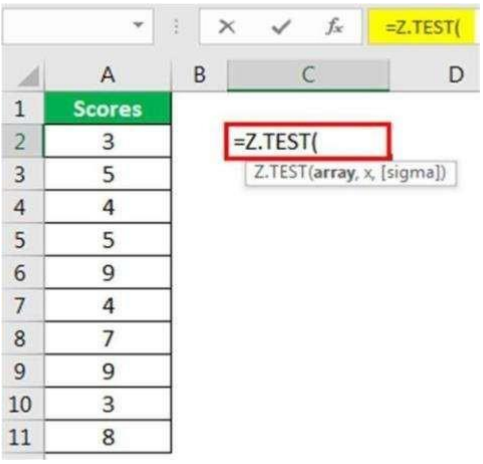
	A
1	Scores
2	3
3	5
4	4
5	5
6	9
7	4
8	7
9	9
10	3
11	8
12	

Look at the below data

We will use this data to calculate the one-tailed probability value of Z.TEST. For this, assume the hypothesis population means 6.

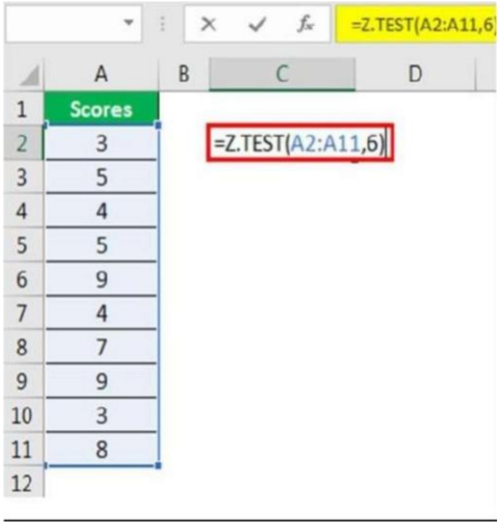
The steps to use the Z.Test formula in Excel are as follows: So,

1) Open the Z.TEST formula in an Excel cell.

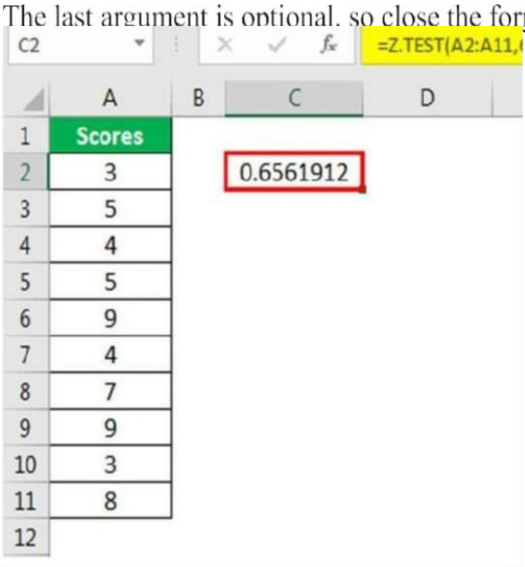


2) Select the array as scores, A2 to A11.

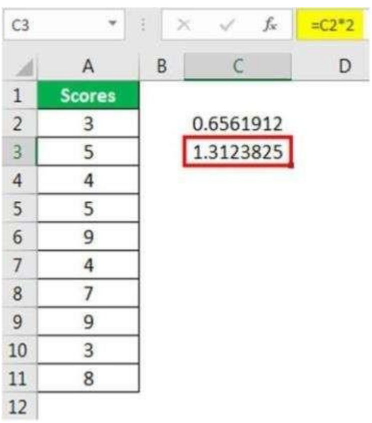
The next argument is "x." Since we have already assumed the hypothesized population mean is 6,apply this value to this argument.



3) The last argument is optional, so close the formula to get the Z.TEST value.



4) It is a one-tailed Z TEST value to get the two-tailed Z.TEST value to multiply this value by 2.



Example #2 – Z TEST Using Data Analysis Option

We can conduct Z.TEST using the "Data Analysis" option in Excel. To compare two means when the variance is known, we use Z.TEST. We can frame two hypotheses here. One is the "Null Hypothesis." Another one is the "Alternative Hypothesis" below is the equation of both these hypotheses.

H0: $\mu 1 - \mu 2 = 0$ (Null Hypothesis)

H1: $\mu 1 - \mu 2 \neq 0$ (Alternative Hypothesis)

The alternative hypothesis (H1) states that the two population means are not equal.

We will use two students' scores in multiple subjects for this example.

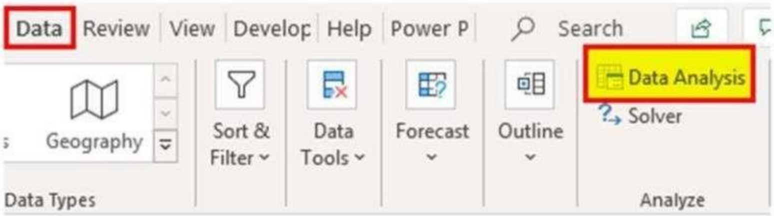
	A	B
1	Student 1	Student 2
2	93	79
3	68	66
4	82	73
5	51	72
6	87	80
7	82	74
8	72	95
9	71	93
10	92	99
11	73	97
12		

Step 1: First, we need to calculate the variables for these two values using the VAR.P function.

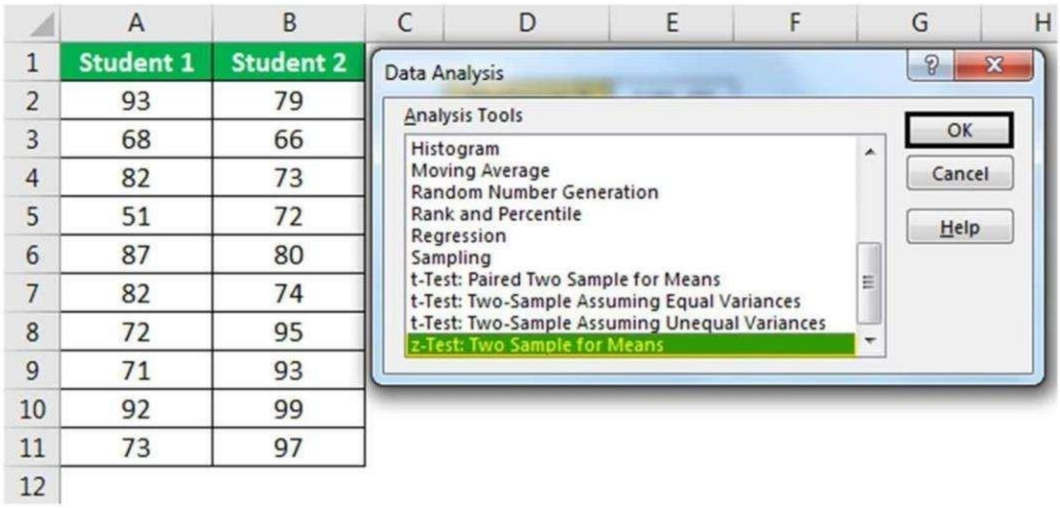
E2 =VAR.P(A2:A11)

	A	B	C	D	E
1	Student 1	Student 2			
2	93	79		Variance 1	146.49
3	68	66		Variance 2	131.16
4	82	73			
5	51	72			
6	87	80			
7	82	74			
8	72	95			
9	71	93			
10	92	99			
11	73	97			
12					

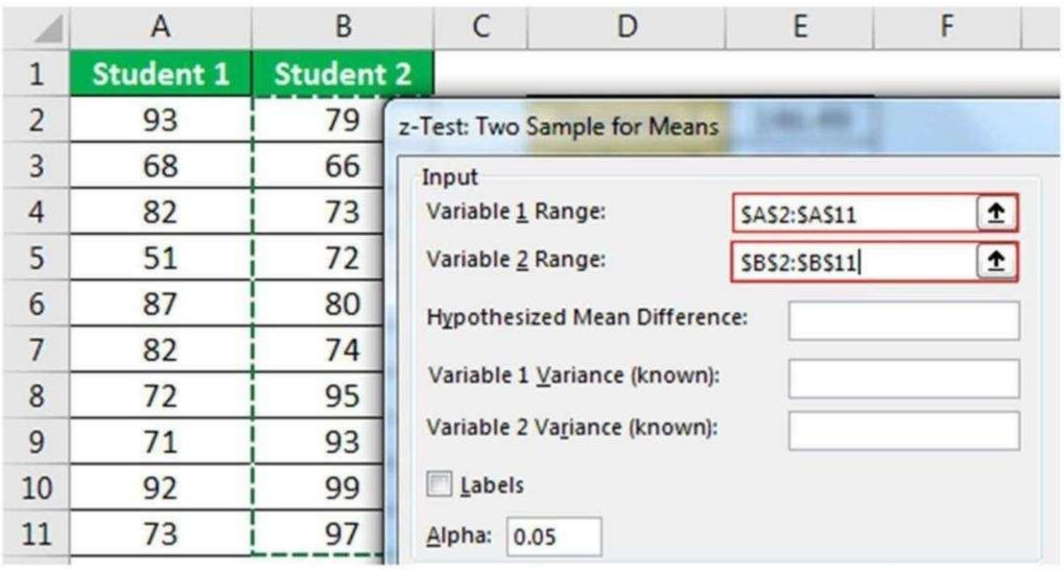
• **Step 2:** Go to the "Data" tab and click "Data Analysis."



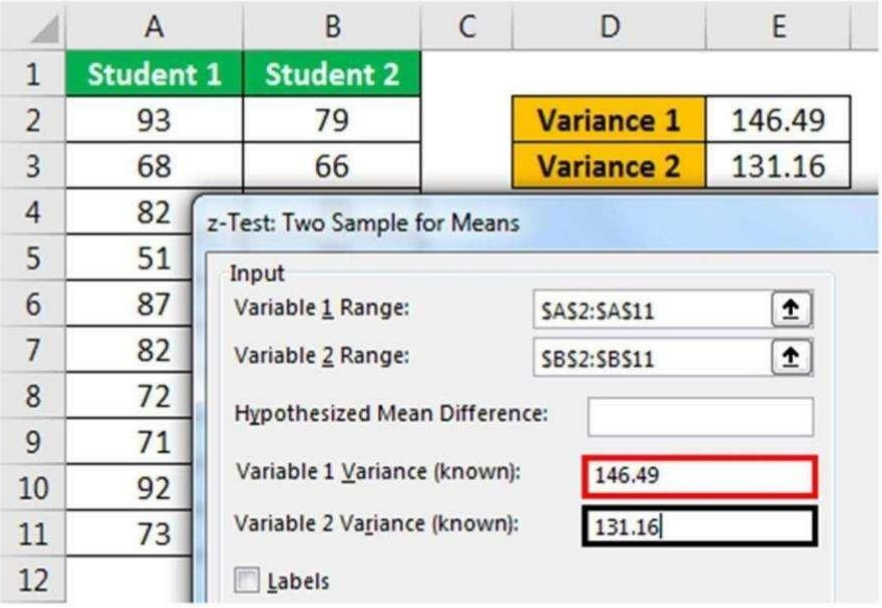
Scroll down and select z-Test: Two Sample for Means" and click on "OK."



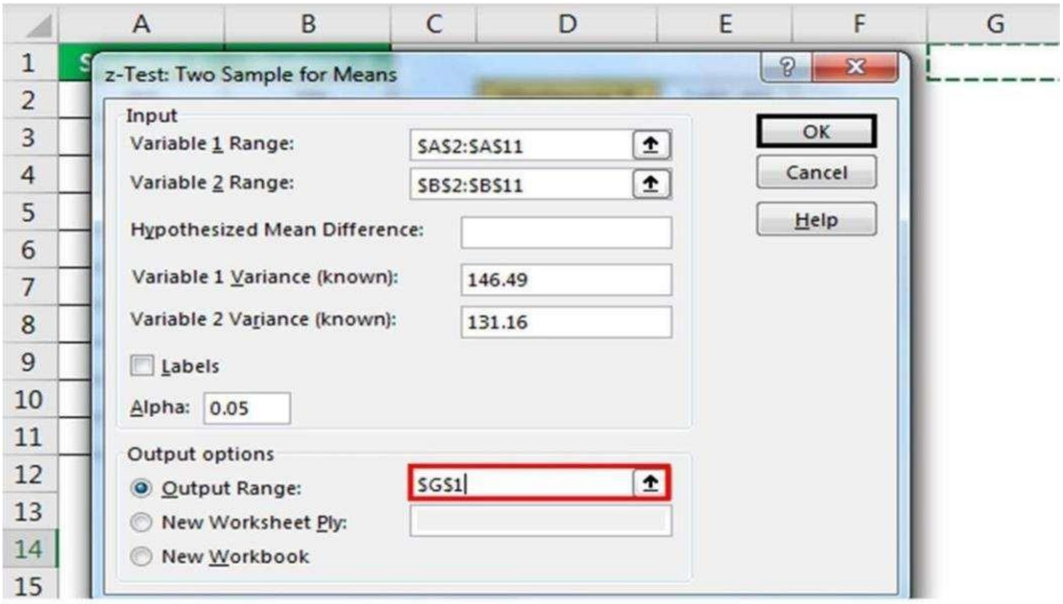
Step 3: For the "Variable 1 Range," select "Student 1" scores. For the "Variable 2 Range," select "Student 2" scores.



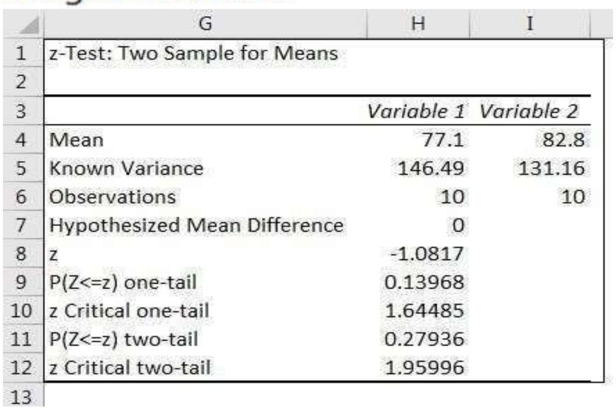
Step 4: For "Variable 1 Variance(known)," select "Student 1" variance score, and for "Variable 2 Variance(known)," select "Student 2" variance score.



Step 5: Select the "Output Range" as a cell and press "OK."



We got the result.



z-Test: Two Sample for Means		
	Variable 1	*Variable 2*
Mean	77.1	82.8
Known Variance	146.49	131.16
Observations	10	10
Hypothesized Mean Difference	0	
z	-1.0817	
P(Z<=z) one-tail	0.13968	
z Critical one-tail	1.64485	
P(Z<=z) two-tail	0.27936	
z Critical two-tail	1.95996	

Anova

This example teaches you how to perform a single factor ANOVA (analysis of variance) in Excel. A single factor or one-way ANOVA is used to test the null hypothesis that the means of several populations are all equal.

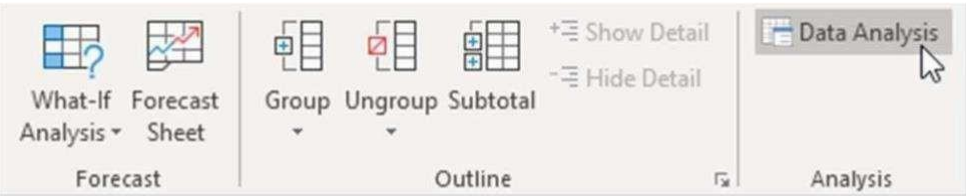
Below you can find the salaries of people who have a degree in economics, medicine or history.

H0: $\mu 1 = \mu 2 = \mu 3$
H1: at least one of the means is different.

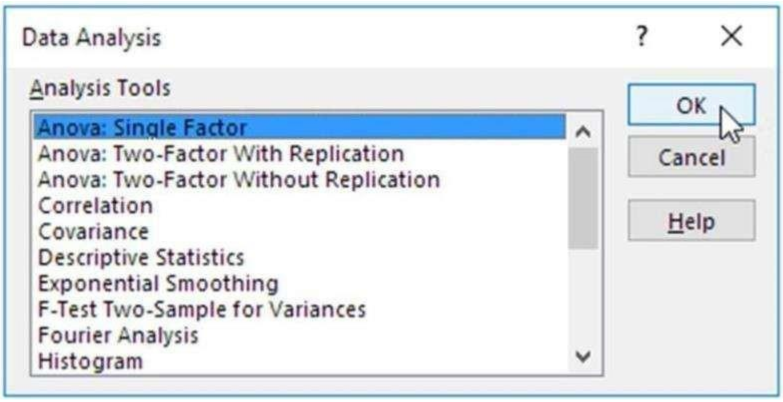
	A	B	C	D
1	economics	medicine	history	
2	42	69	35	
3	53	54	40	
4	49	58	53	
5	53	64	42	
6	43	64	50	
7	44	55	39	
8	45	56	55	
9	52		39	
10	54		40	
11				

To perform a single factor ANOVA, execute the following steps.

1. On the Data tab, in the Analysis group, click Data Analysis.

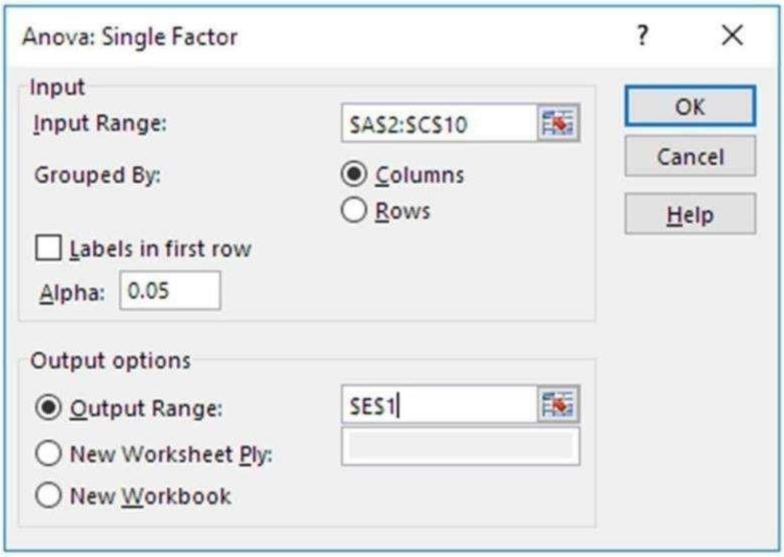


2. Select Anova: Single Factor and click OK.



3. Click in the Input Range box and select the range A2:C10.

4. Click in the Output Range box and select cell E1.



5. Click OK. output:

E	F	G	H	I	J	K
Anova: Single Factor						
SUMMARY						
Groups	*Count*	*Sum*	*Average*	*Variance*		
Column 1	9	435	48.33333	23.5		
Column 2	7	420	60	32.33333		
Column 3	9	393	43.66667	50.5		
ANOVA						
Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	1085.84	2	542.92	15.19623	7.16E-05	3.443357
Within Groups	786	22	35.72727			
Total	1871.84	24				

Result:

Thus t Test, z Test, and anova were successfully performed..

Ex.No.5	DATA PRE-PROCESSING OPERATIONS
DATE:	

a) Handling Missing data

Aim:

To handle missing data in excel using some functions such as ISERROR and VLOOKUP Functions Using NOT, ISNUMBER, MATCH Functions

Procedure

Let's say, we have got a dataset of some people registered for taking a vaccine, their relevant ID, and the person present on the day of taking the dose of the vaccine.

Dataset Overview

ID	Registered Person	Present Person
1602	Mike	Adam
1725	Adam	Antony
1235	David	John
1823	Jonathon	Hamlet
1290	Lilly	Lilly
1692	Bob	Collins
1286	Collins	
1427	John	
1735	Hamlet	
1882	Antony	

You can see that some persons are missing on the day of taking the vaccine and we want to find out the missing persons. Here we will use ISERROR, VLOOKUP functions to deal with the missing data.

In order to do so, proceed with the following steps.

Steps:

First of all, create a column and apply the following formula to the selected cell.

=ISERROR(VLOOKUP(C5,$D$5:$D$14,1,0))

Here,

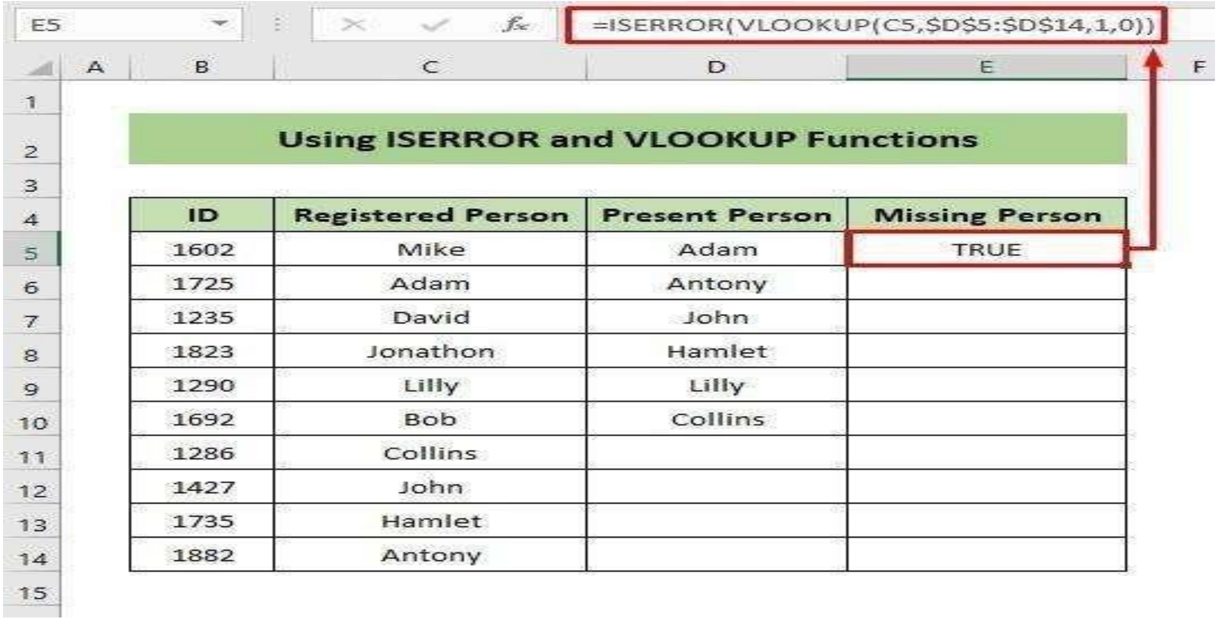
- C5= the data to find
- $D$5:$D$14=Lookup array

Formula Breakdown

1 as col_index_num, and 0 as range lookup.

The VLOOKUP function considers C5 as the lookup_value and $D$5:$D$14 as lookup array,

The ISERROR function returns TRUE as it finds an error or FALSE where it doesn't find an error. So, when the LOOKUP function doesn't find a value and makes an error, the ISERROR function returns TRUE, otherwise FALSE.

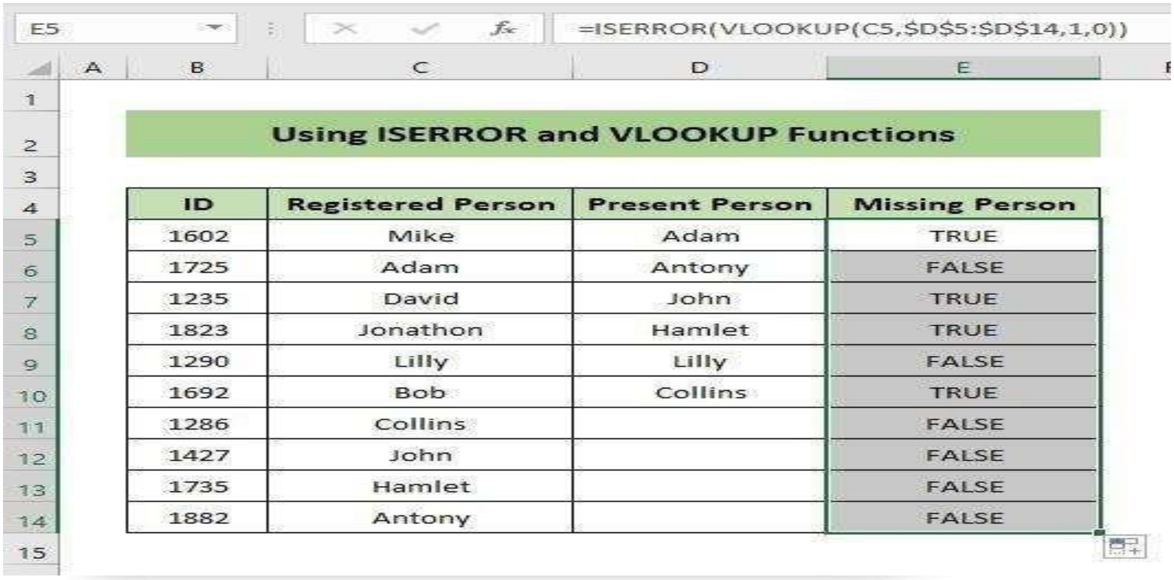


E5 =ISERROR(VLOOKUP(C5,$D$5:$D$14,1,0))

Using ISERROR and VLOOKUP Functions

ID	Registered Person	Present Person	Missing Person
1602	Mike	Adam	TRUE
1725	Adam	Antony	
1235	David	John	
1823	Jonathon	Hamlet	
1290	Lilly	Lilly	
1692	Bob	Collins	
1286	Collins		
1427	John		
1735	Hamlet		
1882	Antony		

Then, use the Fill Handle tool to Autofill the formula for the next cells and you will get the output for the missing data.

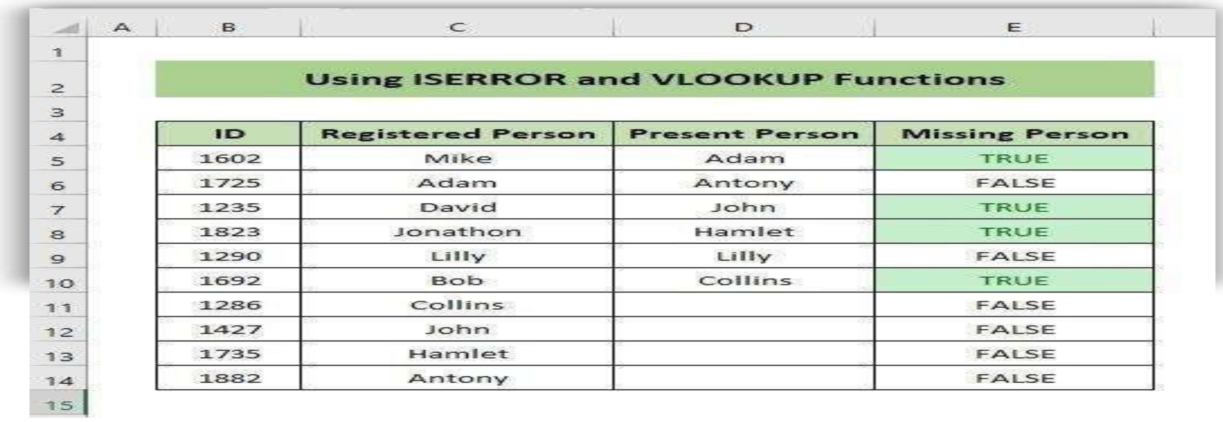


E5 =ISERROR(VLOOKUP(C5,$D$5:$D$14,1,0))

Using ISERROR and VLOOKUP Functions

ID	Registered Person	Present Person	Missing Person
1602	Mike	Adam	TRUE
1725	Adam	Antony	FALSE
1235	David	John	TRUE
1823	Jonathon	Hamlet	TRUE
1290	Lilly	Lilly	FALSE
1692	Bob	Collins	TRUE
1286	Collins		FALSE
1427	John		FALSE
1735	Hamlet		FALSE
1882	Antony		FALSE

Here, you can also apply Conditional Formatting to highlight the missing data. For this, just select the column> go to Conditional Formatting> select Highlight Cells Rules> select Equal to> select the cell (i.e. E6)> choose Fill Color> click OK.



Using ISERROR and VLOOKUP Functions

ID	Registered Person	Present Person	Missing Person
1602	Mike	Adam	TRUE
1725	Adam	Antony	FALSE
1235	David	John	TRUE
1823	Jonathon	Hamlet	TRUE
1290	Lilly	Lilly	FALSE
1692	Bob	Collins	TRUE
1286	Collins		FALSE
1427	John		FALSE
1735	Hamlet		FALSE
1882	Antony		FALSE

Result:

Thus the handling of missing data in excel was executed successfully.

b) Normalization

Aim:

To normalize the data in excel

Procedure:

Normalization refers to scaling data so that the minimum value is equal to 0 and the maximum value is equal to 1. This process is often referred to as min-max scaling. Standardization is often also referred to as normalization – in this case, the dataset is modified so that the mean is equal to 0 and the standard deviation is equal to 1.

Standardization and normalization allow you to work with data of different scales more effectively. Imagine having a dataset that compares a person's income and height in centimeters. Likely, these values will be in completely different scales. Because of this, looking at their means or standard deviations would not be comparable.

Standardizing data is quite simple using built-in operators and functions, such as the MIN() and MAX() functions.

Mathematically, we normalize value using the formula below:

$$x_{scaled} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

The formula for min-max scaling normalization

In the example above, you scale each value in a dataset by subtracting the minimum value from the value and dividing it by the difference between the max and minimum values. Let's break this down, step-by-step:

1. Subtract the minimum value in the dataset from the value
2. Divide this difference by the difference between the maximum and minimum values of the dataset
3. Repeat this for each item in the dataset Let's see how we can normalize data in Excel:

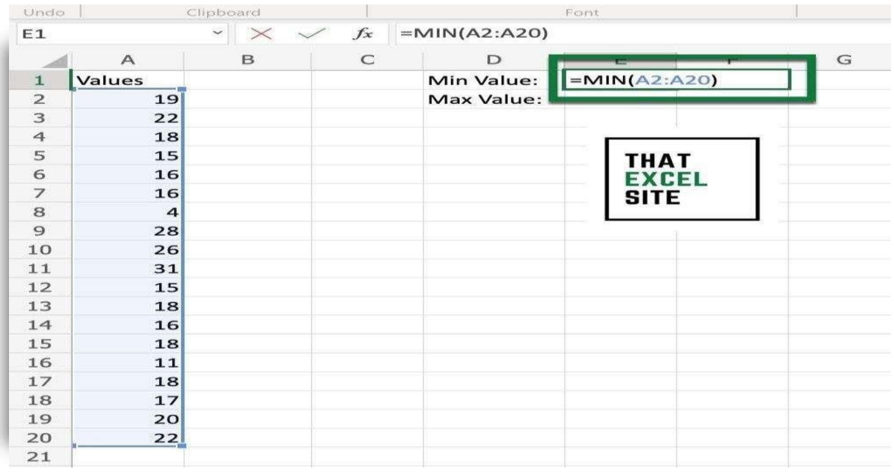
	A	B	C
1	Values		
2	19		
3	22		
4	18		
5	15		
6	16		
7	16		
8	4		
9	28		
10	26		
11	31		
12	15		
13	18		
14	16		
15	18		
16	11		
17	18		
18	17		
19	20		
20	22		
21			

Step 1: Load the dataset you want to normalize

The dataset for the tutorial

In our sample dataset, we have values ranging from 4 to 31. By using min-max scaling, all of our values will range from 0 to 1.

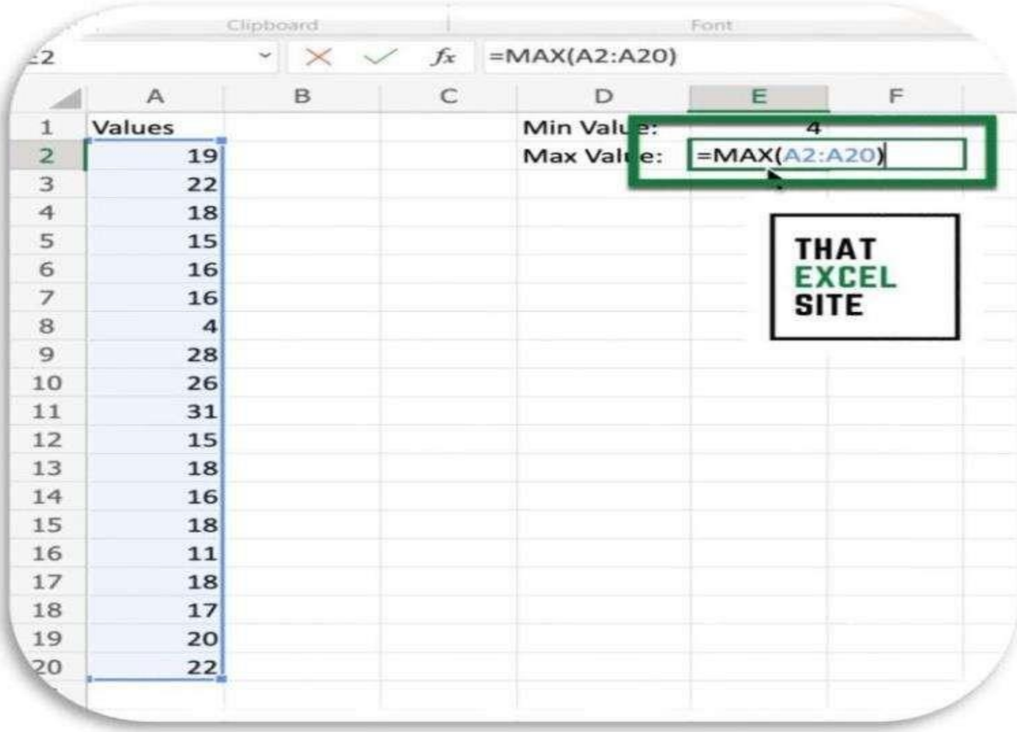
Step 2: Calculate the minimum value in the dataset using the MIN() function



Calculate the minimum value of the dataset

Use the MIN() function to calculate the minimum value of the dataset. In this case, this returns 4.

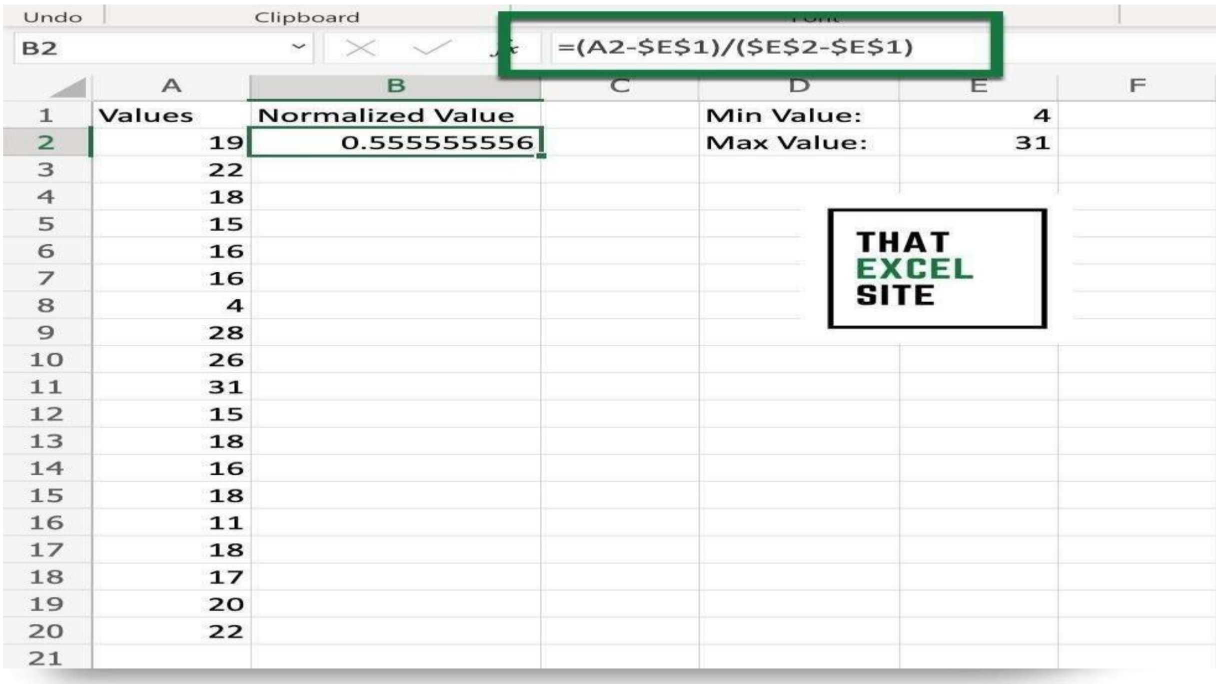
Step 3: Calculate the maximum value in the dataset using the MAX() function



Calculate the maximum value of the dataset

Use the MAX() function to calculate the maximum value of the dataset. In this case, this returns 31.

Step 4: Normalize the first record by subtracting the minimum value from the observation and dividing by the range of the dataset



Undo | Clipboard | Font

B2 =(A2-$E$1)/($E$2-$E$1)

	A	B	C	D	E	F
1	Values	Normalized Value		Min Value:	4	
2	19	0.555555556		Max Value:	31	
3	22					
4	18					
5	15					
6	16					
7	16					
8	4					
9	28					
10	26					
11	31					
12	15					
13	18					
14	16					
15	18					
16	11					
17	18					
18	17					
19	20					
20	22					
21						

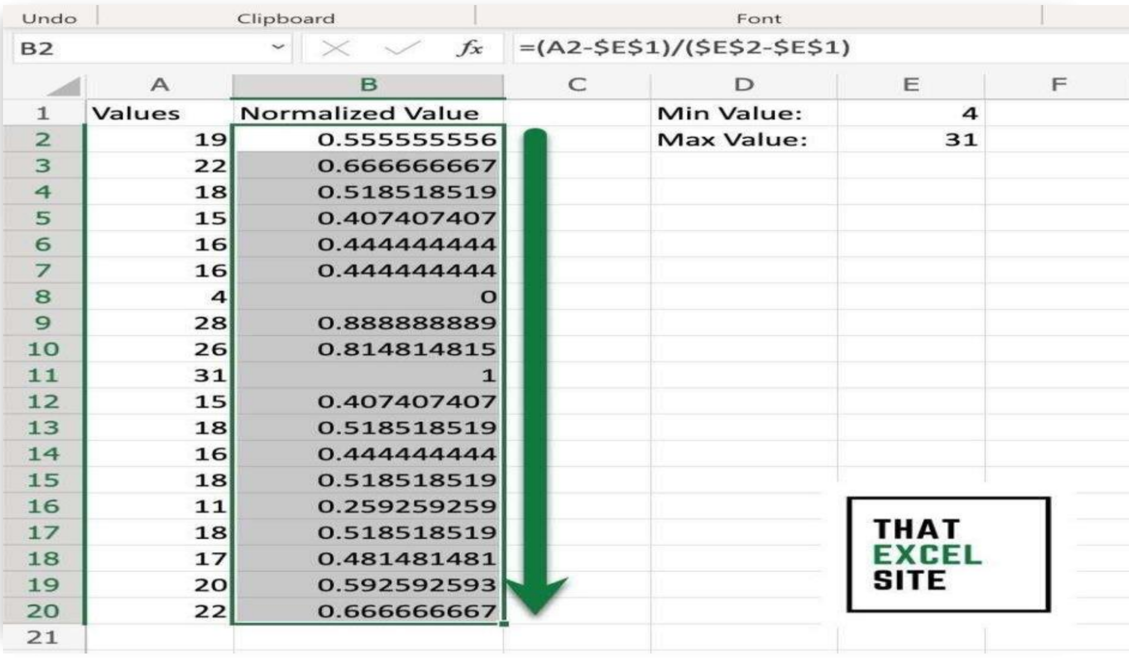
Subtract the minimum value from the record and divide by the difference between the maximum and minimum values

Step 5: Drag the fill handle down to normalize all records in the dataset

To normalize a single value, subtract that minimum value from the record itself. Then, divide this number by the range of the data (the maximum value subtracted by the minimum value).

Drag the fill handle all the way down to normalize all values in Excel

To normalize the entire dataset, simply drag the fill handle all the way down. Because we're using absolute references for the minimum and maximum values of the dataset, we don't have to worry about values changing through relative references.



Undo | Clipboard | Font

B2 =(A2-$E$1)/($E$2-$E$1)

	A	B	C	D	E	F
1	Values	Normalized Value		Min Value:	4	
2	19	0.555555556		Max Value:	31	
3	22	0.666666667				
4	18	0.518518519				
5	15	0.407407407				
6	16	0.444444444				
7	16	0.444444444				
8	4	0				
9	28	0.888888889				
10	26	0.814814815				
11	31	1				
12	15	0.407407407				
13	18	0.518518519				
14	16	0.444444444				
15	18	0.518518519				
16	11	0.259259259				
17	18	0.518518519				
18	17	0.481481481				
19	20	0.592592593				
20	22	0.666666667				
21						

Result:

Thus the normalizing of data in excel was executed successfully.

Ex.No.6	**Perform dimensionality reduction operation using PCA, KPCA & SVD**
DATE:	

Aim:

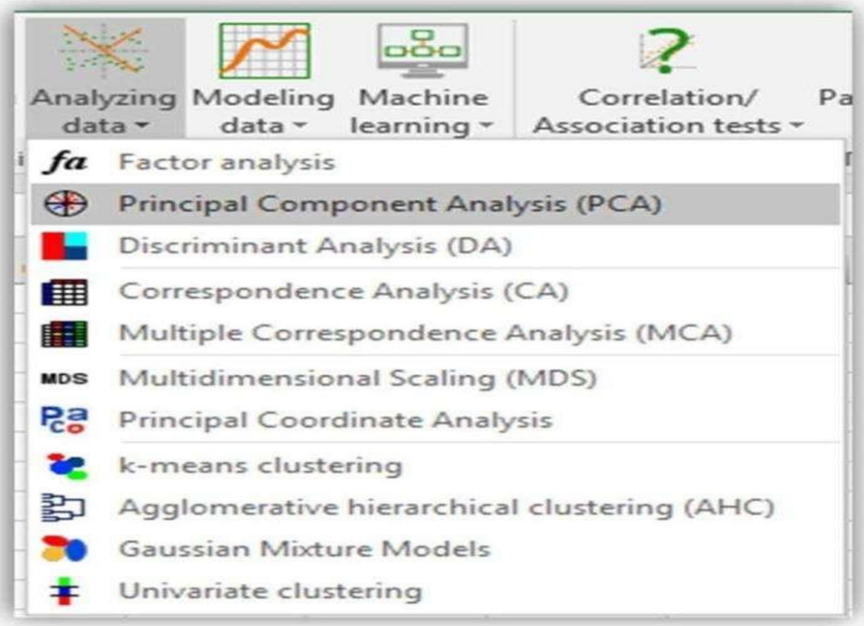
To Perform dimensionality reduction operation using PCA, KPCA & SVD

Procedure:

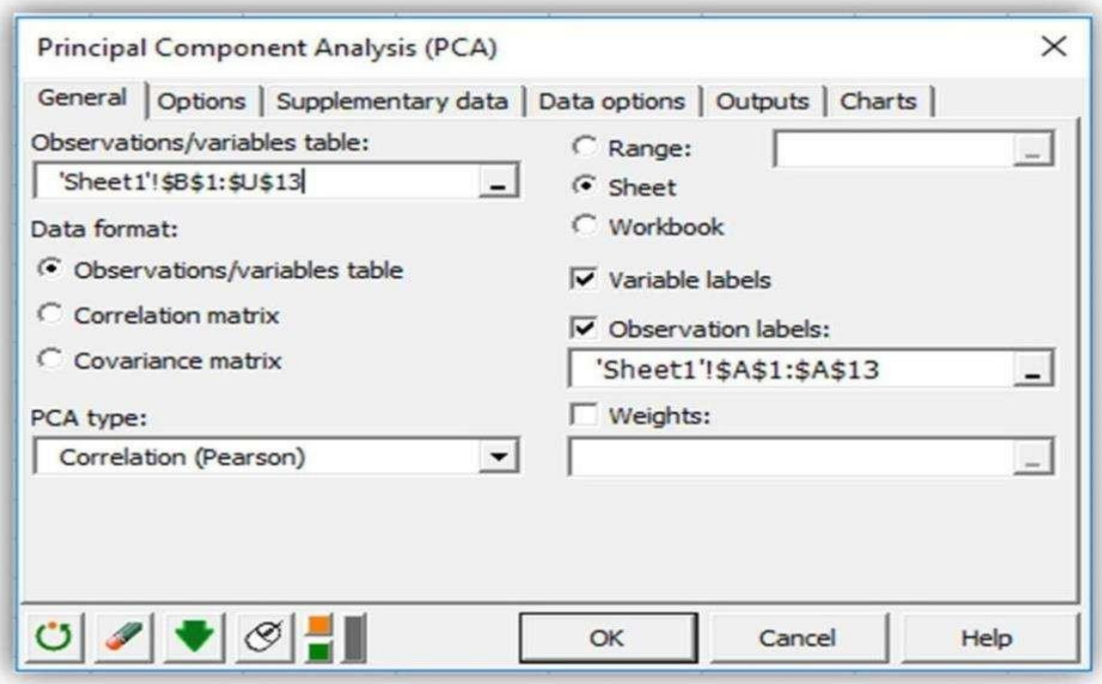
How to build a PCA in XLSTAT?
Five simple steps:

- Import your data into Excel(download excel 6 xlsm file)
- Launch the PCA dialog box under Analyzing data menu
- Select your data
- Choose the options/output of your interest
- Click OK to run the PCA

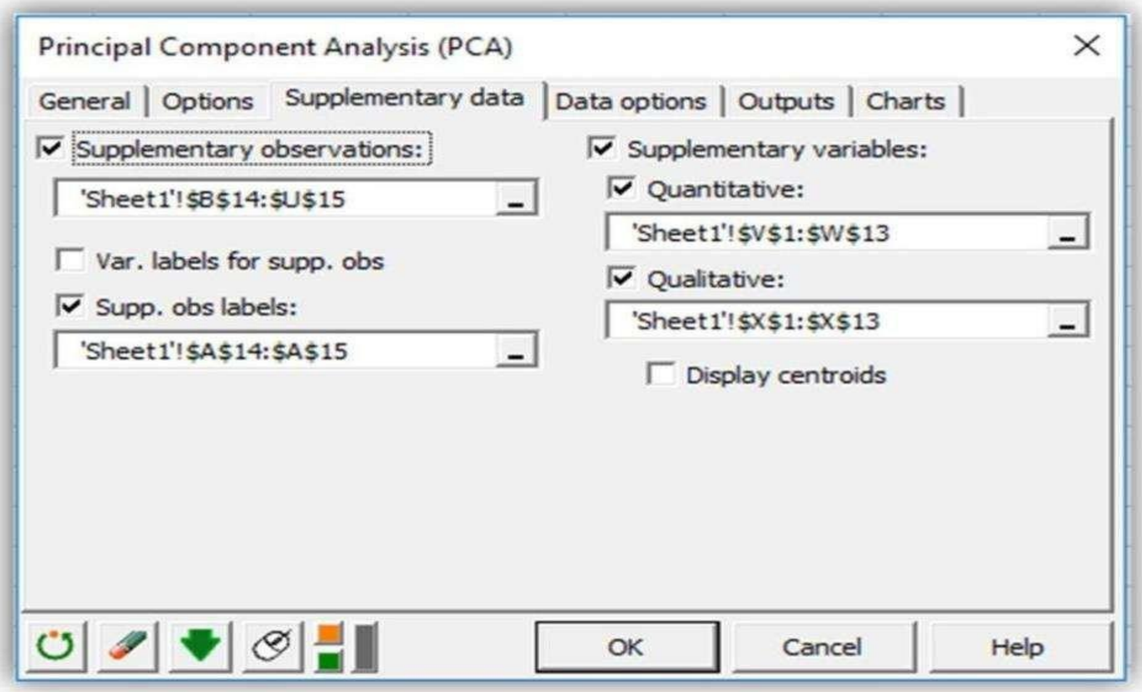
Setting up a Principal Component Analysis in Excel using XLSTAT
Once XLSTAT is activated, select the XLSTAT / Analyzing data / Principal components analysis command (see below).



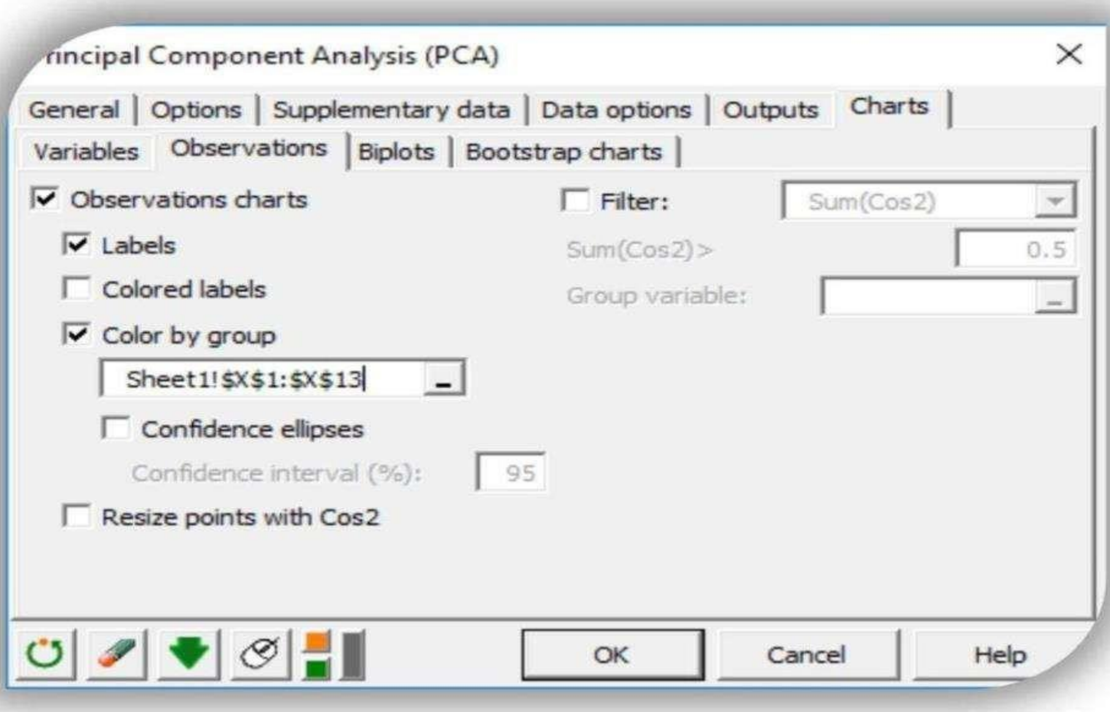
The Principal Component Analysis dialog box will appear
In this example, the data start from the first row, so it is quicker and easier to use columns selection.
This explains why the letters corresponding to the columns are displayed in the selection boxes.
The Data format chosen is Observations/variables because of the format of the input data.



In the Supplementary data tab, we choose the two last rows as supplementary observations, Acidity and Sweetness as quantitative supplementary variables and Texture as a qualitative supplementary variable. We also can check the display centroids option to display the centroids of each category on the observation graph. Here, we will see how to color observation according to their category.

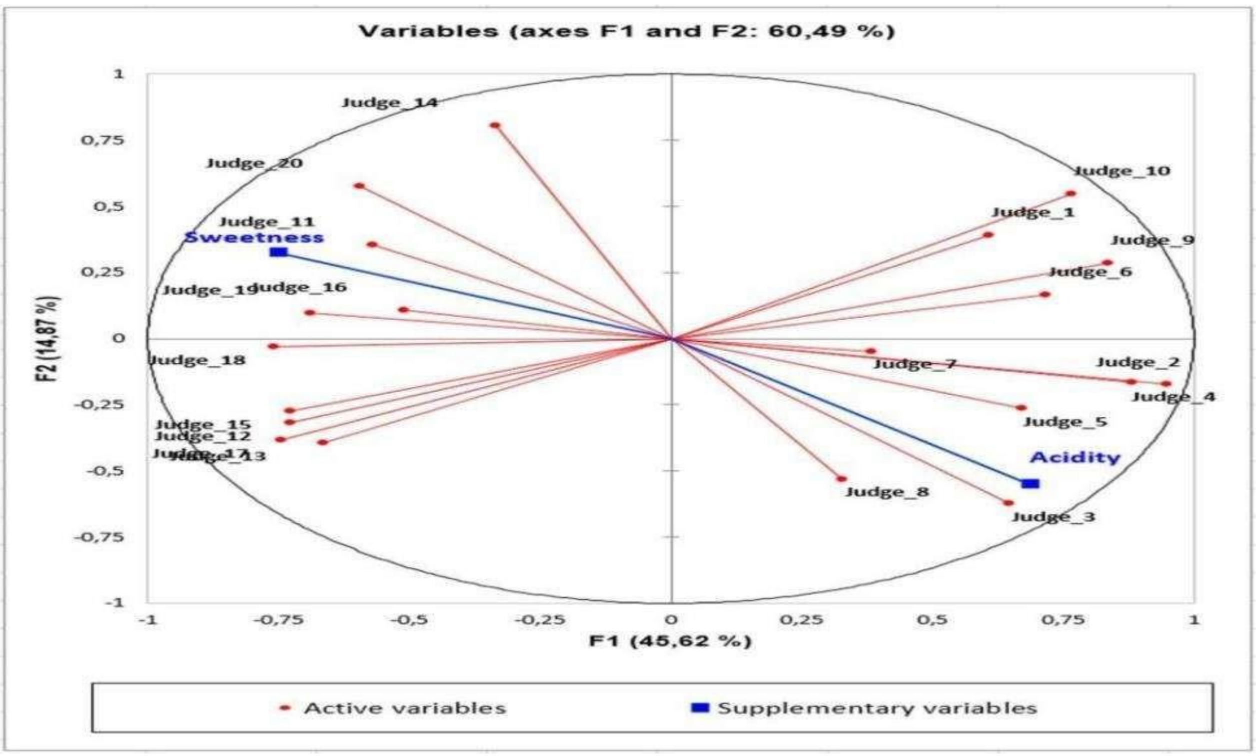


In the Charts tab, and the Observations sub-tab, we can check the Color by group option and select our qualitative supplementary variable Texture to color observations according to the category they belong to.

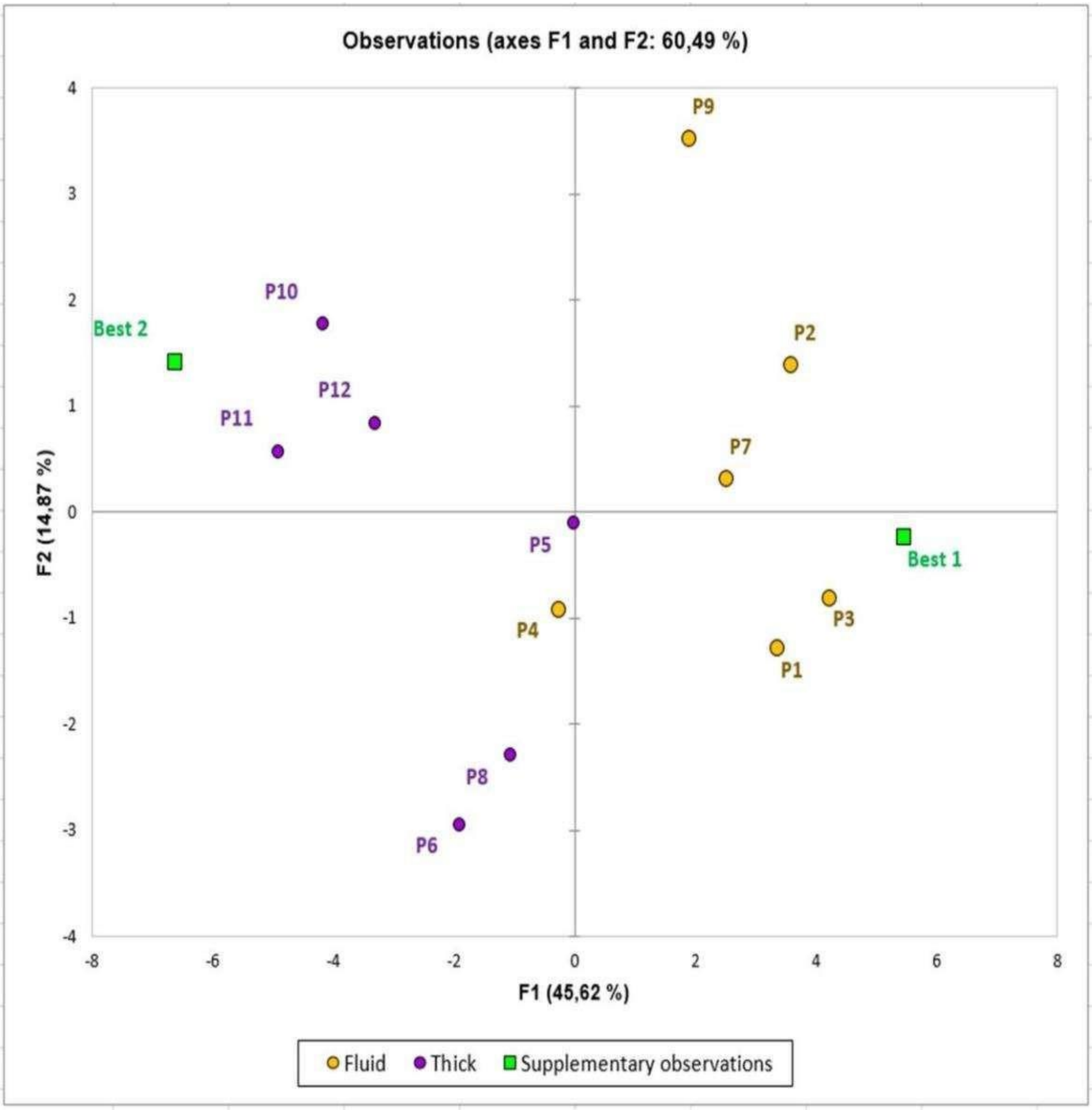


Interpreting the results of a Principal Component Analysis in Excel using XLSTAT

Here, the supplementary variables Acidity and Sweetness allow us to identify two kind of consumers: those who prefer products characterized by an acidity and those who prefer products by a sweet taste. Without this information, we would not be able to explain the differences between those two clusters of consumers. Products on the right-side of the of the observations graph will be characterized by an acidity and preferred by the right cluster of judges. Products on the left of the observation graph will be characterized by a sweet taste and preferred by the left cluster of consumers.



On this graph, we can see that one group of products (P1, P2, P3, P7) is preferred by the judges situated in the right half of the correlation plot. These consumers like the acidity of these products. The Best 1 prototype seems to be a good product for this kind of consumers expectations. Moreover, this group of products is characterized by a fluid texture. On the other hand, products P10, P11, and P12 are preferred by judges in the left half of the correlation plot. This kind of consumers seems to like products with a sweet taste and a thick texture. Best 2 looks like a good candidate for this kind of consumers.



Result:

Thus the dimensionality reduction operation was performed successfully.

Ex.No.7	Perform bivariate and multivariate analysis on the dataset
DATE:	

Aim:

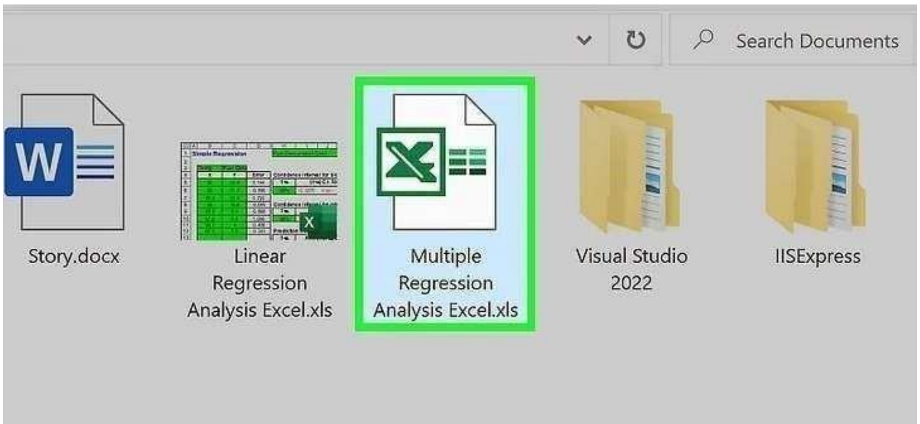
To Perform bivariate and multivariate analysis on the dataset.

Procedure 1: (multivariate)

1 Enable the data analysis add-in (if needed). Whether you're studying statistics or doing regression professionally, Excel is a great tool for running the analysis. Excel has a built-in data analysis add-in called "Analysis ToolPak." You can check to see if it's active by clicking the Data tab. If you don't see the Data Analysis option, you will need to enable it:[1]

- Windows:
- Open the File tab (or press Alt+F) and select Options (Windows).
- Click Add-Ins on the left side of the window.
- Select Excel Add-ins next to "manage" and click Go.
- In the new window, check the box next to "Analysis ToolPak", then click OK. This will enable the built-in data analysis add-in.

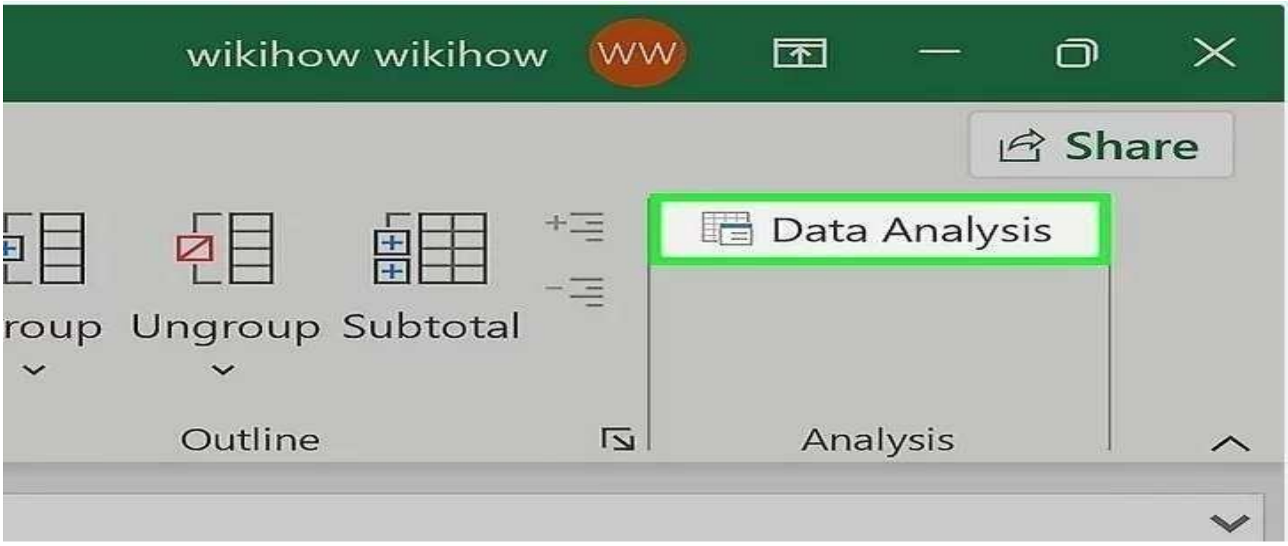
2 Enter your data or open your data file. Data must be arranged in immediately adjacent columns and labels should be in the first row of each column. This is a typical format for databases.



3 Click the Data tab and click Data Analysis. This option is in the "Analysis" section near the far right of

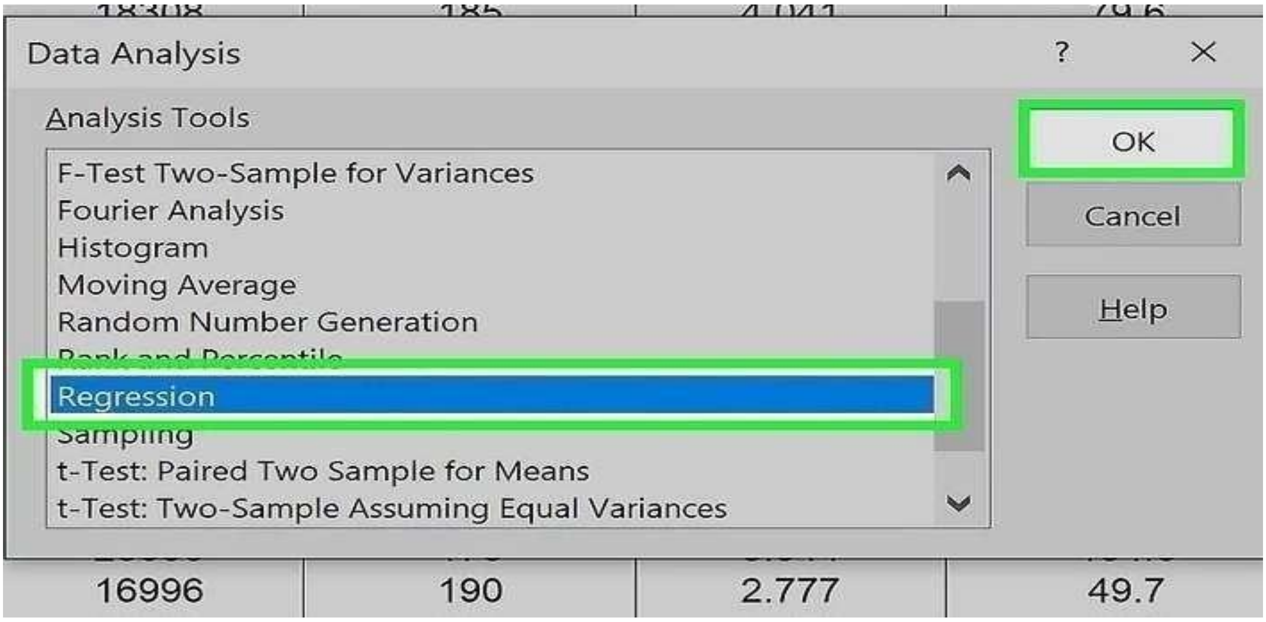
Data tab options.

Another powerful tool is Excel's Solver function, an optimization model feature.



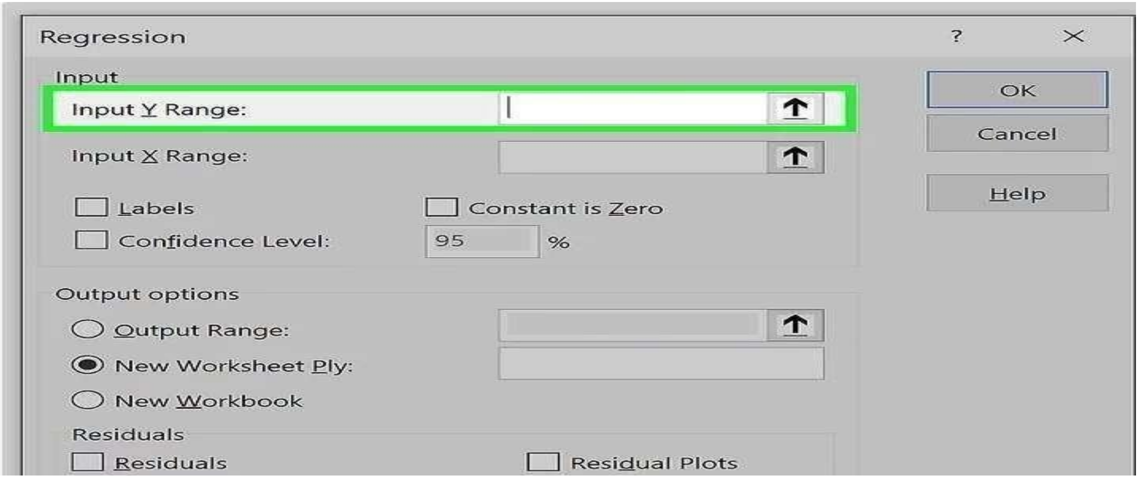
4 Click Regression and then OK. This will open a new window for inputting the parameters of the regression

model.



5 Input the dependent (Y) data range. To do so:

- Click the "Input Y Range" field.
- Highlight the column containing your dependent variable values.
- Click the Labels checkbox if your data has a header row

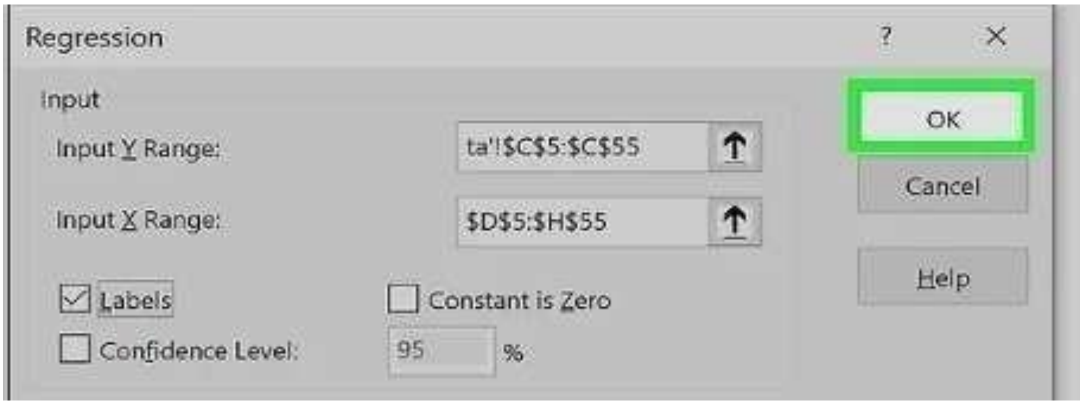


6 Input the independent (X) data range. To do so:

- Click the "Input X Range" field.
- Highlight the column or columns containing your dependent variable values. This can include multiple columns if you have more than one independent variable.
- Note: The independent variable data columns must be adjacent to one another for the input to work.

7 Adjust the regression options (if needed). You can change the following parts of the analysis in the Regression window:

- The default confidence level is 95%. If you want to change this value, click the box next to Confidence Level and modify the adjacent value.
- Under "Output options," select where you want the regression results to output.
- Select the desired options in the "Residuals" category. Graphical residual outputs are created by the Residual Plots and Line Fit Plots options.



8 Click OK and the analysis will be created. You'll see the following information:

- Regression Statistics includes the correlation values, standard error, and number of observations.

Procedure 2: (Bivariate)

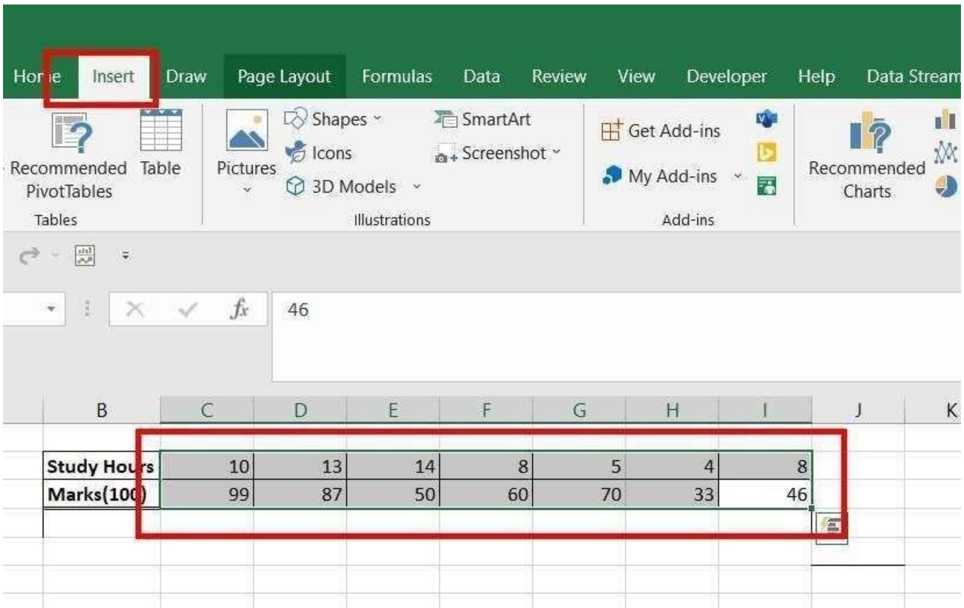
What is Bivariate Data :

Bivariate data is data which have two variable dependencies. The data can be either Quantitative or Qualitative. The value of one variable changes accordingly with the value of the second variable. The Quantitative bivariate data can be represented in the form of a scatter plot, and Qualitative variable data can be represented in the form of a frequency distribution table. A correlation can exist in bivariate data, the correlation value ranges from -1 and 1.

For example, Arushi is the class teacher of the tenth (X)th class, she collected data from her students, how many hours they use to study in a day, and added the marks achieved corresponding to that student, her task is to analyze this bivariate data in excel, by plotting it in a scatter plot.

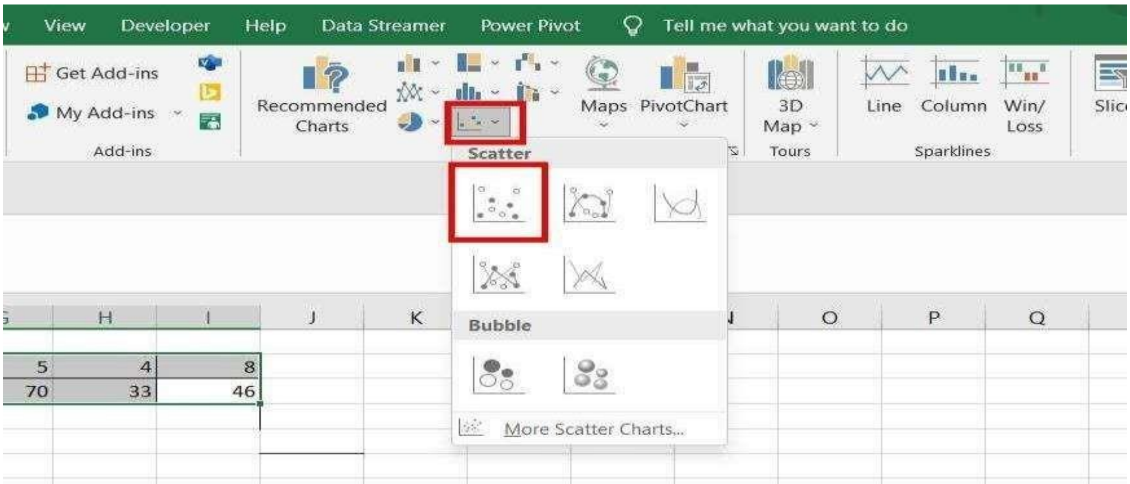
Step 1: Access the Insert Tab

Select the data you entered, and go to the Insert Tab. After selecting Insert Tab you have to add a scatter chart.



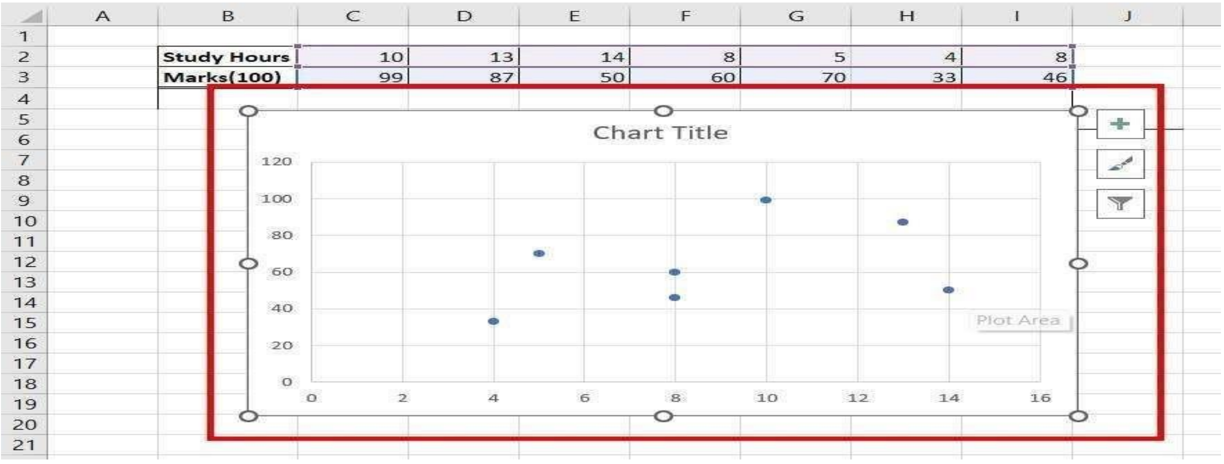
Step 2: Select the Scatter Chart

Find the Charts section, and under the section go to the scatter chart and select it.



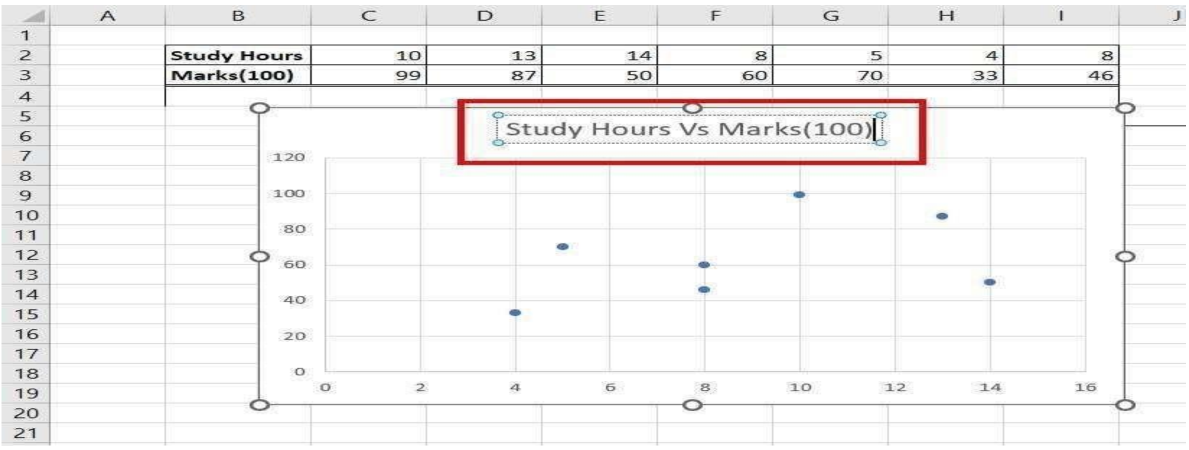
Step 3: Chart is Successfully Created

As you click on Scatter charts option, a chart is created, between Study Hours (X-axis) and Marks Scored (Y-axis).



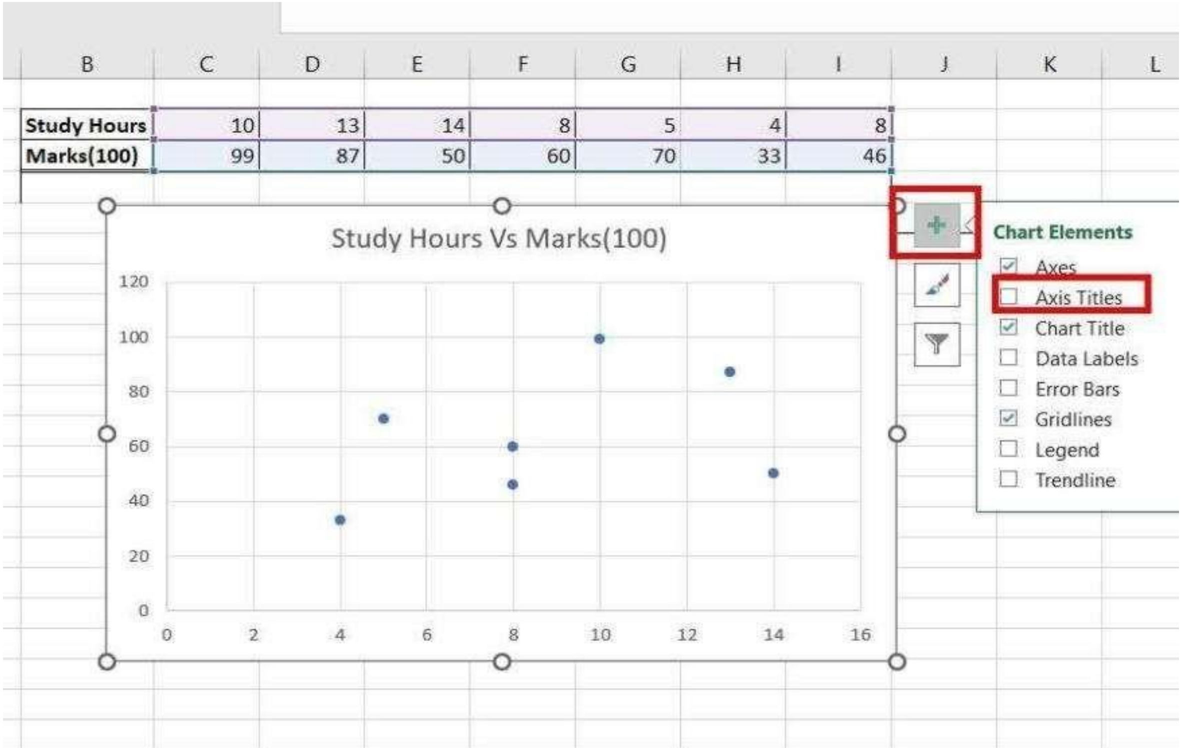
Step 4: Adding Custom Title

Add a custom title to the prepared chart i.e. Study Hours Vs Marks(100) by double clicking on the Title Box.



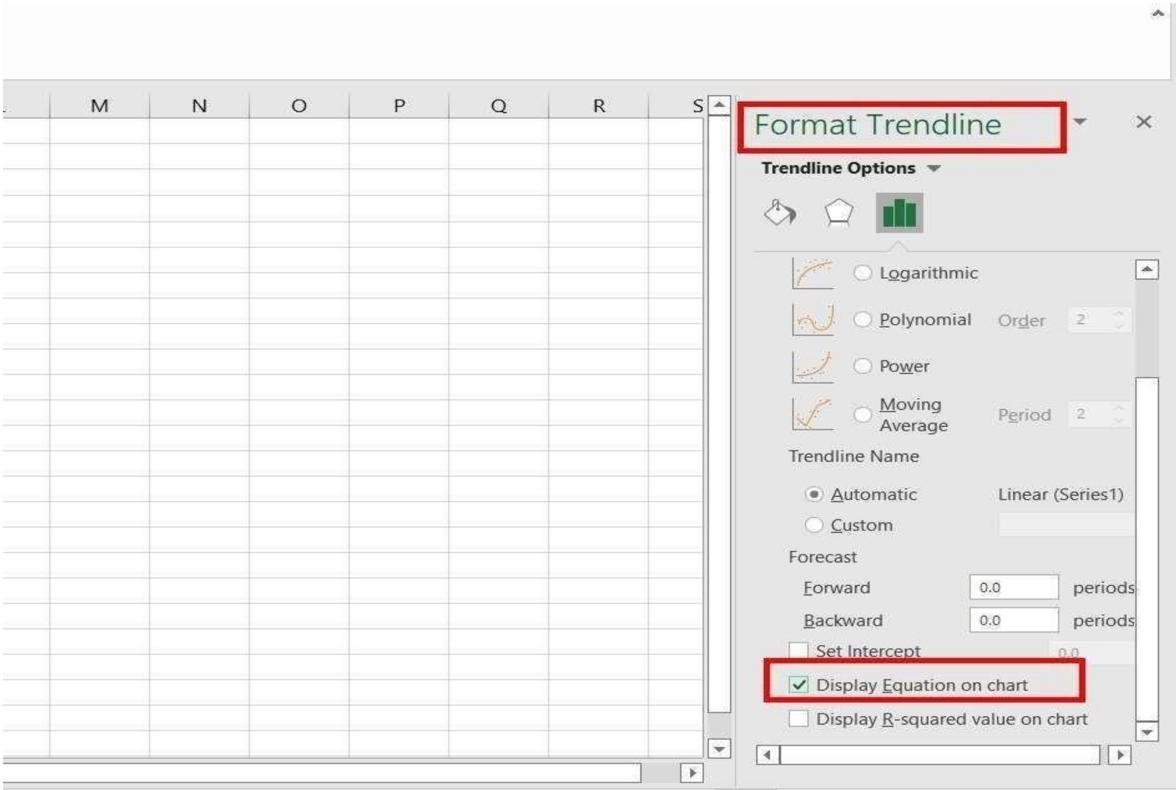
Step 5: Add Titles to X and Y Axis

Select the chart, click on the plus(+) button, and check the box Axis Titles. This adds the name of the X-Axis and Y-Axis.



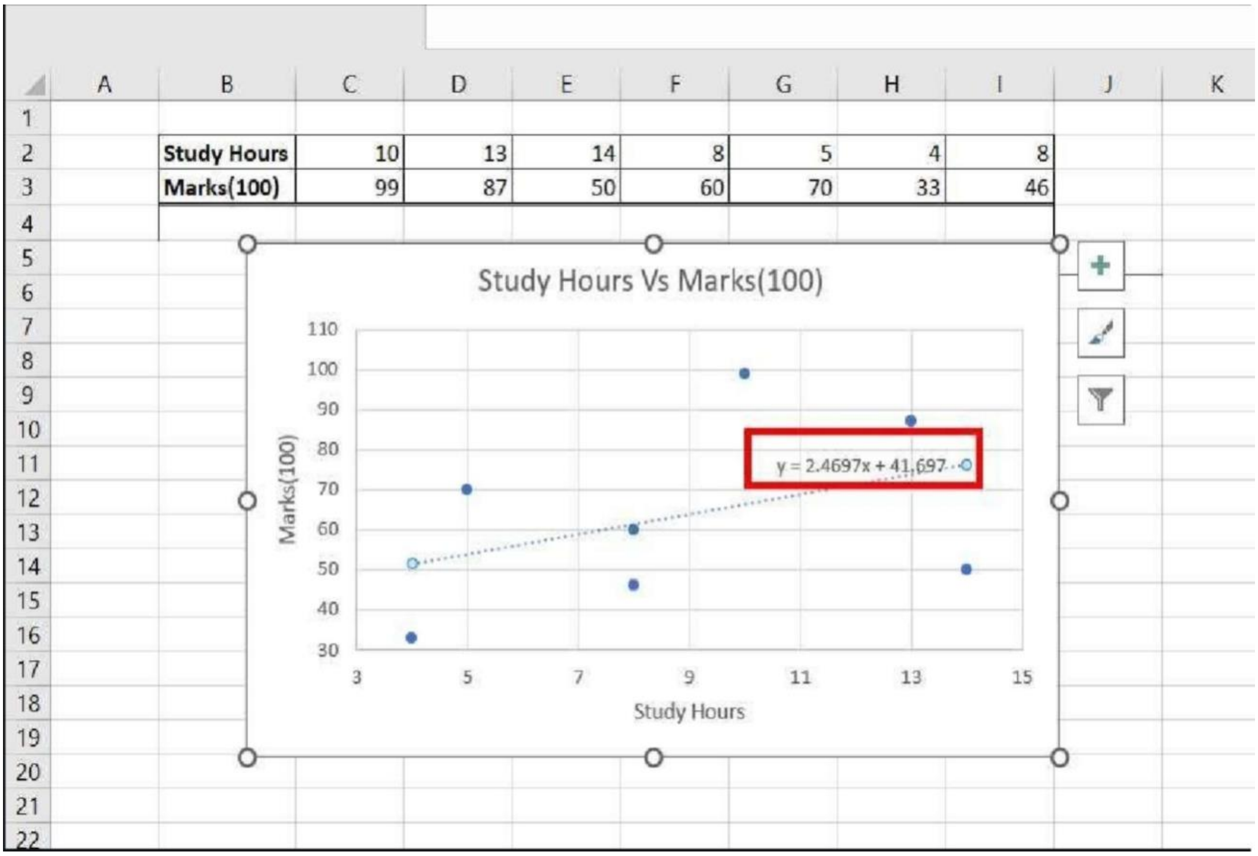
Go to Format Trendline and Select Display Eaquation On Chart

Also, a Format Trendline dialogue box appears, at the right-most side of your screen. Under the Trendline Options, check the box, Display equation on the chart.



Preview Added Trendline

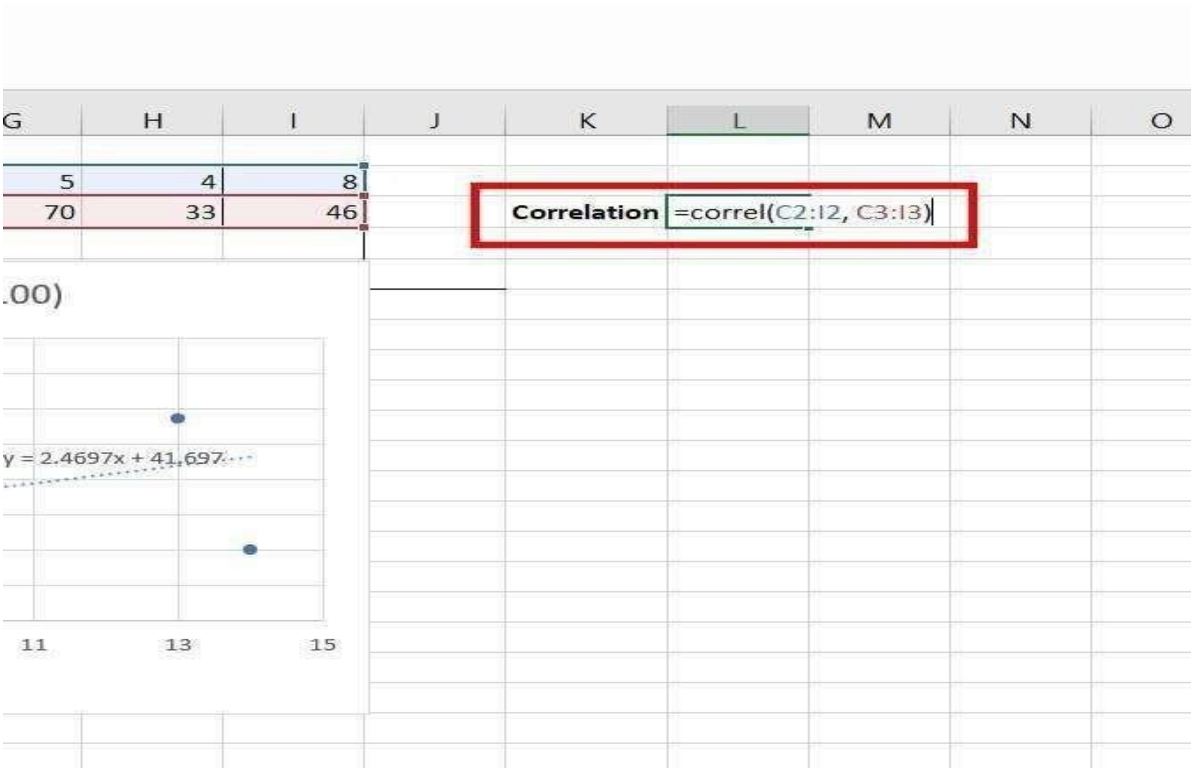
A Trend line and its equation is added to your chart.



Trend line is added

Step 6: Use CORREL Function

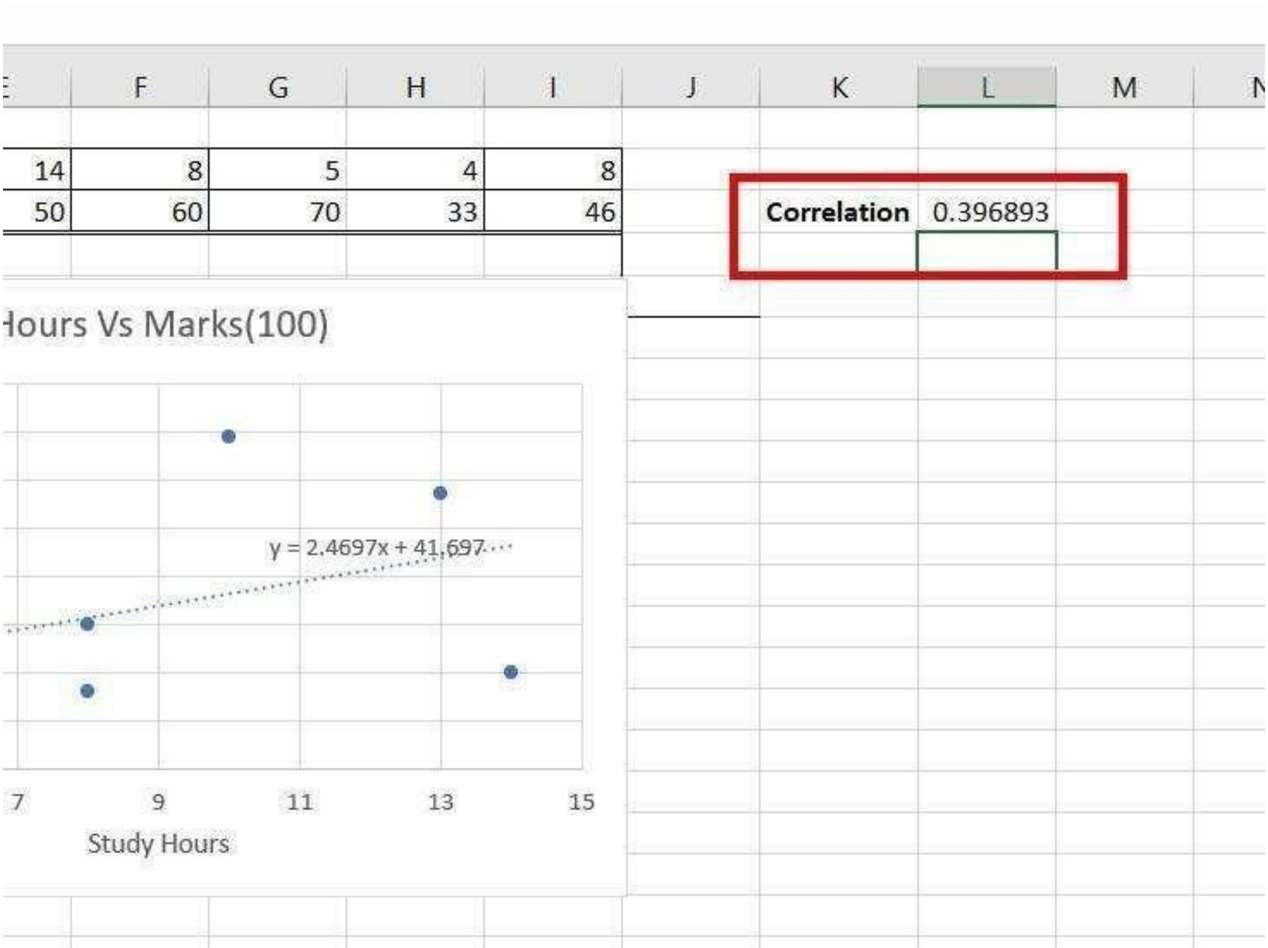
Our Bivariate data is plotted, efficiently for analytical purposes. The only work left is to add the correlation value between the two variables.



Using CORREL Function

Step 6: Value is Positive

The correlation value is 0.396 i.e. positive, which signifies that as the study hours increase, the marks of students also increase.



Correlation Value is Positive

Result:

Thus the bivariate and multivariate analysis was performed successfully.

Ex.No.8	Apply and explore various plotting functions on the data set
DATE:	

Aim:

To Apply and explore various plotting functions on the data set.

Procedure:

Set up your data

A line graph requires two axes, so your table should contain at least two columns: the time intervals in the leftmost column and the dependent values in the right column(s).

In this example, we are going to do a single line graph, so our sample data set has the following two columns:

	A	B
1	Month	Sales
2	Jan	$120
3	Feb	$150
4	Mar	$125
5	Apr	$140
6	May	$110
7	Jun	$100
8	Jul	$30
9	Aug	$60
10	Sep	$25
11	Oct	$160
12	Nov	$90
13	Dec	$80

1. **Select the data to be included in the chart**

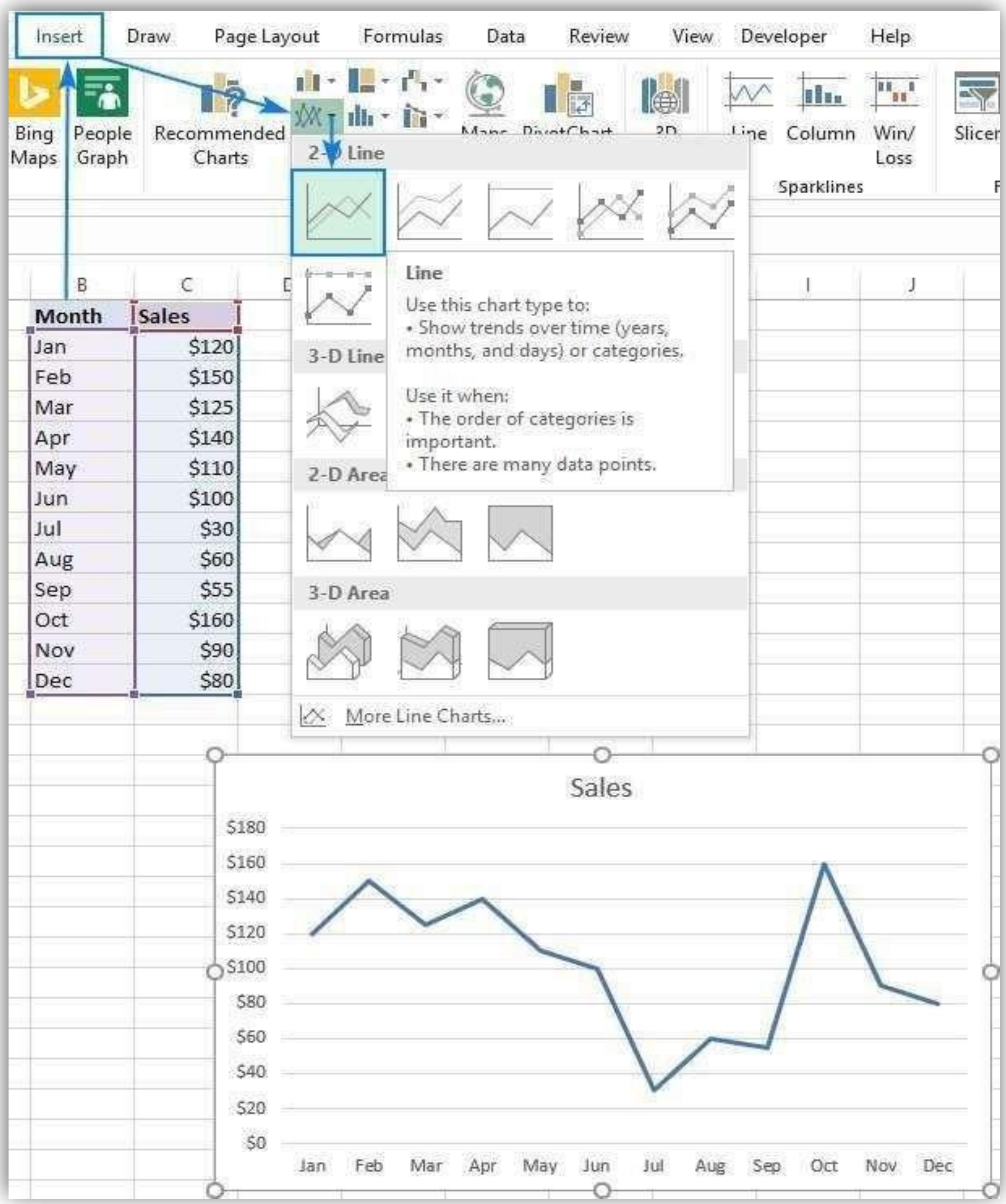
   In most situations, it is sufficient to select just one cell for Excel to pick the whole table automatically. If you'd like to plot only part of your data, select that part and be sure to include the column headers in the selection.

2. **Insert a line graph**

   With the source data selected, go to the Insert tab > Charts group, click the Insert Line or Area Chart icon and choose one of the available graph types.

As you hover the mouse pointer over a chart template, Excel will show you a description of that chart as well as its preview. To inset the chosen chart type in your worksheet, simply click its template.

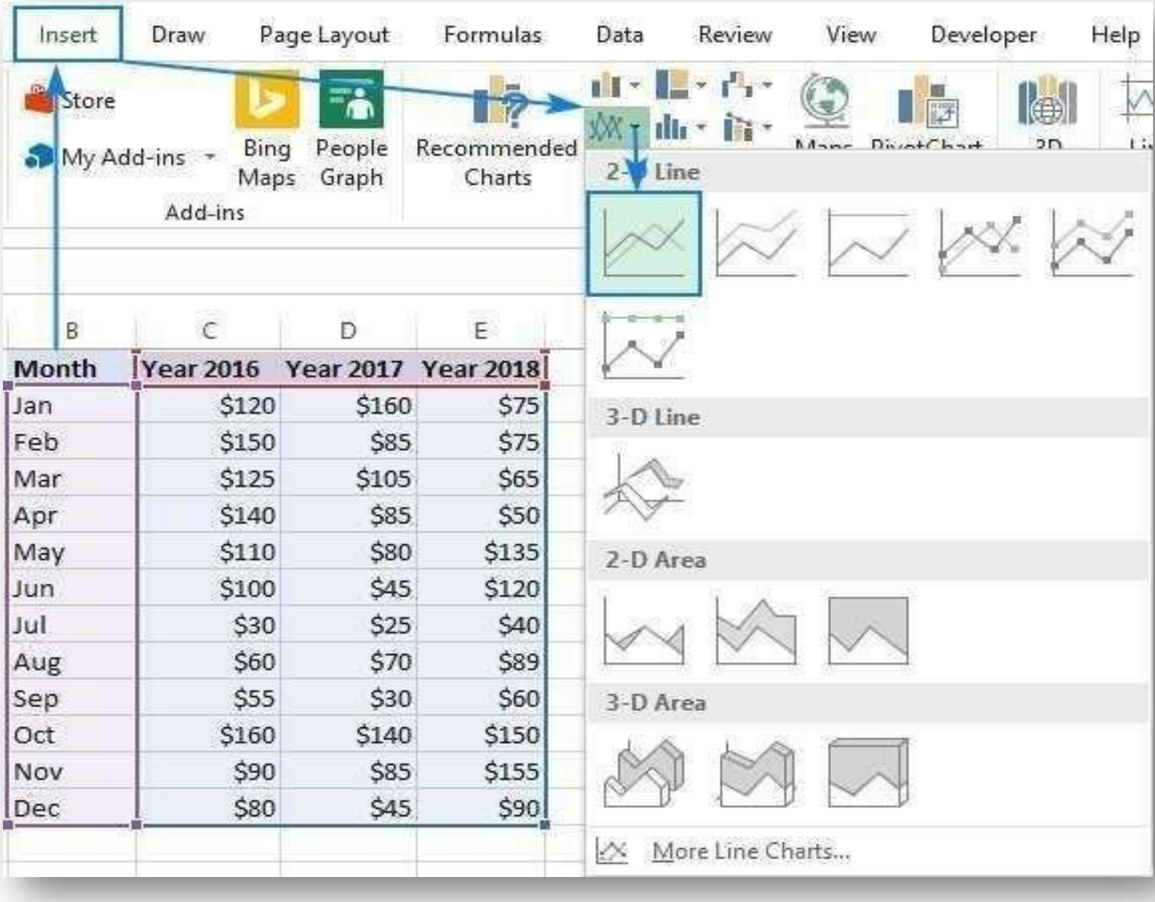
In the screenshot below, we are inserting the 2-D Line graph:



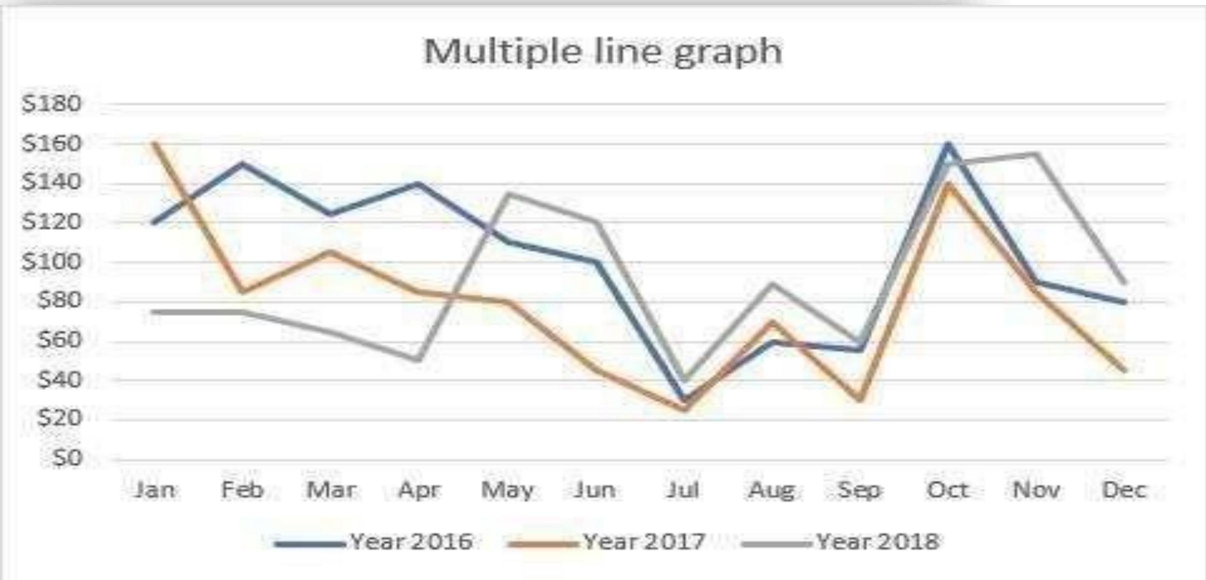
How to graph multiple lines in Excel

To draw a multiple line graph, perform the same steps as for creating a single line graph. However, your table must contain at least 3 columns of data: time intervals in the left column and observations (numeric values) in the right columns. Each data series will be plotted individually.

With the source data highlighted, go to the Insert tab, click the Insert Line or Area Chart icon, and then click 2-D Line or another graph type of your choosing:



A multiple line graph is immediately inserted in your worksheet, and you can now compare the sales trends for different years to one another.

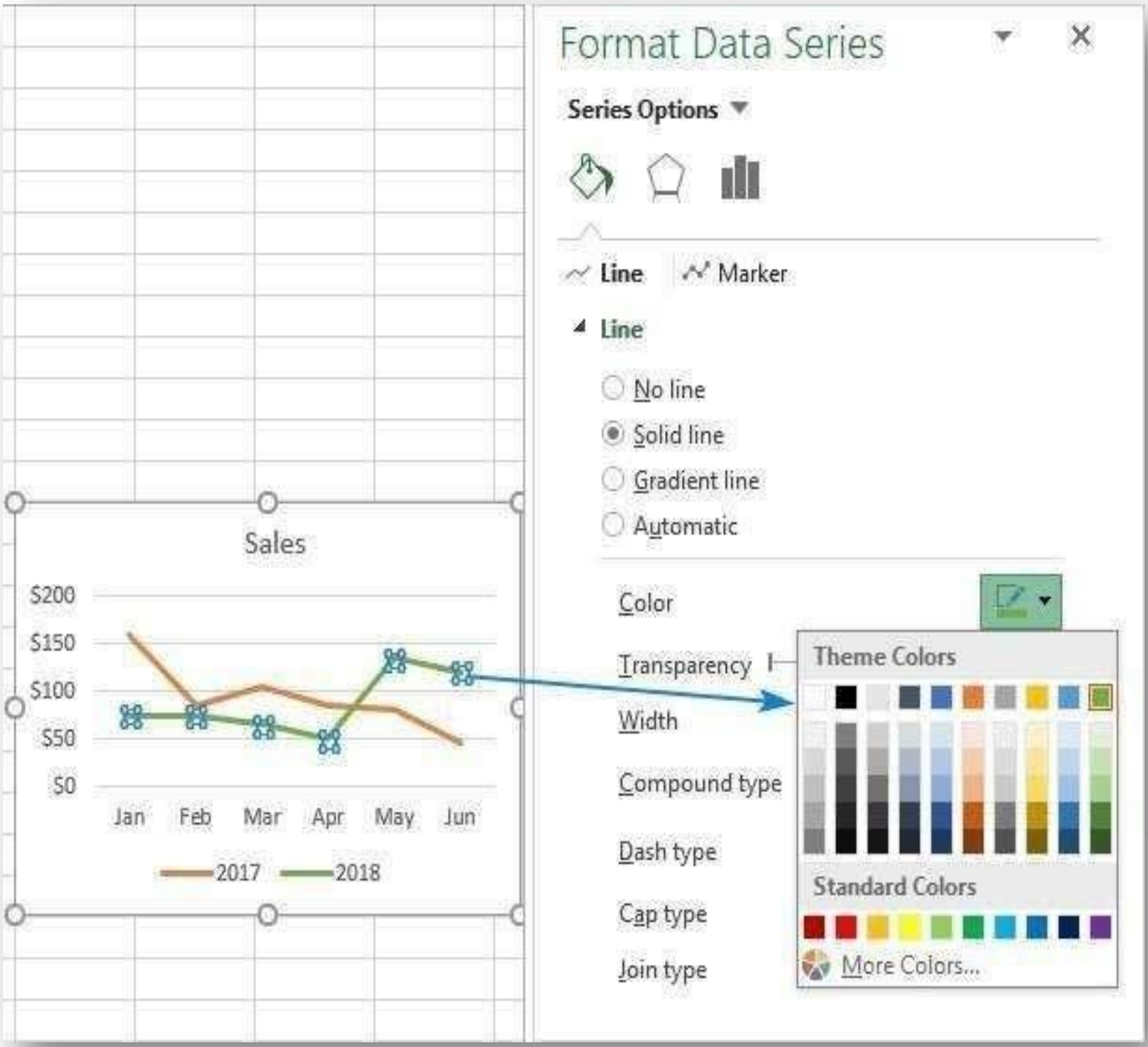


When creating a multiple line chart, try to limit the number of lines to 3-4 because more lines would make your graph look cluttered and hard to read.

Change color and appearance of a line

If the default line colors do not look quite attractive to you, here's how you can change them:

1. Double-click on the line you want to re-color.
2. On the Format Data Series pane, switch to the Fill & Line tab, click on the Color drop box, and choose a new color for the line.



Result:

Thus the various plotting functions on the data set was explored successfully.

Ex.No:09	Explore the features of power BI Desktop
DATE:	

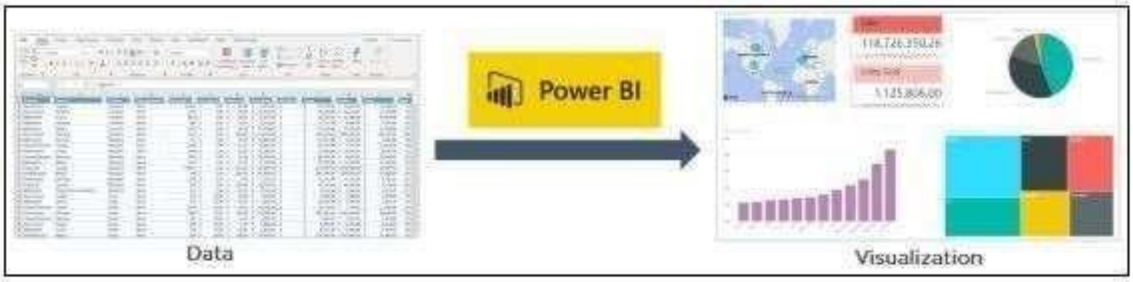
AIM:

To Explore the features of Power BI Desktop

PROCEDURE:

What is Power BI?

Power BI is a business analytics service provided by Microsoft that lets you visualize your data and share insights. It converts data from different sources to build interactive dashboards and Business Intelligence reports.



As you see above, we have some sales data in an excel sheet. Using this data, Power BI helps you build different charts and graphs to visualize the data.

Now that you have learned what is Power BI, let us now understand why Power BI is required.

Advantages Of Power BI

1. **User-friendly interface**: Power BI has an intuitive interface allowing users to visualize and analyze data easily.
2. **Data integration:** Power BI allows users to easily integrate data from various sources, including Excel, SQL Server, and cloud-based sources like Azure and Salesforce.
3. **Customizable dashboards:** Users can create customized dashboards and reports to display data in a way that is meaningful to them.
4. **Real-time data**: Power BI supports real-time data processing, which means users can view up-todate data in their dashboards and reports.
5. **Collaboration**: Power BI allows users to share their dashboards and reports with others, making collaborating on data analysis projects easy.

Disadvantages Of Power BI

1. Limited data processing capabilities: Power BI is not designed for heavy-duty data processing and may struggle with large datasets or complex queries.

2. Limited customization options: While Power BI offers a range of customization options, users may find that they are limited in their ability to create truly unique visualizations and reports.

3. Cost: Power BI is not a free tool, and users may need to pay for additional features or storage space.

Salient features of Microsoft Power BI

Microsoft Power BI is a collection of BI tools for creating complex visualizations and shareable reports.

The following are some of Power BI's outstanding features:

1. Analytics and data management

Microsoft Power BI provides powerful analytics to assist users in gaining valuable insights, transforming data into powerful components capable of providing ideas and breakthroughs to resolve business problems.

Advanced analytics in Power BI allows business users to track key performance parameters. It helps the business figure out the signs which lead to additional opportunities and success. Power BI creates stunning interactive dashboards and is a complex data management solution worthy of being termed. It uses data mining and BI technologies to identify data trends and provides tools to aid advanced analytics.

2. Quick Insights

Using powerful algorithms, users can obtain exciting insights from various subsets of the data. Quick Insights gives people better and faster access to analytics results with a single click.

3. Ask a Question

With this functionality, users can ask questions and get instant responses in the form of visuals such as charts, tables, or graphs.

4. Intuitive reports

Power BI reports are a collection of dashboards that are comprehensive and well organized, with various types of visualizations and formats that are relevant to a specific business issue by presenting valuable insights. Users may quickly share Power BI reports with other users.

5. Integration of Azure Machine Learning

Because of the inclusion of Machine Learning in Power BI, users can now visualize the outcomes of Machine Learning algorithms by dragging, dropping, and joining data modules.

6. Data Analysis Expression

Data Analysis Expression helps extract one or more values from a data set by analyzing the results generated by applying multiple formulae to the dataset under observation. It works similarly to Microsoft Excel, except without the numbers and rows. Data Analysis reports are simple to comprehend and create.

7. Real-time data analytics

As data is sent or streamed in, Power BI dashboards refresh in real-time, allowing viewers to solve problems and spot possibilities instantly. Real-time data and images are available in any report or dashboard. Factory sensors, social media sources, and anything else that may gather or transmit time-sensitive data are potential sources of streaming data into Power BI dashboards/reports.

8. Visualization and Reporting

Using Power BI will result in a more profound knowledge of data and the discovery of business insights in real-time. Power BI has many pre-built visualizations and the possibility to customize existing ones or choose from an ever-growing library of in-built visualizations.

9. Lively Dashboards

Provides a single point of reference with a complete visual picture of business-critical information.

10. Pre-existing Ready to use templates

Microsoft and the community have offered hundreds of out-of-the-box graphics, including charts, cards, KPIs, maps, and matrices.

Result:

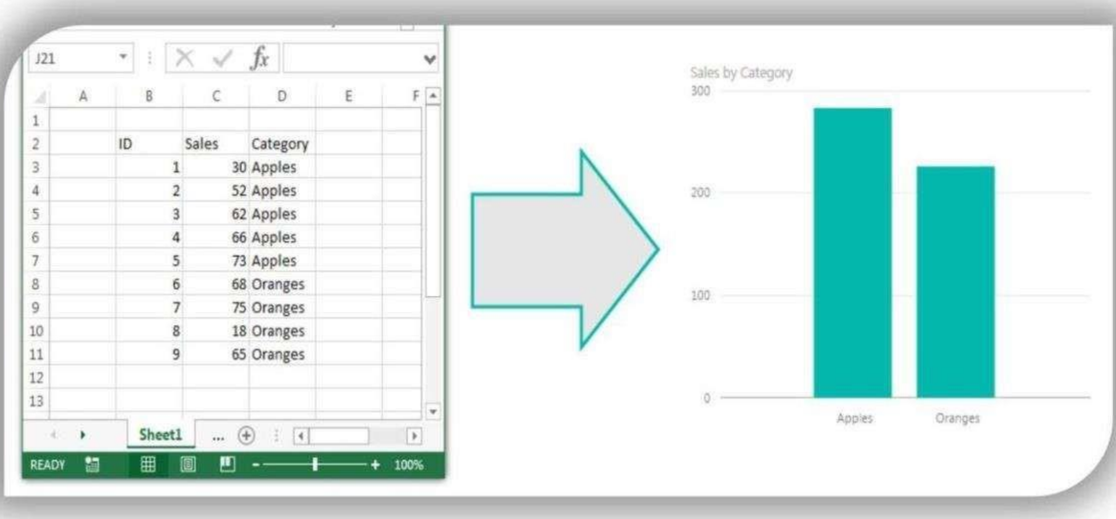
Thus the features of power bi was verified successfully.

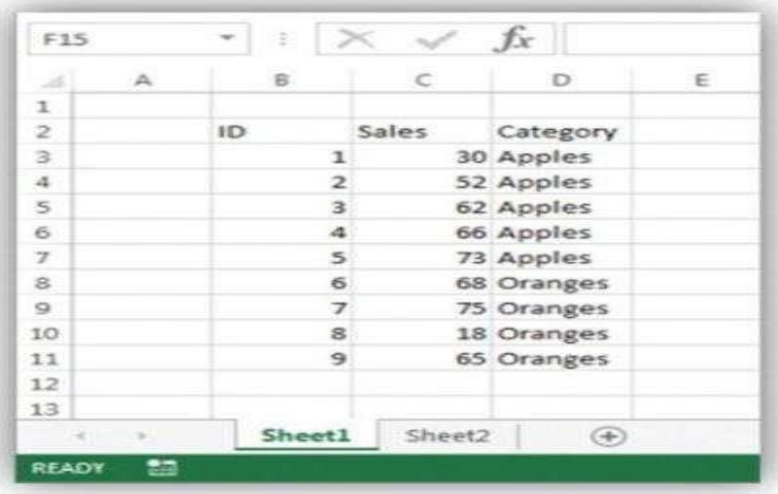
Ex.No.10	Prepare & Lead data Using Power BI
Date:	

AIM:

To Prepare & Load data Using Power Bi

PROCEDURE:





	A	B	C	D	E
1					
2		ID	Sales	Category	
3		1	30	Apples	
4		2	52	Apples	
5		3	62	Apples	
6		4	66	Apples	
7		5	73	Apples	
8		6	68	Oranges	
9		7	75	Oranges	
10		8	18	Oranges	
11		9	65	Oranges	
12					
13					

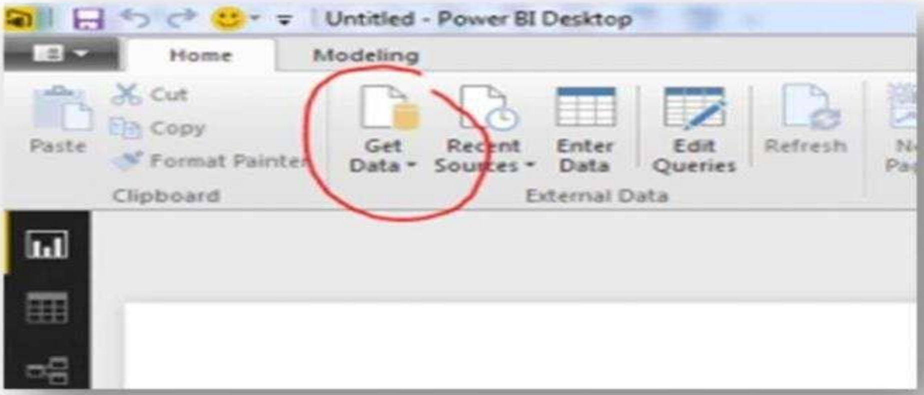
Import an Excel file into PowerBI

Lets start off with some simple data in excel:

Sample of Data in Excel

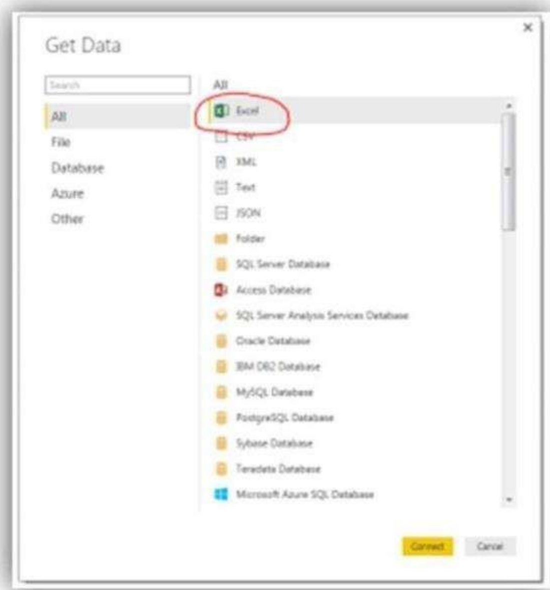
We have three columns of data, two have number in it and one has text values.

For now we will close out of excel and jump over to Power BI Desktop. Once the program loads we will click the Home ribbon then select the Get Data button.



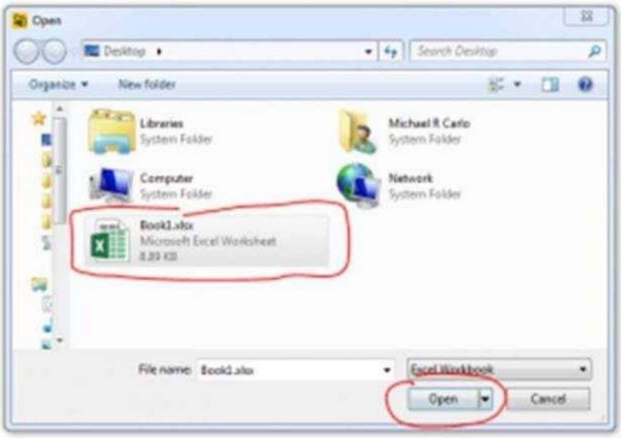
Button for Get Data

After pressing the button a new menu will pop up showing us all the sources where data can be ingested from. The very first item in the list is Excel. Click the Excel then click the Connect button in the lower right hand corner.



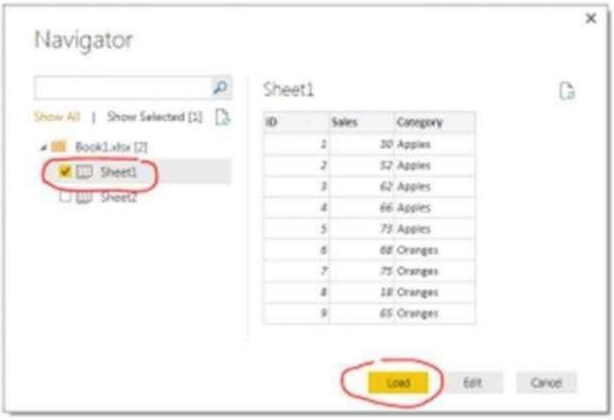
Select Excel as Data Source

After clicking Connect a new window will pop up asking for the location of the Excel file. Navigate to our sample data called Book1.xlsx you can down load the actual file used here: Book1 I saved my Book1.xlsx file on the desktop of my computer. Select Book1 and then Click Open.



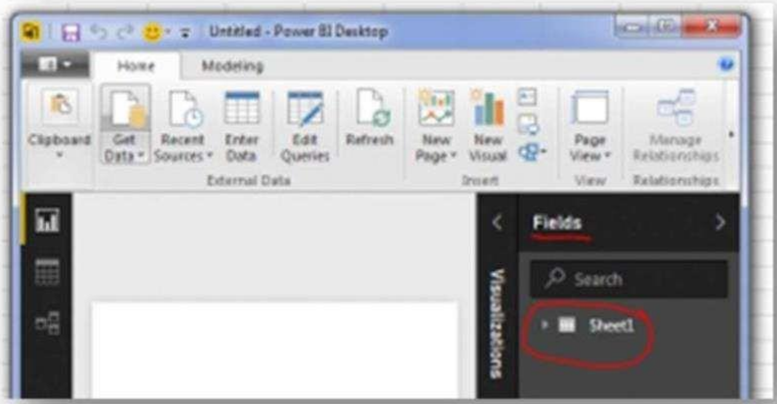
Open Excel File Dialog Box

Next we are presented with the Navigator screen that reveals what is inside the workbook. There are two sheets. For now we are only interested in the data on Sheet1. Select Sheet1 and then click Load. This will load our data from Sheet1 into the Power BI Desktop data model.



Navigator Selection Screen

Now our data has been added to the Power BI Desktop data model. The data and the various columns we loaded can be found in the tool bar at the far right of PBI called Fields.



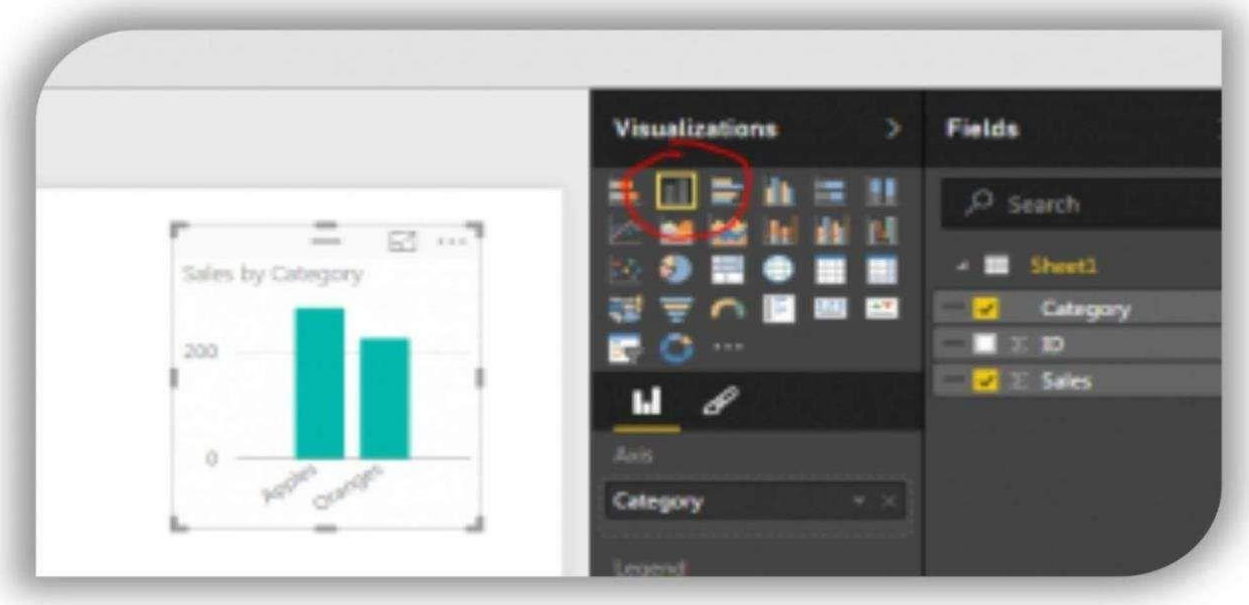
]

Location of Loaded Excel

DataTech Tip: Power BI Desktop (PBI) opening the file and loading the relevant data into the memory of the computer. This has an approximate 4 to 1 compression ratio. In practical terms this means that a 100MB file will only consume 25MB of file size in PBI when it is saved. This is extremely useful as the data model can be quite large when loading multiple data files but the PBI file will compress down to a manageable size.

Make a Data from Column Sales and Category

Finally, the Sheet1 data table can be expanded into is respective columns by clicking the triangle next to the table icon. Finally you can drag and drop the column names into the visualization page to begin making visualizations. For this demo I used the Category Column and the Sales column to make a table.By selecting a different visualization in the visualizations bar you can change your data table into a Bar Chart.



Data Transformed into a Bar Chart Well that is it for the first tutorial. Share your thoughts and comments below. Let me know if you have any suggestions on what you would like to see next.

Result:

Thus the prepare & load data operations were executing successfully using power bi tool

Ex.No.11 **DATE:**	**Make a data model using power BI**

Aim:

To Make a data model using power bi

Procedure:

Step 1: Creating Model Relationships

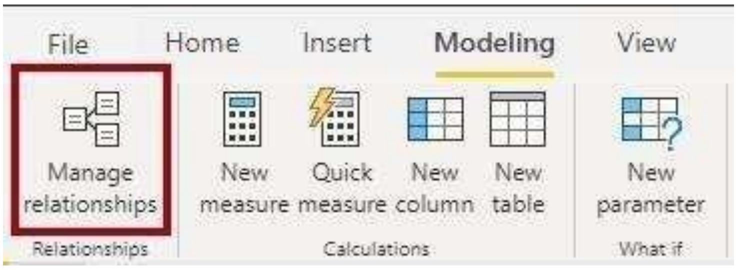
Open the Power BI Desktop application and click on the “File” ribbon.
Click on the “Open” option and import the data for making Power BI Data Model.
For this tutorial, sample data from Sales Analysis will be used.
Click on the “Model View” option from the left side of the Power BI, as shown in the image below.

- Here, you can view each table and relationship.
- In the “Fields” pane rightclick on an empty area and click on the “Expand All” option to view all the table fields.
- Now, create a visual table by selecting the “Category” field inside the “Product” table, as shown in the image below.

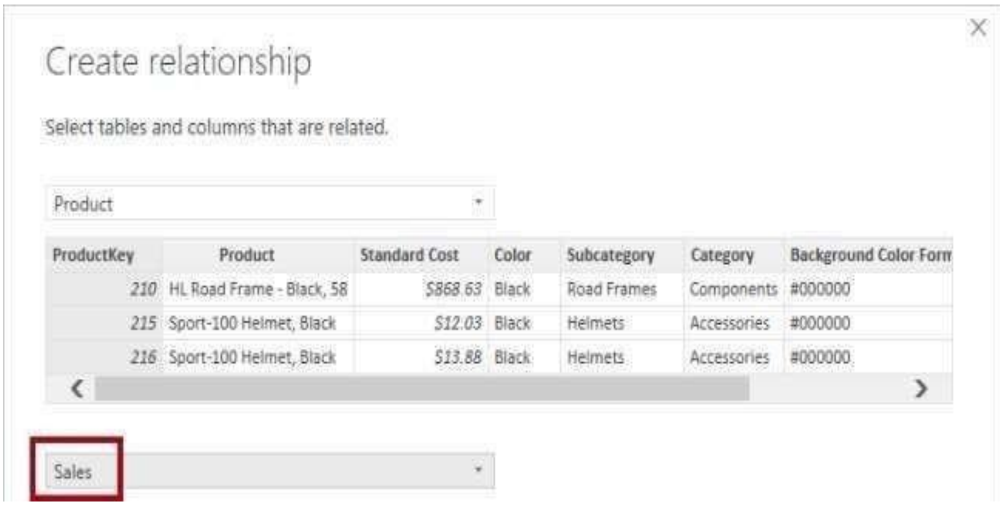
- For adding a column to the table, go to the “Fields” pane and select the “Sales” from the “Sales” table, as shown in the image below.

Category	Sales
Accessories	$77,548,570.2
Bikes	$77,548,570.2
Clothing	$77,548,570.2
Components	$77,548,570.2
Total	**$77,548,570.2**

- There is no relationship between tables, so it will not filter the "Sales" table.
- Navigate to the "Modeling" ribbon, and from the "Relationships" group, select the "Manage Relationships" option, as shown in the image below.

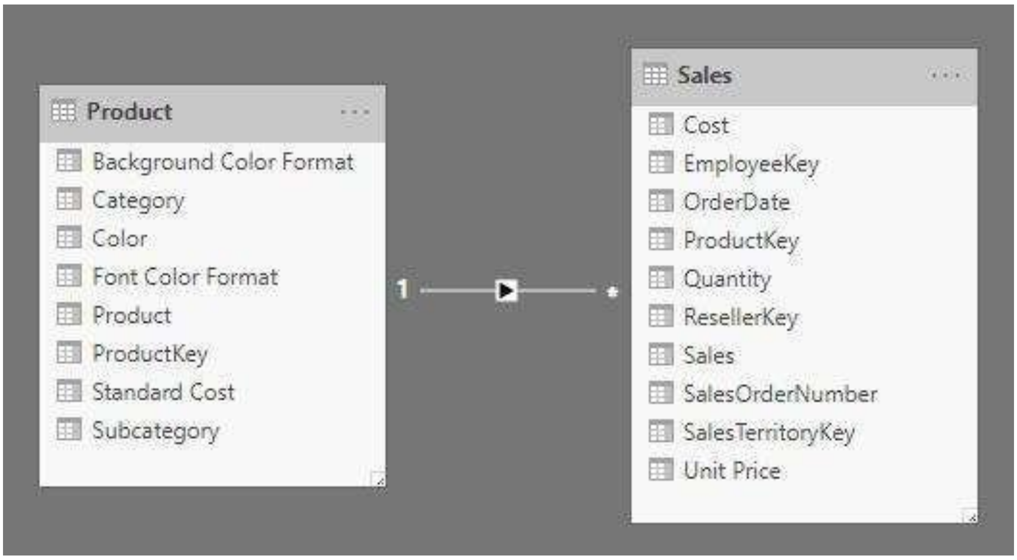


- Click on the "New" button to create a new relationship.
- It will open the "Create Relationship" window.
- Select the "Product" table from the first drop-down list, then select the "Sales" table from the second drop-down list, as shown in the image below.

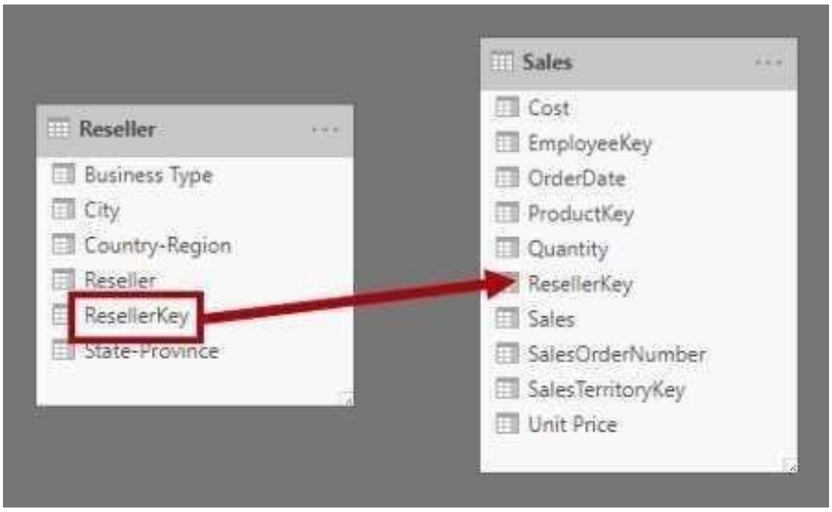


The "ProductKey" columns from both the tables are automatically selected as common columns because it has the same data type and name.

- Click on the "Ok" button.
- Now notice that the table visual has been updated to display values for each product category.
- If you open the "Model View", you will find a connector between the two tables that were not present before, as shown in the image below



- Another way to create a relationship between tables is by dragging one column from one table to another.
- In this Model View, select the “ResellerKey” column from the “Reseller” table and drag it onto the “Resellerkey” column of the “Sales” table, as shown in the image below.

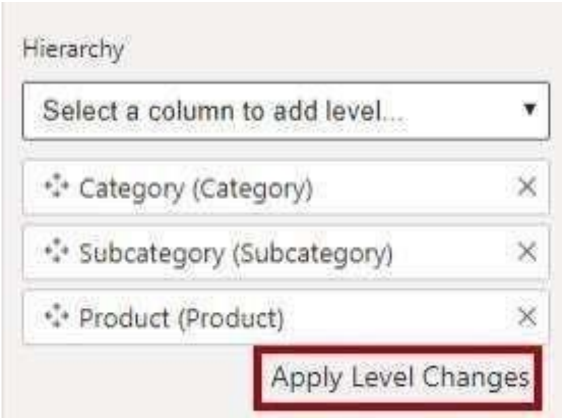


- Similarly, create the following relationships by dragging columns. listed below:
  - “SalesTerritoryKey” column from the “Region” table to “SalesTerritoryKey” column of “Sales” table.
  - “EmployeeKey” column from the “Salesperson” table to the “EmployeeKey” column of the “Sales” table.
- Now save the Power BI file.

Step 2: Configuring Tables

- Expand the “Product” table from the “Fields” pane.
- Now, right-click on the “Category” column and select the “Create Hierarchy” option to create a hierarchy.
- Now, from the “Properties” pane (located left to the Fields pane), replace the text with “Products” in the “Name” box.
- To add the second level to the hierarchy, in the “Properties” pane, in the “Hierarchy” drop-down list, select the “Subcategory” option.

- Similarly for adding the third level to the hierarchy, select the “Product” table.
- Click on the “Apply Level Changes” option, as shown in the image below.



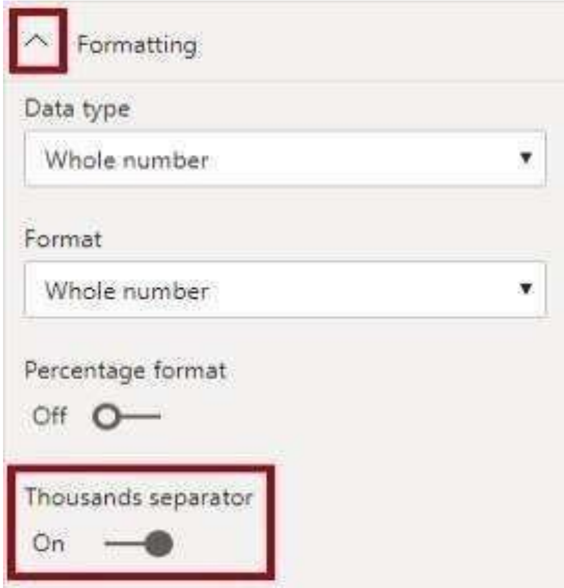
- Similarly, configure the “Region” table by creating a hierarchy named “Regions” with the 3 levels of hierarchy as listed below.
    - Group
    - Country
    - Region

- Now, select the “Country” table and open the advanced properties.
- Here, in the “Data Category” drop-down option select the “Country/Region” option, as shown in the image below.

- Now, let's configure the "Reseller" table by creating a hierarchy named "Resellers" with 2 levels, "Business Type" and "Reseller", as shown in the image below.

- Create another hierarchy named "Geography" with 4 levels of hierarchy listed below:
  - Country- Region
  - State-Province
  - City
  - Reseller

- Coming to the "Sales" table, here select the "Cost" column and type "Based on standard cost" in the description box from the "Properties" pane.
- Now, select the "Quantity" column, and from the "Properties" pane go to the "Formatting" section and slide the "Thousands Separator" property to "Yes", as shown in the image below.



- Next, select the “Unit Price” column and set the “Decimal Places” to 2 under the “Formatting” section.
- Then, from the advanced properties, select the “Average” option from the “Summarize By” drop-down list, as shown in the image below.

- Now, press the “Ctrl” key and select the following columns listed below.
  - ProductKey column from the Product table
  - SalesTerritoryKey column from Region table
  - ResellerKey column from the Reseller table
  - EmployeeKey column from the Sales
  - ProductKey column from the Sales table
  - ResellerKey column from the Sales table
  - SalesOrderNumber column from the Sales table
  - SalesTerritoryKey column from the Sales table
  - EmployeeID from Salesperson table
  - EmployeeKey column from the Salespersontable
  - UPN column from Salesperson table
  - EmployeeKey columnfrom SalespersonRegion table
  - SalesTerritoryKey columnfrom SalespersonRegion table
  - EmployeeID column from Targets table

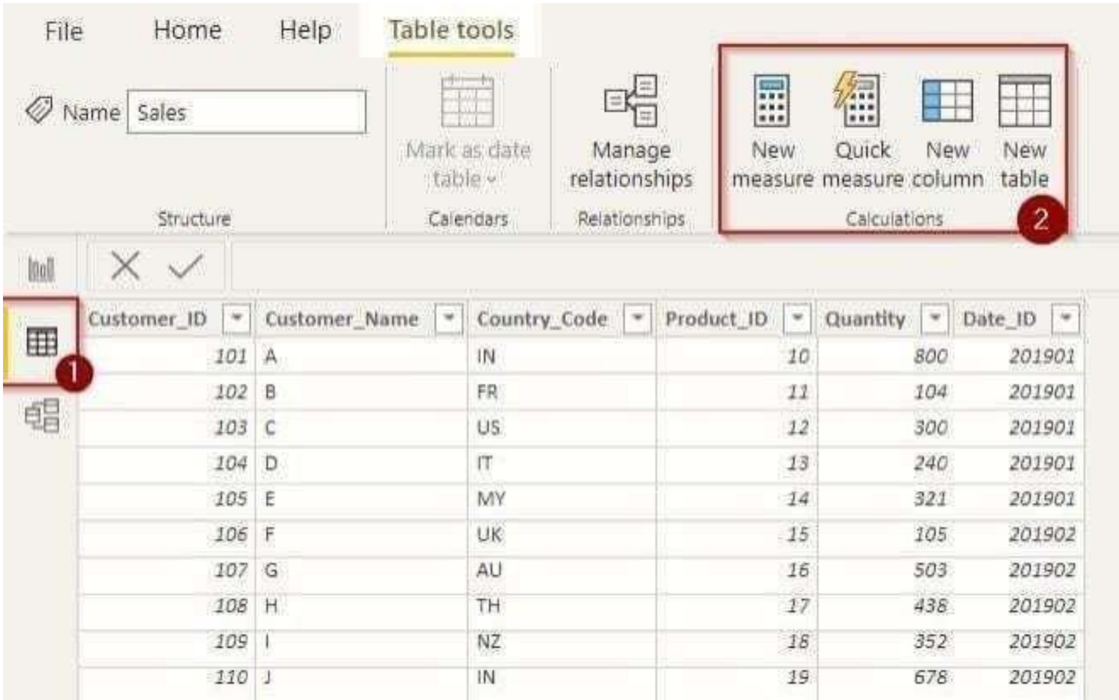
- From the “Properties” pane, toggle the “Is Hidden” property to “On“.
- Now, switch to the “Report View” and review the designed Power BI Data Model.

That’s it! You have successfully built a Power BI Data Model.

Step 3: Calculate And Measure Data

- Create Table
- Create Column

Let's make some computations using any Power BI reference data that is available. Visit the 'Data' tab from the left menu as highlighted in the image below. You can find various tools to calculate your data right here. These will be utilized by Power BI.



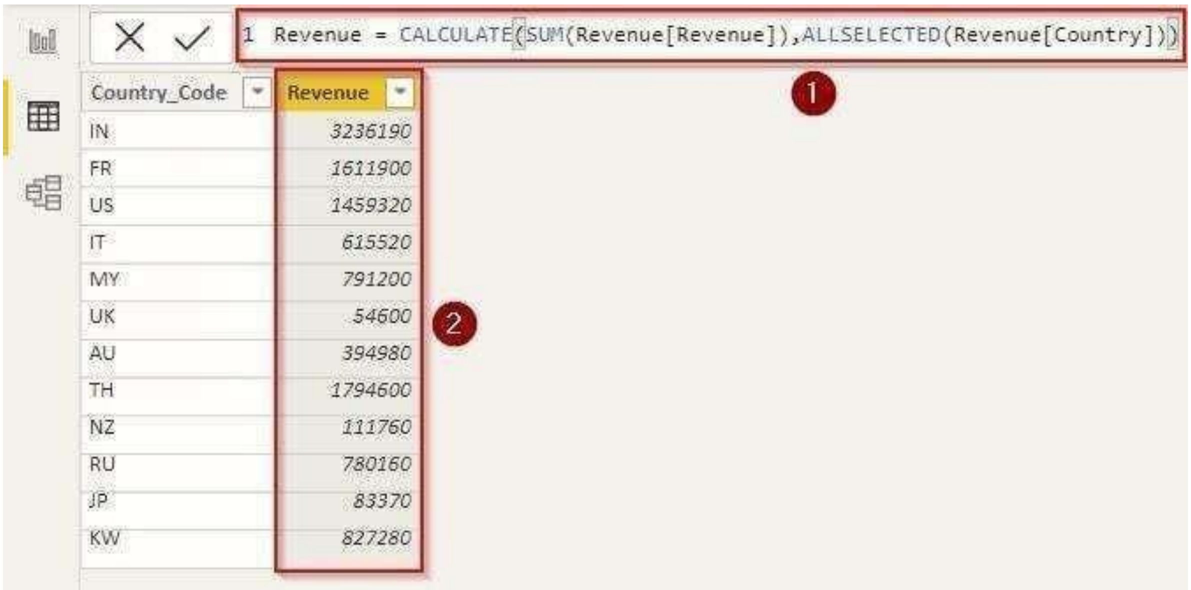
Create Table

We must insert the DAX expression given in the following image after clicking "New Table."

- The name of the table is specified in the expression's first component.
- The second step is the filter; the 'DISTINCT' function will only choose the column's singular values.
- The arguments that we must supply into the "DISTINCT" function are the locations from which we will extract our data. We have now reached the table and column names that include our nation codes. Click "Enter" once your expression is complete.
- We will receive our new column with the default name and results after applying the expression. You can double-click on a column to rename it.

Create Column

To create a calculated column, select "New Column" from the top menu.

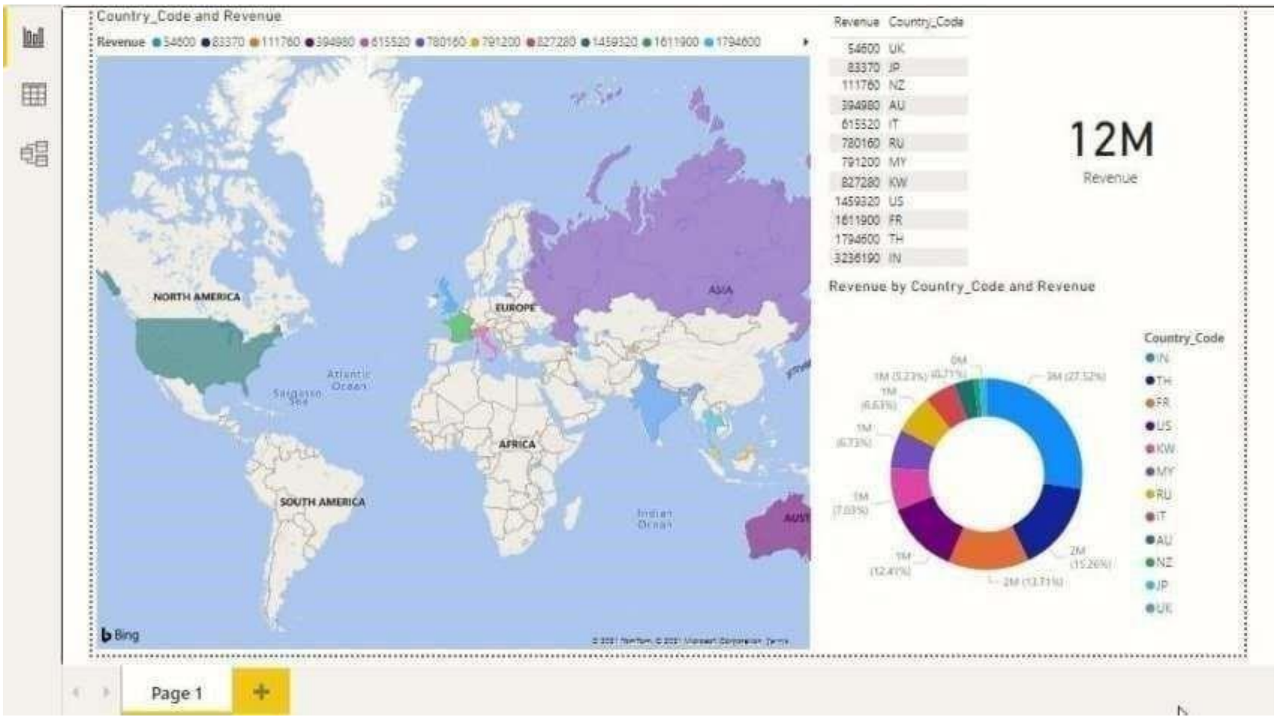


- For instance- in the above example shown the DAX expression will total up all of the revenue from the "Revenue" table with the "Country" filter. Without this phrase, we might have spent hours figuring out how much money each person brought in from the nation.
- We obtain this outcome from the expression.

Even though Power BI advises you to write an expression, it could be challenging to remember them all. Use the Quick Measure tool if this is the case. For calculations, you only need to enter the parameters and function. Depending on your choice, this tool will then automatically generate the expression. When you need swift calculations for your reports, these measurement tools come in handy.

Step 4: Create Visualization

Here, for instance, we've used various visualizations to display the country's revenue on a map of the globe. In the same manner, Power BI lets you construct and manage data models.



Benefits of Power BI Data Model

Some advantages of using a Power BI Data Model are listed below:

- Power BI Data Modeling makes it easier for other users to navigate through the data.
- Power BI Data Model helps in connecting and building relationships between different data sources.
- Power BI Data Model optimizes the query and aggregates data in volume.

Result:

Thus the data modelling was explored successfully.

Ex.No.12 DATE:	**Perform DAX calculations using power BI Tool**

Aim:

To Perform DAX calculations using power bi tool.

Procedure:

	A	B	C	D	E
1	City	State	Units Sold	Sale Price	Sale Value
2	Auburn	Alabama	$ 143	$ 14	$ 2,002
3	Auburn	New York	$ 181	$ 22	$ 3,982
4	Auburn	Washington	$ 219	$ 18	$ 3,942
5	Columbia	Maryland	$ 363	$ 13	$ 4,719
6	Columbia	South Carolina	$ 309	$ 15	$ 4,635
7	Columbia	South Carolina	$ 436	$ 24	$ 10,464
8	Columbia	Georgia	$ 154	$ 17	$ 2,618
9	Columbus	Georgia	$ 544	$ 16	$ 8,704
10	Columbus	South Carolina	$ 123	$ 14	$ 1,722
11	Columbus	Georgia	$ 241	$ 11	$ 2,651
12	Concord	California	$ 272	$ 18	$ 4,896
13	Concord	New Hampshire	$ 270	$ 21	$ 5,670
14	Concord	California	$ 433	$ 10	$ 4,330
15	Des Moines	Iowa	$ 473	$ 16	$ 7,568
16	Des Moines	Washington	$ 129	$ 18	$ 2,322
17	Des Moines	Iowa	$ 382	$ 24	$ 9,168
18					

You can directly upload the data table to the Power BI file. We have already uploaded the table to the Power BI Desktop file.

City	State	Units Sold	Sale Price	Sale Value
Auburn	Alabama	143	14	2002
Auburn	New York	181	22	3982
Auburn	Washington	219	18	3942
Columbia	Maryland	363	13	4719
Columbia	South Carolina	309	15	4635
Columbia	South Carolina	436	24	10464
Columbia	Georgia	154	17	2618
Columbus	Georgia	544	16	8704
Columbus	South Carolina	123	14	1722
Columbus	Georgia	241	11	2651
Concord	California	272	18	4896
Concord	New Hampshire	270	21	5670
Concord	California	433	10	4330
Des Moines	Iowa	473	16	7568
Des Moines	Washington	129	18	2322
Des Moines	Iowa	382	24	9168

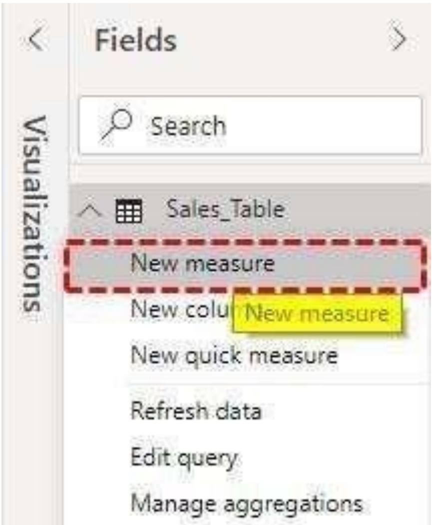
Now, we will experiment CALCULATE function to arrive at different sets of results.

Arrive at one particular city sales total

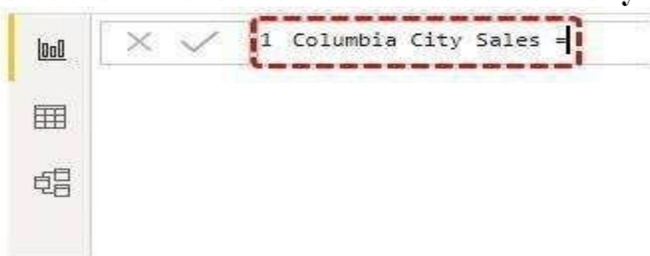
For example, assume you need to create a "New Measure," which gives one particular city total, for example, "Columbia" city.

The steps to use the DAX calculate function in Power BI is as follows.

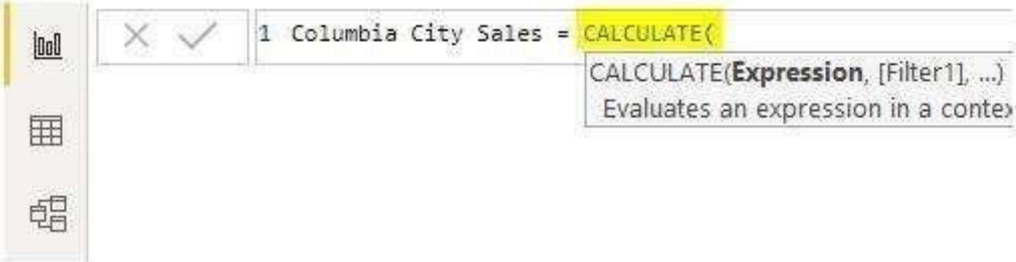
1. **Right-click on the table, and choose the "New measure" option.**



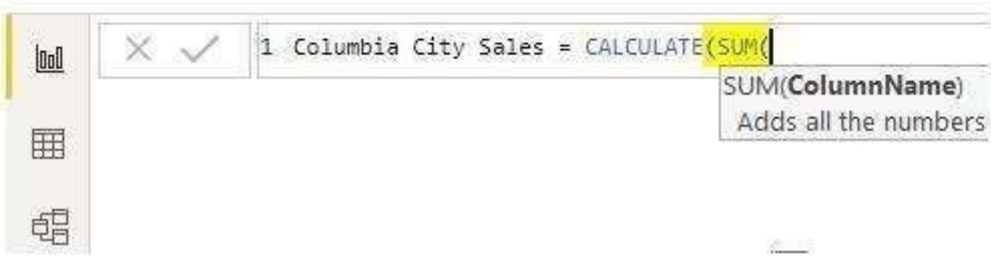
2. **Give the name to this measure "Columbia City Sales."**



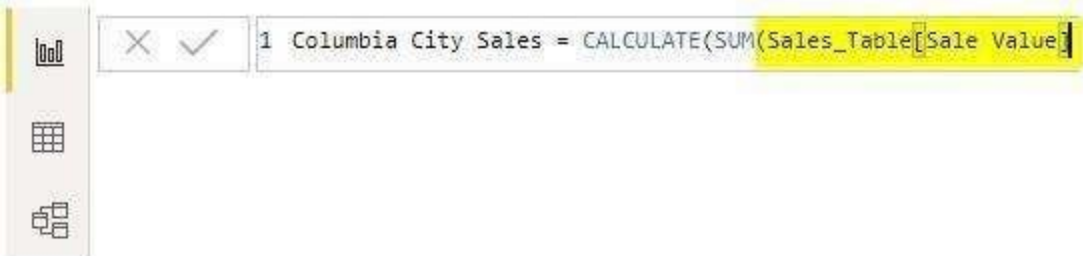
3. **Now, open the CALCULATE function.**



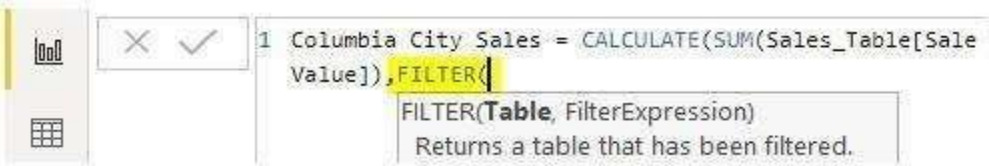
4. **An expression is the first option. We need to add the “Columbia” city total in this example, so open the SUM function.**



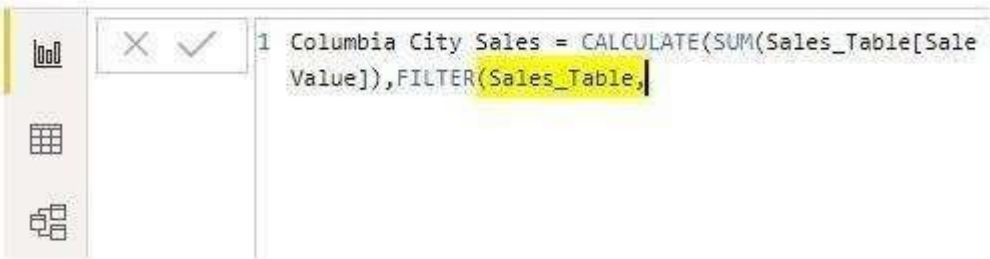
5. **The “ColumnName” we need to SUM is the “Sales Value” column, so choose the respective column**.



6. **The “ColumnName” we need to SUM is the “Sales Value” column, so choose the respective column.**

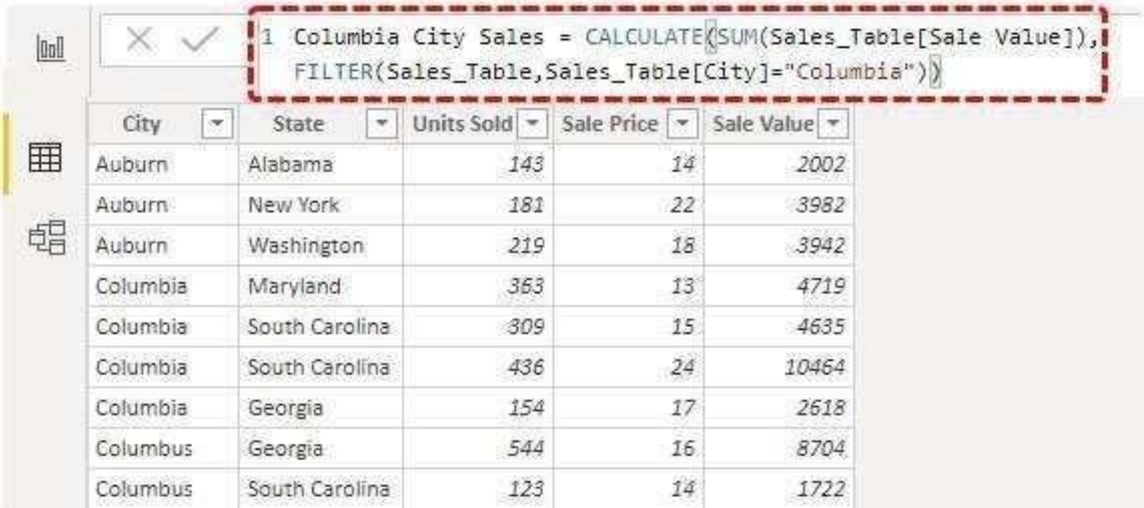


7. **The table we refer to is the “Sales_Table.” So first, choose the table name.**

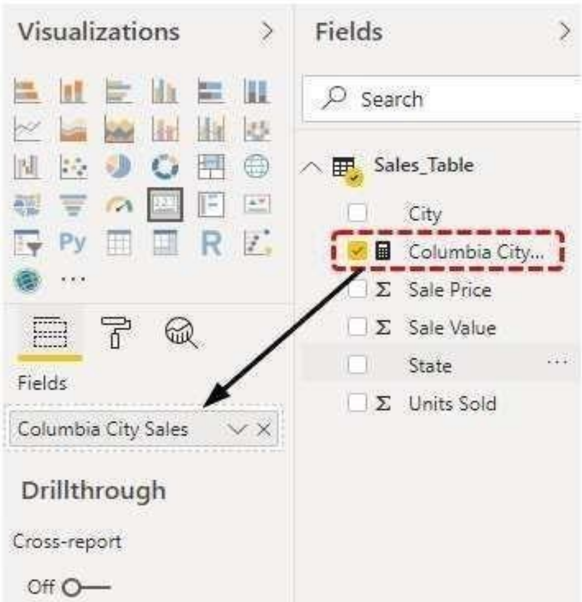


8. We need to select the "City" column for FilterExpression and give the criteria as "Columbia."

9. We are done. Close two brackets and press the "Enter" key to get the new measure.



10. Drag the "Columbia City Sales" to "Fields" to see the new measure.



11. **Now, this measure only gives the total sales of the city "Columbia."**

12. **You can cross-check the "Columbia" city total in Excel as well.**

	A	B
1		
2		
3	**City**	**Sum of Sale Value**
4	Auburn	9,926
5	Columbia	22,436
6	Columbus	13,077
7	Concord	14,896
8	Des Moines	19,058
9	**Grand Total**	**79,393**
10		

Like this, we can use the CALCULATE function to arrive at different results.

Result:

Thus the about DAX calculations was executing successfully using power bi tool.

Ex.No.13	Power BI used to Design a report
Date:	

Aim:

To design a report using power bi.

Procedure:

Preparing Your Data

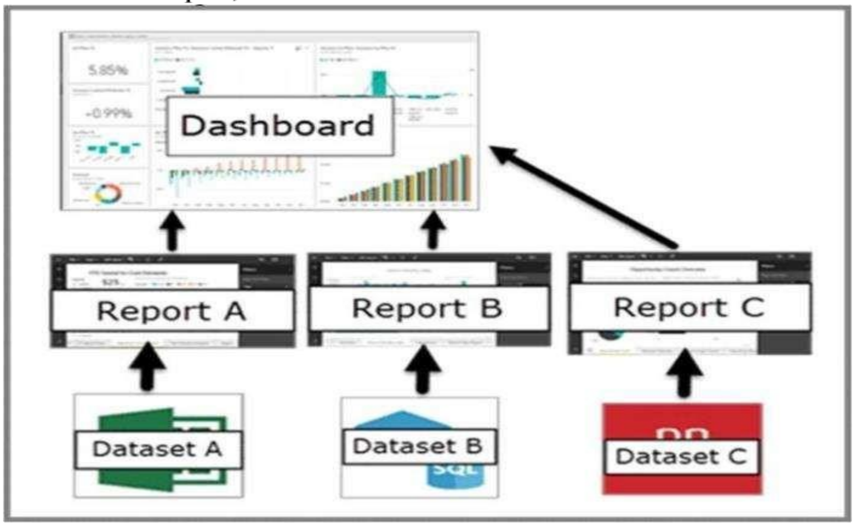
Before you can create any reports in Power BI, you need a dataset and a blank report canvas.

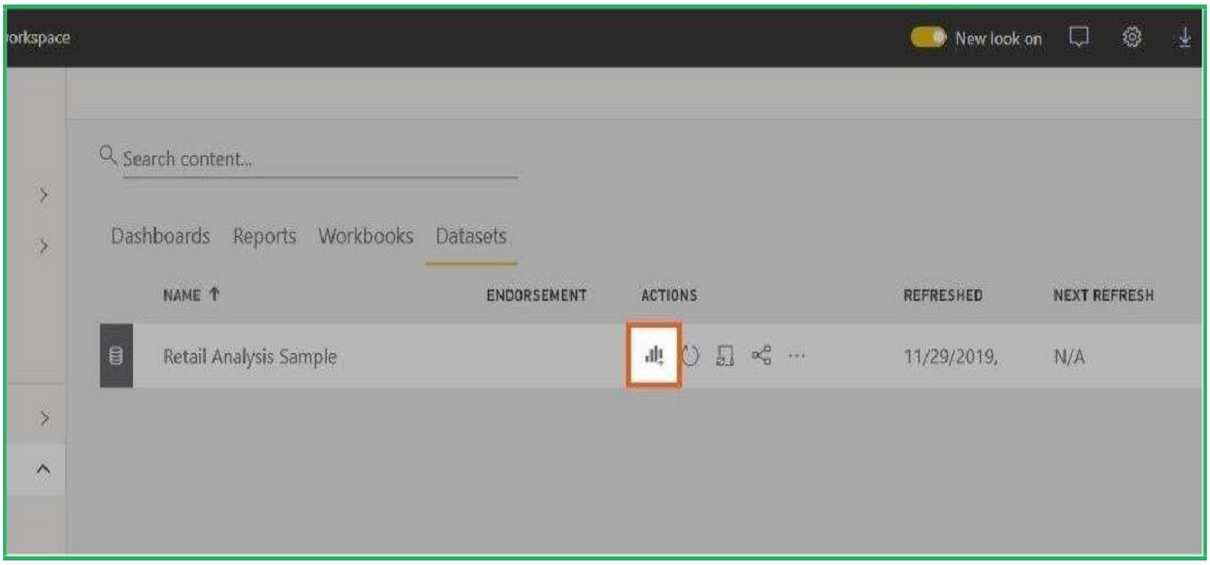
For this exercise, we'll be using the "Retail Analysis Sample" prepared by Microsoft and Obvience.

If you don't know how to get sample data, check out this tutorial: How to Use the Included Sample Data in Power BI (+Examples).

Once you have your dataset, go to the "Datasets" section in your workspace and click the 'Create report' icon.

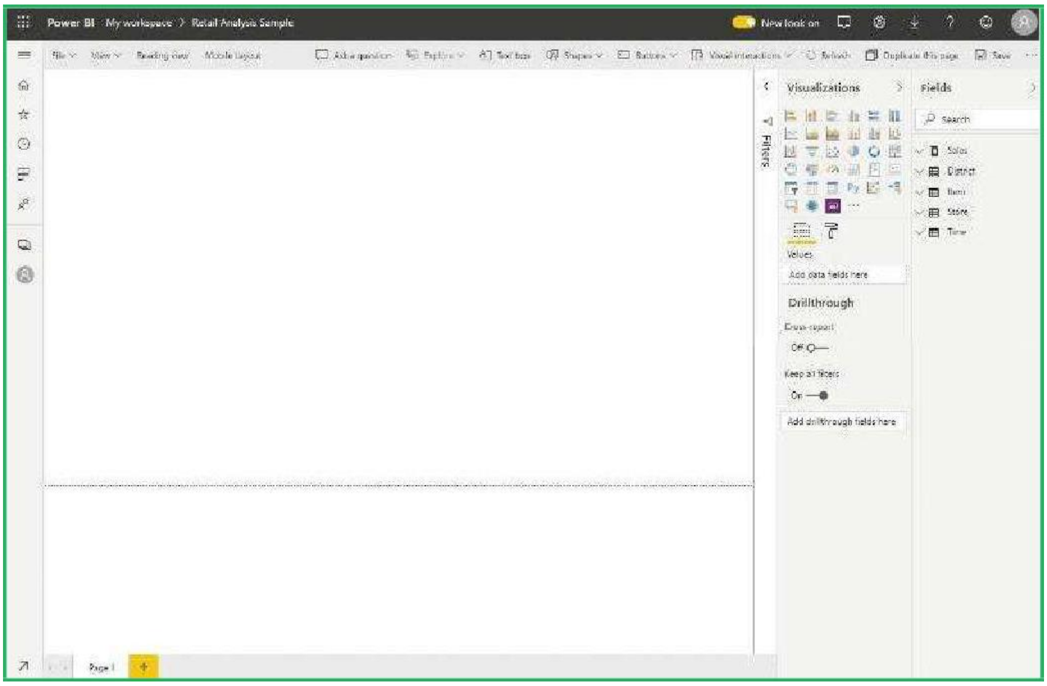
In our example, the available dataset is the "Retail Analysis Sample":





the Power BI Report Builder

The steps you took from the previous section will bring you to the reports builder:



There are three primary sections you have to know:

- Canvas
- Fields pane
- Visualizations pane

Obviously, the canvas is the empty white space at the center where the visuals will be made.

The fields pane contains the different fields of your dataset (think of it as the columns of your dataset).

The visualizations pane is where you can edit and modify your visualizations like the type of visualization, the format, and specific options for the values of your visual.

Creating Your First Power BI Report

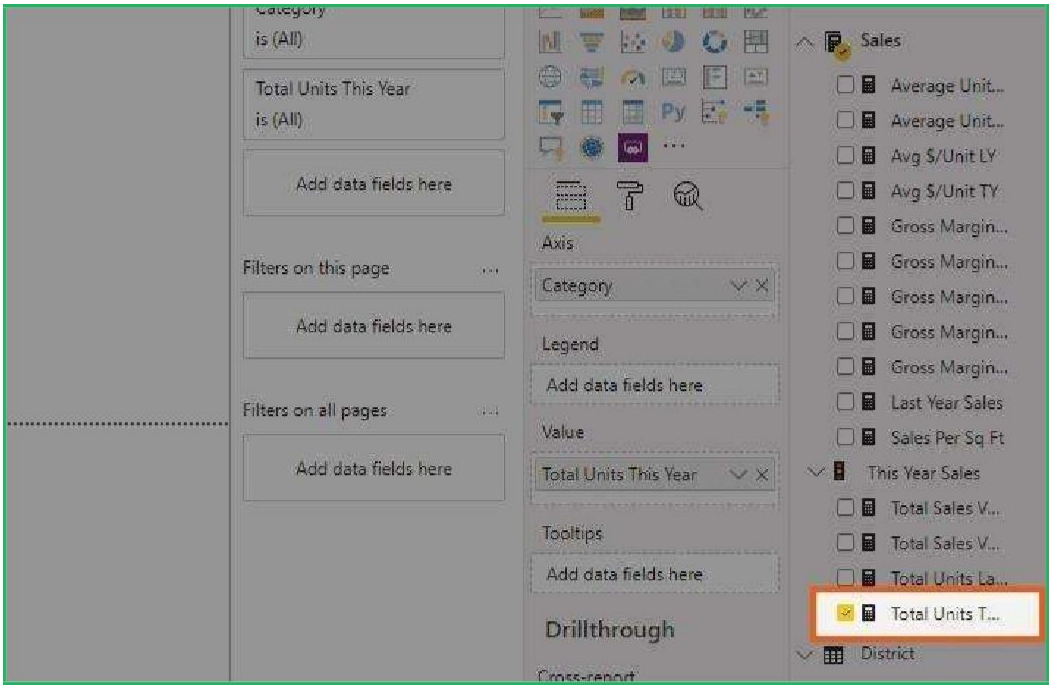
If you have tried creating charts and graphs for your data in Excel, then creating reports in Power BI is even easier.

- Select the fields first then visualizations after
- Or if you have the specific visualization in mind, select the type of visual first then the fields after

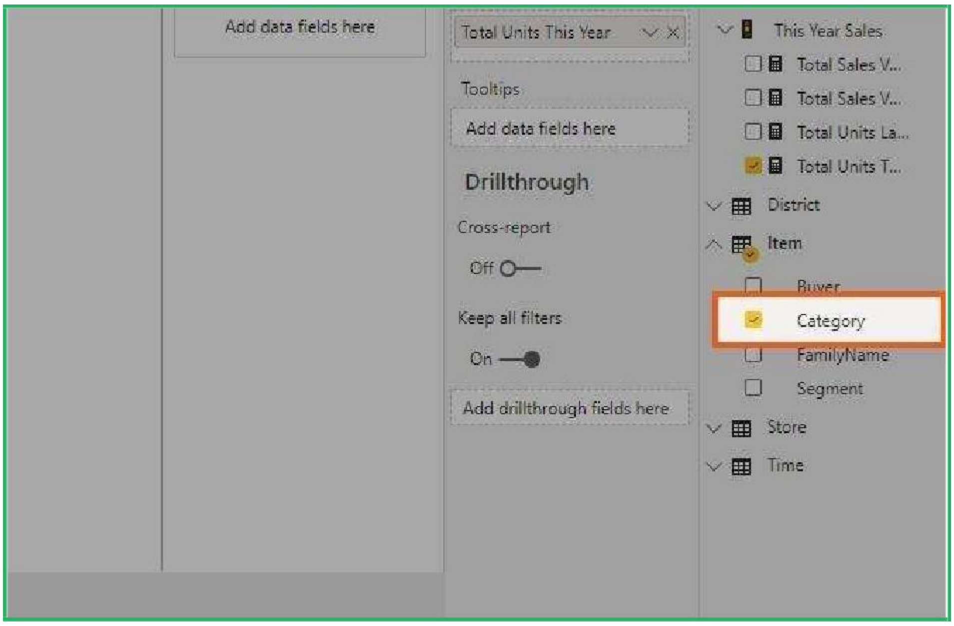
We'll go with the first method.

To start, focus your attention on the fields pane.

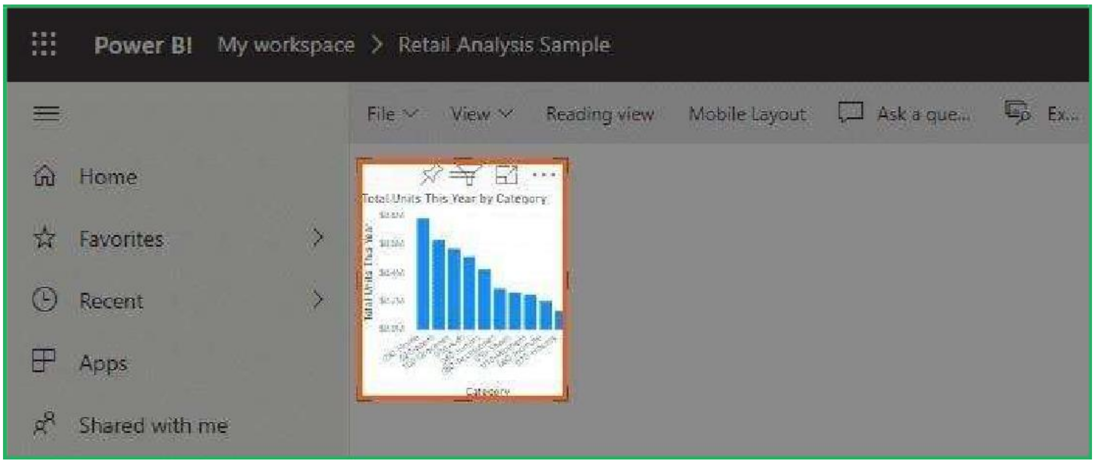
Select "Sales" and then "Total Units This Year":



Select “Item” and then “Category”:



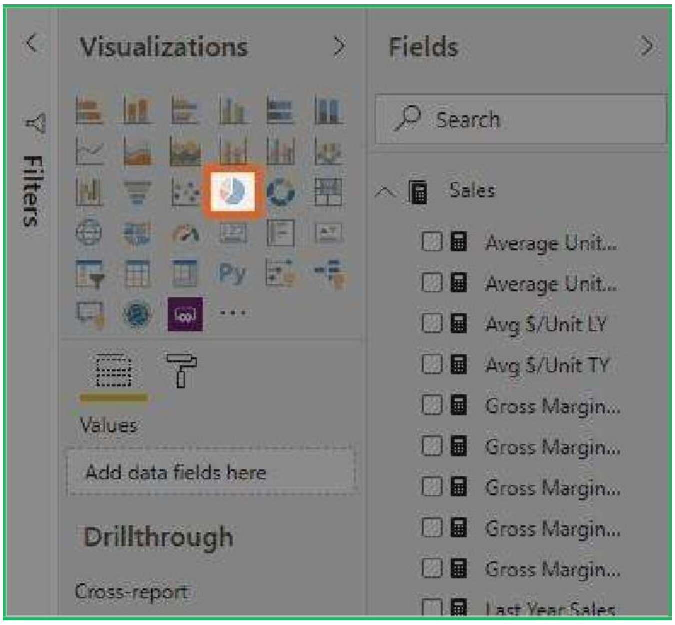
While clicking those fields, a visual will appear on the canvas:



Cool, right? Creating a report has never been this easy.

But now, you think that a pie chart is a better visual for your data.

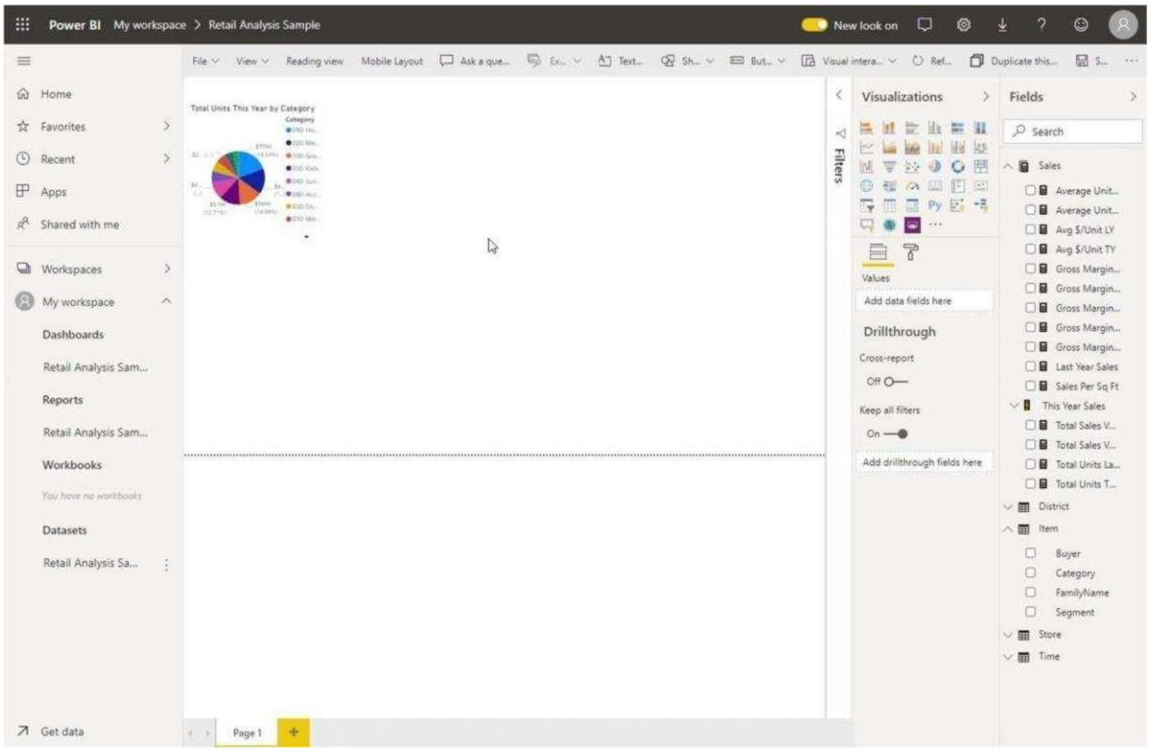
To change the type of visual, just head over to the visualizations pane and click the pie chart icon:



Now you got your report in the visual type you preferred.

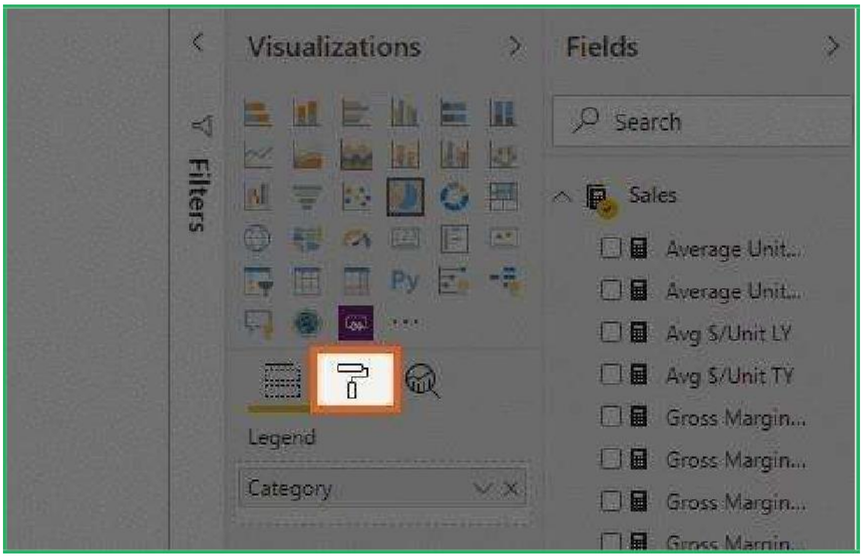
Modifying Your First Power BI Report

For example, feel free to enlarge your visual to the size you want. Simply click on the visual and drag its corners:



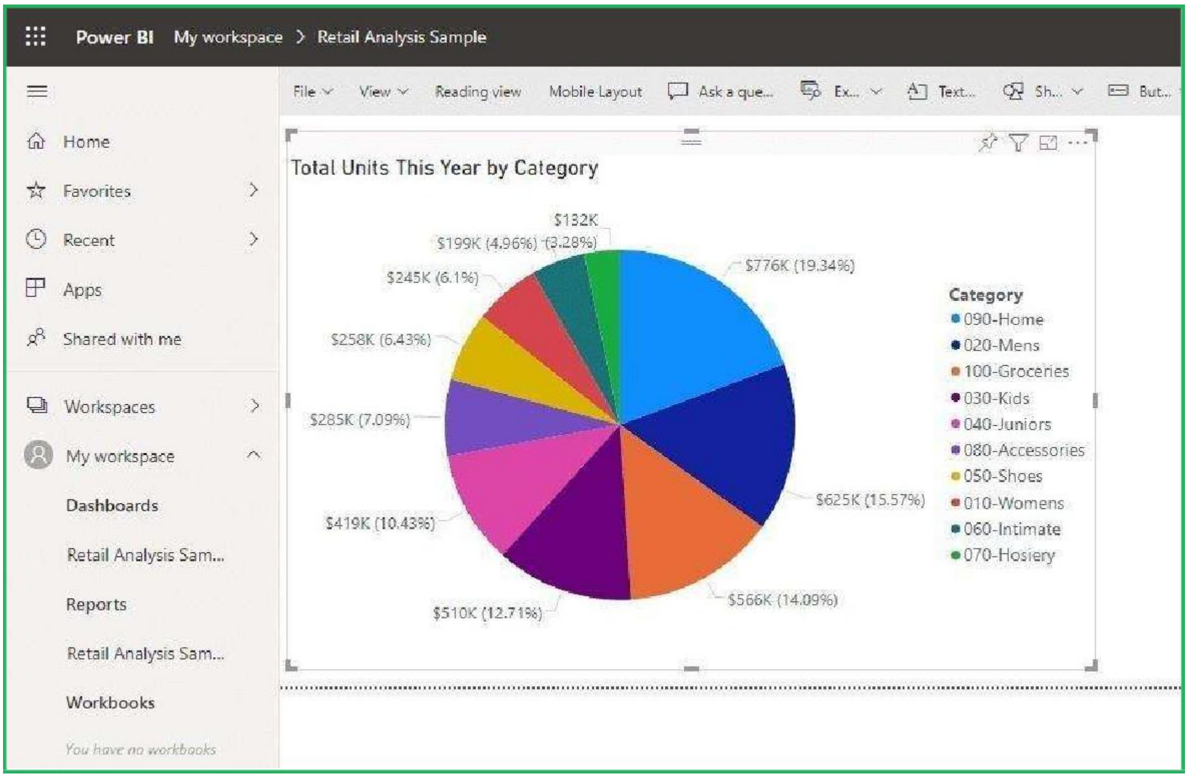
The next part is enlarging the texts — legend, detail labels, and title — of your report.

On the visualizations pane, go to the 'Format' section:



Here, you can change the format of your visuals. Each type of visual has different sets of options so an option on another might not be available on another.

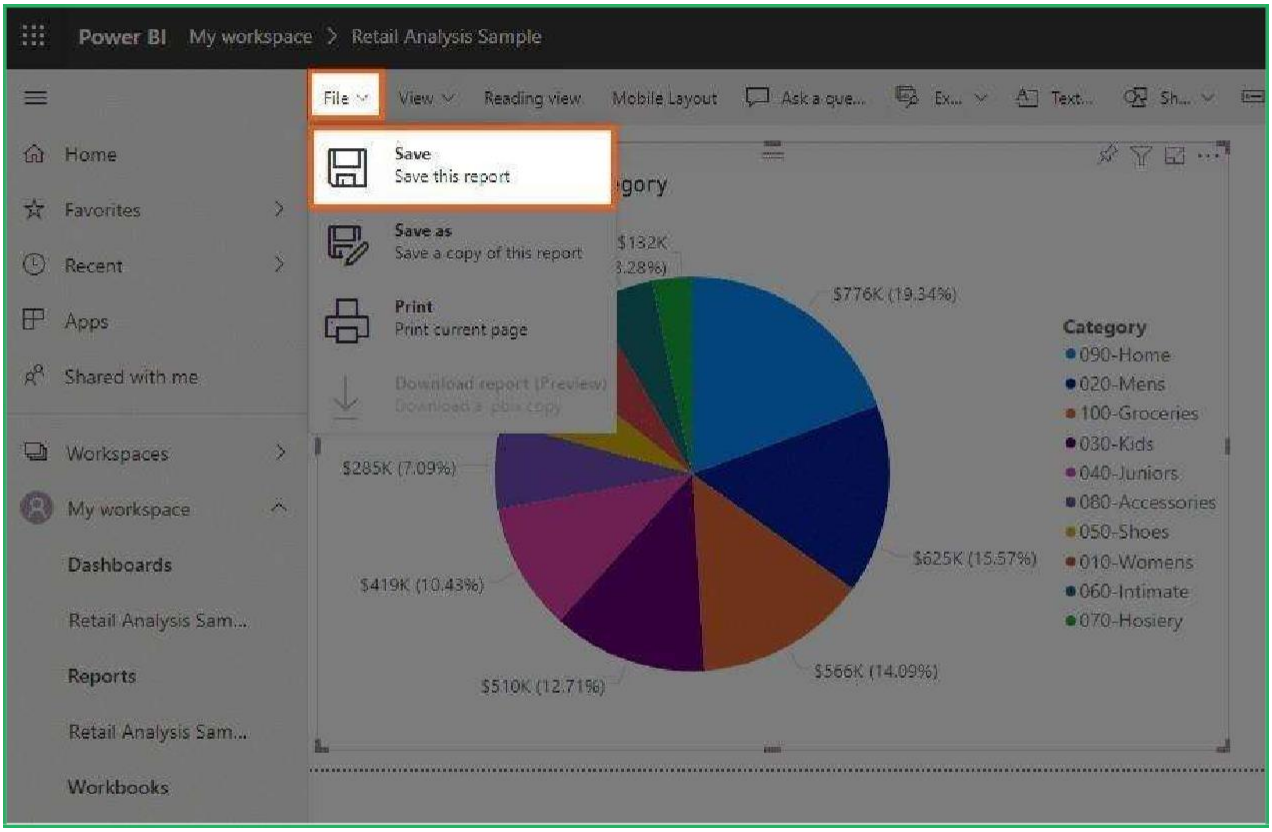
To enlarge the text, simply go through the 'Legend', 'Detail labels', and 'Title' and then adjust the text size:



Saving Your First Power BI Report

If you move out of the report builder without saving your report, the work you did would be for nothing.

To save your report, click 'File' from the tab list and select 'Save':



Result:

Thus the report was designed successfully using power bi tool.

Ex.No.14	**Create a dashboard and perform data analysis**
DATE:	

AIM:

To create a dashboard and perform data analysis

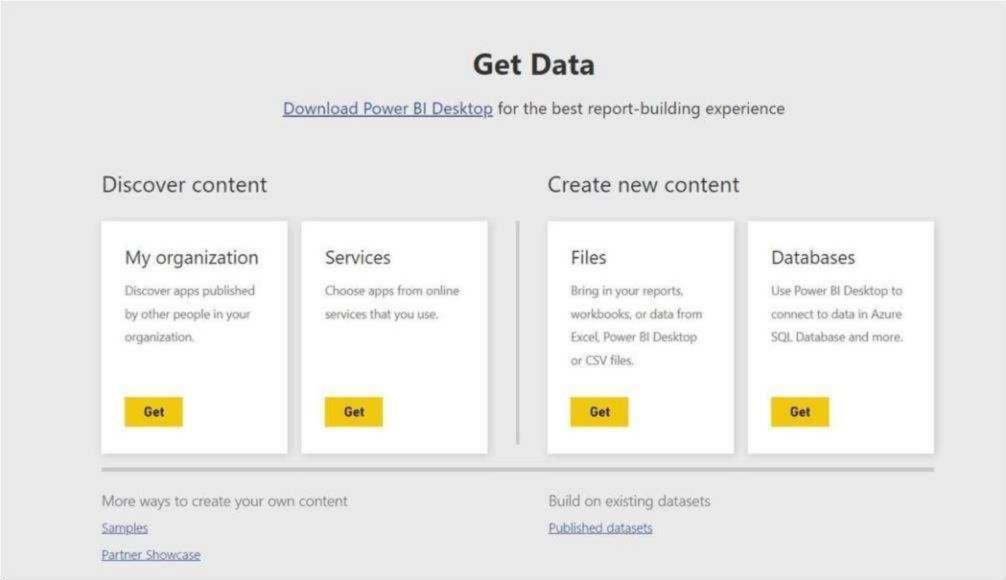
PROCEDURE:

Importing Data

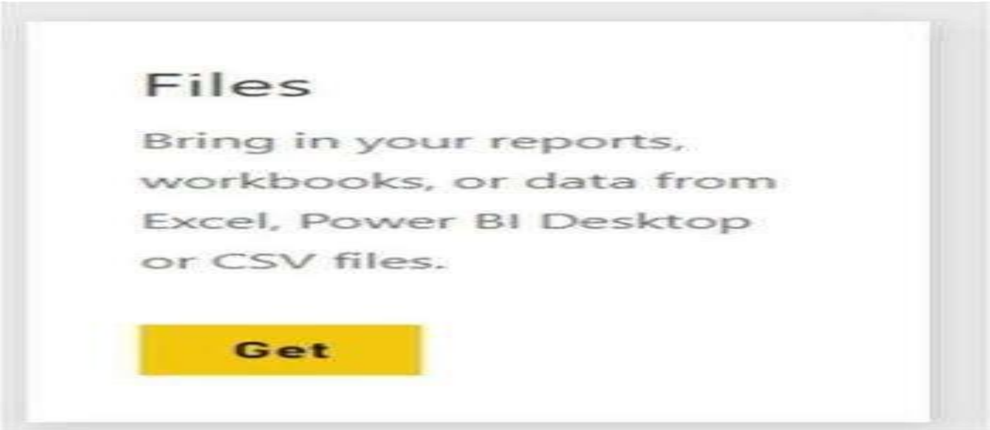
The first step in building a Power BI dashboard is to import the dataset that will be used to build the report. You can connect to a variety of data sources, including Excel worksheets, databases, the web, and cloud services.

On the homepage of the Power BI service, you can click on the 'Get Data' button found at the bottom right of your window.

A new 'Get Data' window will come up, and we should be able to preview some of the major data sources available for building our dashboard.

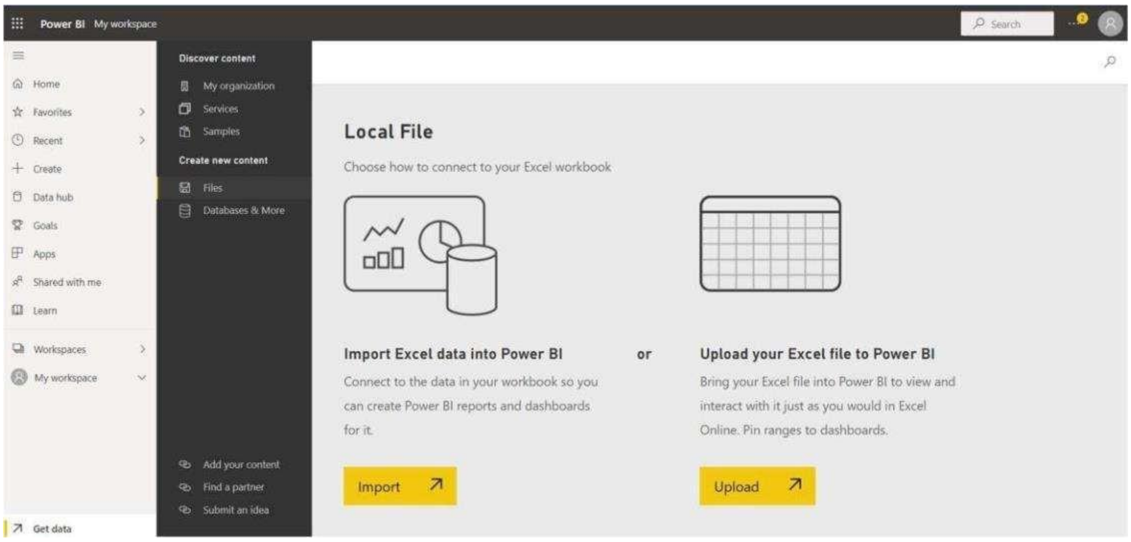


Our file type is an Excel workbook. Next, we will click on the 'Get' button in the 'Files' section.



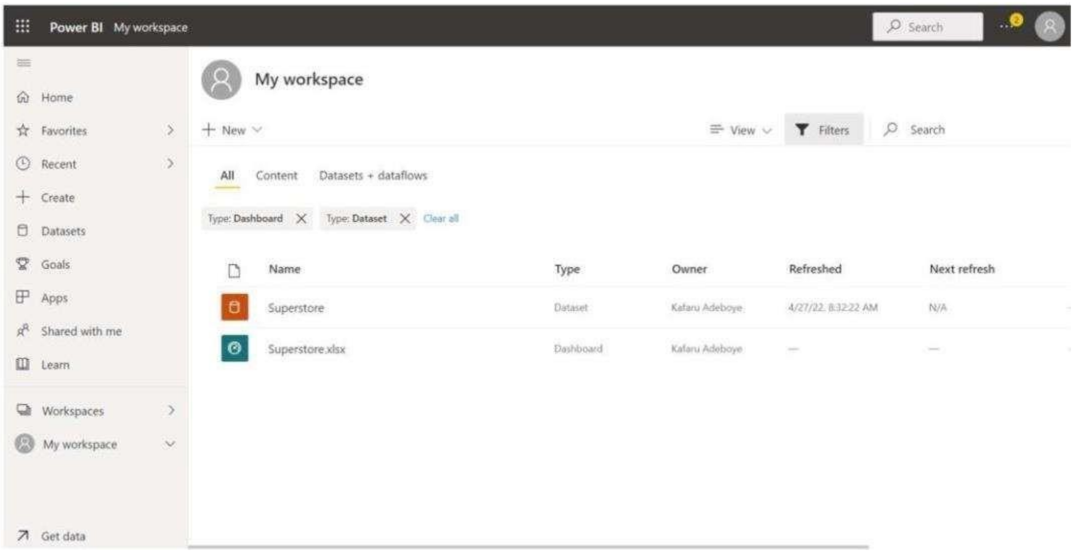
Choose the 'Local File' option from the list of options on the next page and upload the Excel file.

- Note: If we are importing an Excel workbook, we will have an extra window as shown below. Select the 'Import' option. A 'Success!' pop-up will then appear.



The Superstore dataset upload (downloadable here) should be visible on the workspace page. If not, reload your browser page.

You can also make use of other datasets provided by DataCamp.

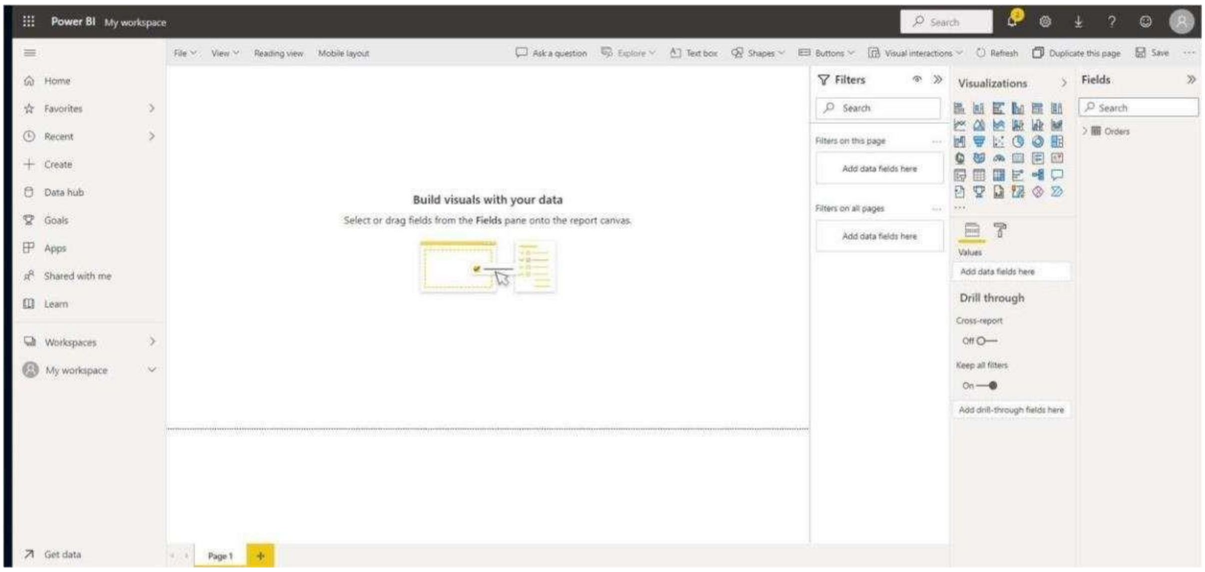


Opening a Report from the Uploaded Data

Notice that from the workspace image, there are two file types of the same name. One is the dataset and the other is the dashboard.

Click on the Superstore.xlsx dashboard. On the blank canvas that pops up, click on the dataset name 'Superstore.xlsx'.

Finally, a Reporting canvas will appear, similar to the view on Power BI Desktop.



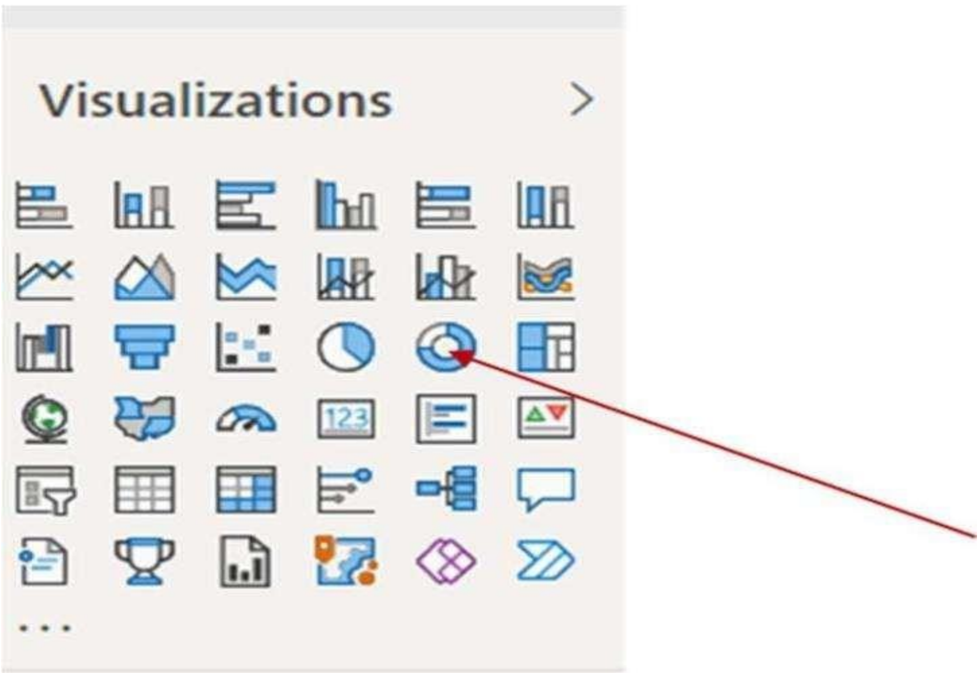
Creating a Tile and 'Pin to a Dashboard'

Power BI report service, just like the desktop version, includes a variety of page formatting options, including visuals, shapes, and images, that can help your report stand out. One of the most efficient ways to identify and communicate insights is to use Power BI to create visuals. You can begin creating your visuals after you have decided on the data that will form the basis of your report, and designed the layout of this report.

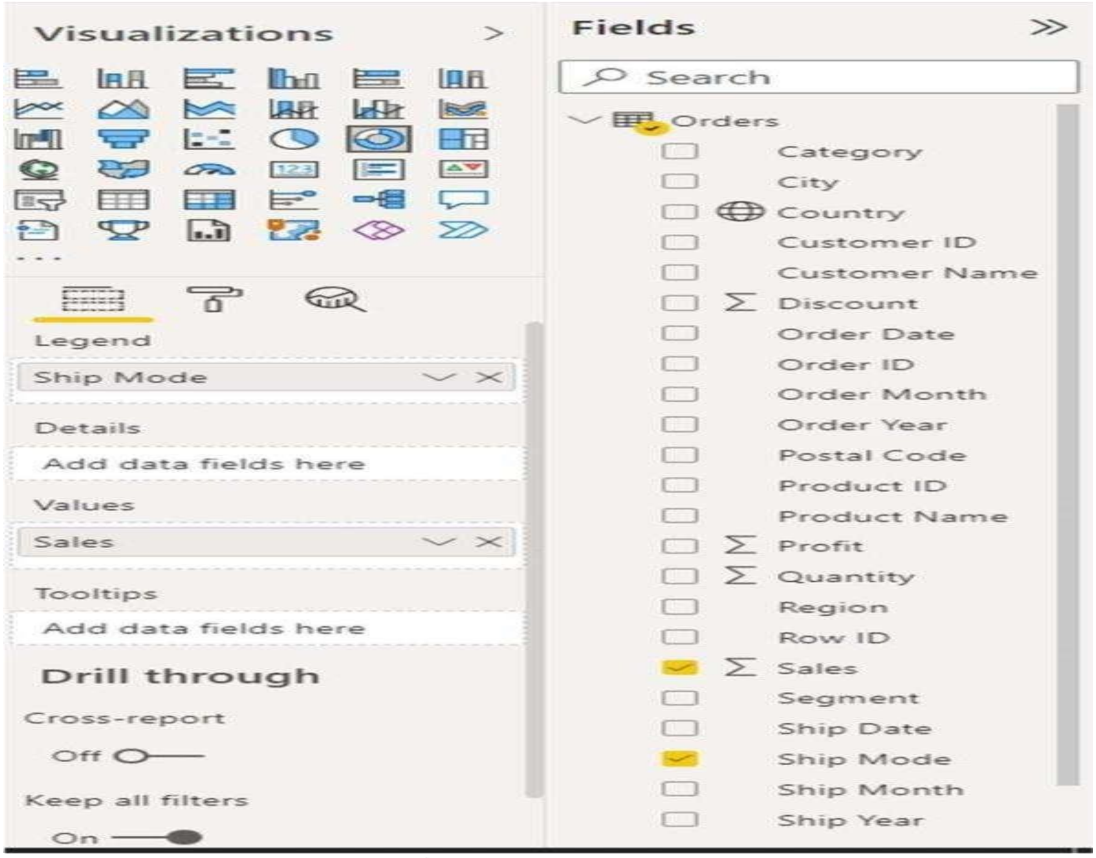
However, before you start working on your data visualization, you must first decide what insights you want to garner, and then develop and personalize visualizations to display the data in an engaging and informative manner.

You can work on how to visualize these insights after you have identified them in order to bring value to your audience. We will create a simple donut chart that shows the 'Sales across the Shipmode'.

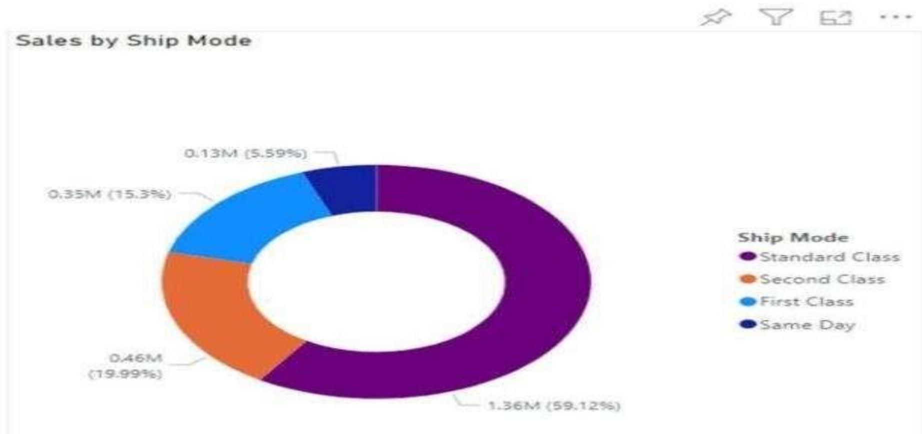
- Select the donut chart option from the visualization pane.



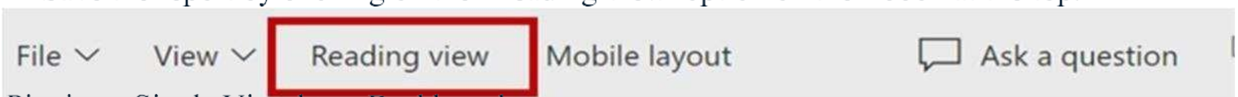
- Drop the Sales in the Values field and Ship Mode to the legend.



- We now have a donut chart.



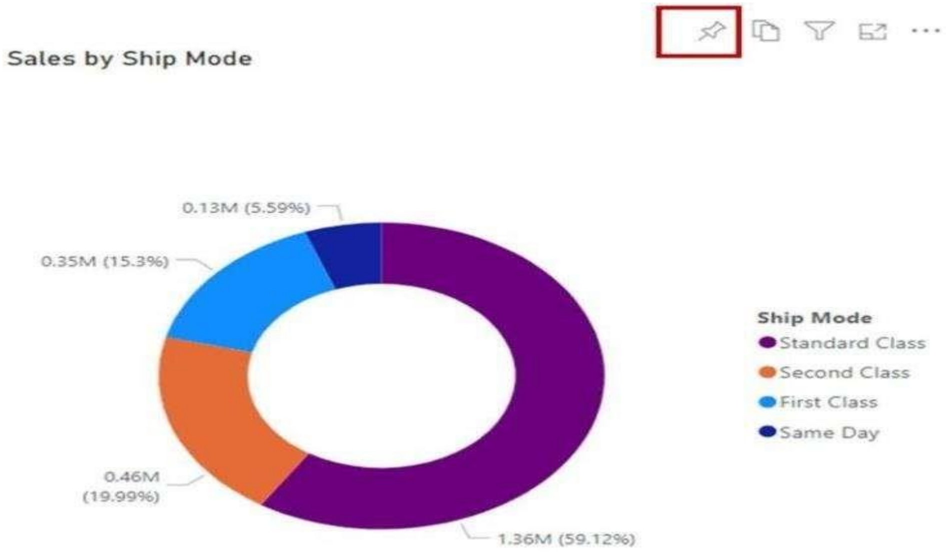
Save the report by clicking on the 'Reading view' option on the ribbon at the top.



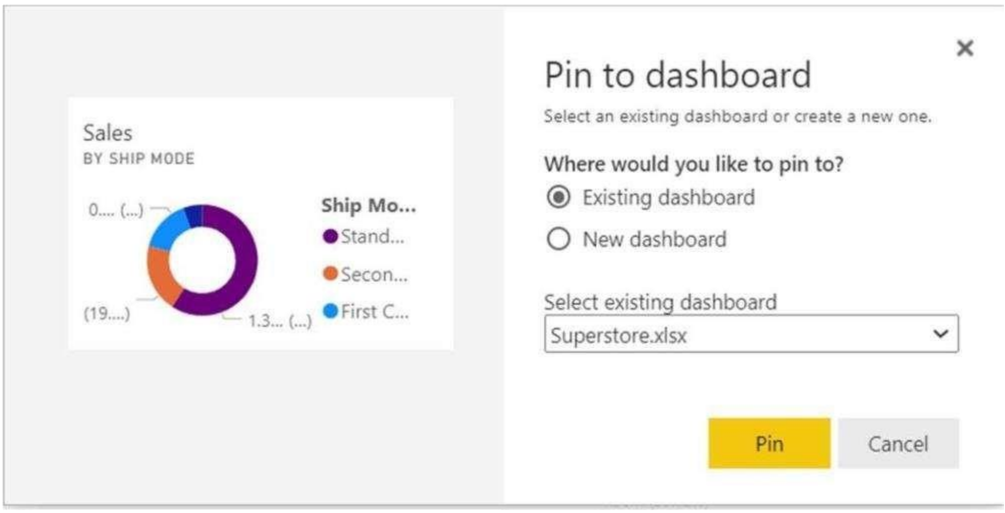
Pinning a Single Visual to a Dashboard

Now that we have created a tile, we can pin this to a dashboard.

-

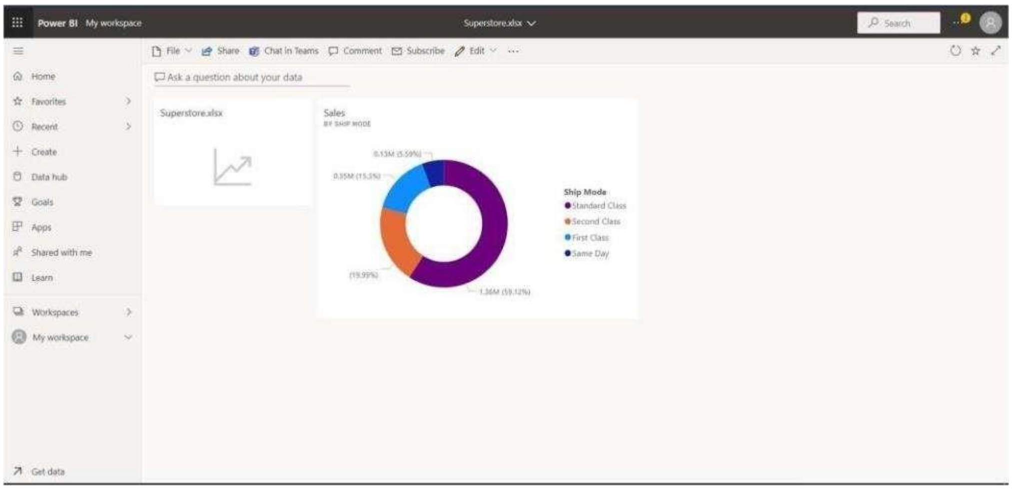


- Hover over the recently created donut chart and click on the pin icon that appears.
- A 'Pin to dashboard' window will appear. You can pin to an existing dashboard that we created while importing the dataset or create a new one. In our case, we will pin to the existing one and then click on the 'Pin' button.



A pop-up will appear on the top right. Click on 'Go to dashboard'.

We will be taken to a new window, the Superstore Dashboard.



Organizing and Updating Tiles

Positioning of Tiles

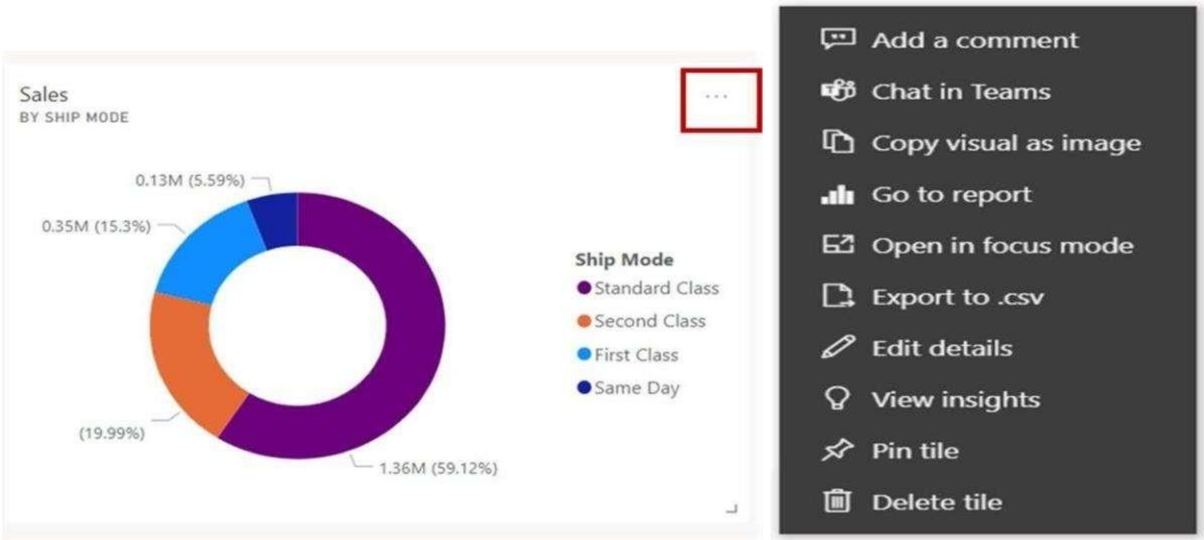
You can move the tile around on your dashboard to a convenient position just by holding a left click on the tile and moving it.

Tile Size

You can increase or decrease your tile size by dragging the little arrow at the bottom right of the tile.

Other Tile Options:

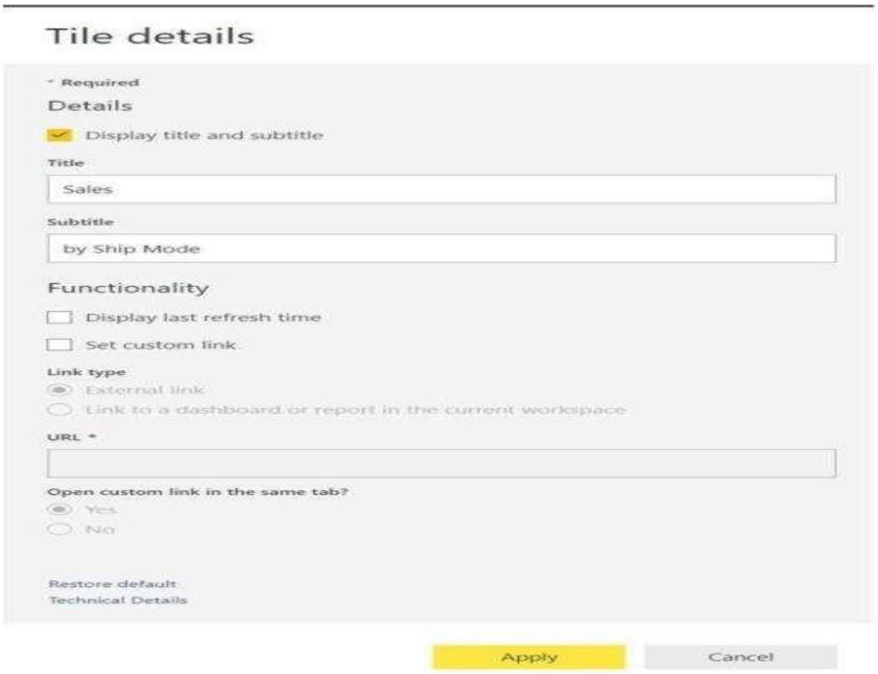
By clicking on the kebab (...) menu on the tile, more options are shown around 'Tile settings'.



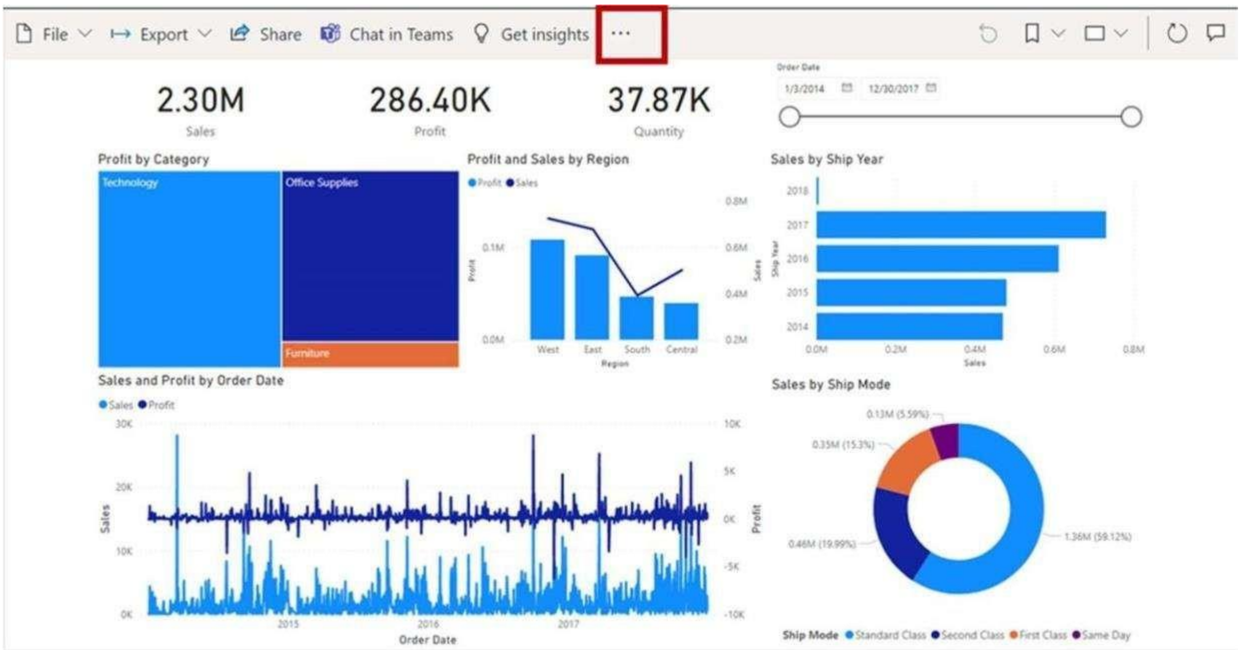
•

Edit Details

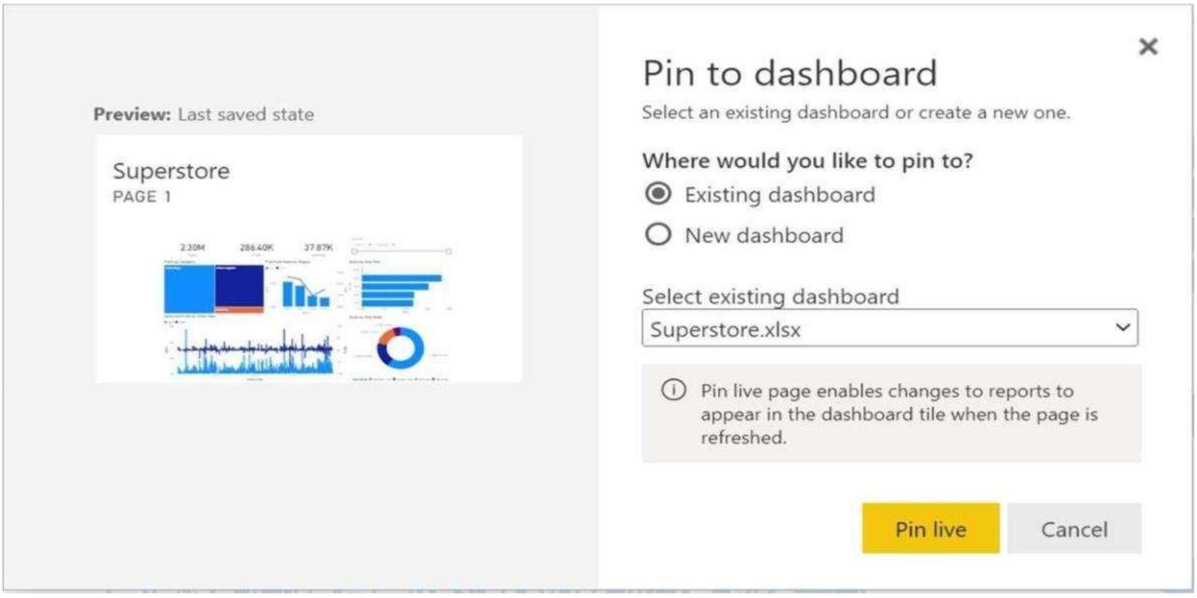
From the option above, we can click on 'Edit details'.



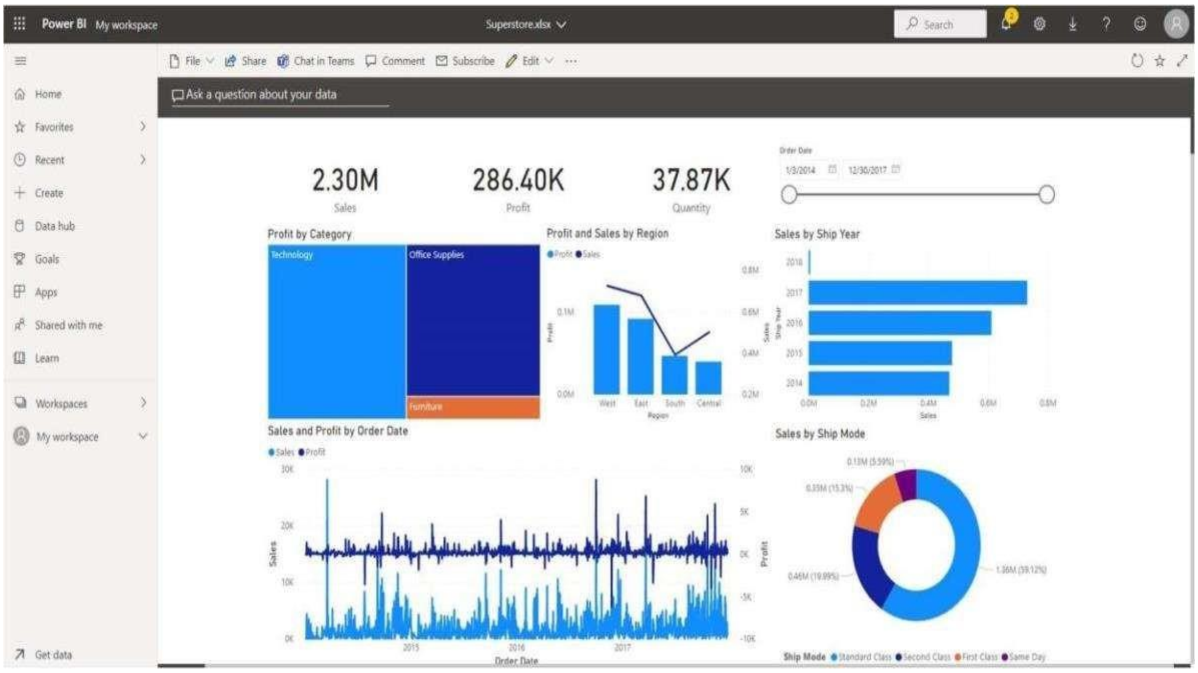
Pin an Entire Report to a Power BI Dashboard



Choose the 'Pin to a dashboard' option.



- Click the ‘Pin live’ option. Then select the ‘Go to dashboard’ option.
  Note: when pinning at a report level, any changes that happen to the report will also be taken into effect on the dashboard.
- You should now see your report page pinned to a dashboard.



Result:

Thus the dashboard was created and data analysis was performed successfully.